

Probability and applications

Facilitator guide and complete solutions / Remaining atlas entries / September 2026

Give one student page at a time; allow exploration before hints. The accompanying check programs verify finite instances and identities. General statements need the arguments or supplied facts in the keys. Preparation, timing and classroom engagement remain untested.

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Print selected pages, single-sided, on US Letter at actual size. Keep student pages and solutions separate. The original random-ten book remains a separate volume.

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The listed tools are prerequisite gates, not age labels. More detailed gates appear in the facilitator guide.

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Print selected pages, single-sided, on US Letter at actual size. Keep student pages and solutions separate. The original random-ten book remains a separate volume.

The same clue, a different experiment

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

First-pilot candidate for a group comfortable with fractions. The protocol distinction creates the surprise; avoid turning the repeated-draw page into long simulation or presenting posterior fractions before learners filter the cards.

Two correct but different answers to the red clue, proved from the mechanism; then a complete two-draw evidence comparison.

Prerequisites and preparation

Core: Counting equally likely labeled outcomes and fractions; distinguish a random selection from a deliberate report. No algebra is needed for the complete finite certificate.

Extensions: Multiply simple fractions or count a two-draw grid; prior odds and likelihood ratios are an optional algebraic continuation.

Reading and representation: Short protocol cards; labels A1,A2,A3,B1,B2,B3. An adult can read the protocol, but the learner must keep its two choices distinct.

Materials: Two cups; A has labeled counters A1=R,A2=R,A3=B, and B has B1=R,B2=B,B3=B; fair coin; three equally likely bag-ticket cards A,B,B; six outcome cards and two blank 3×3 outcome grids.

Preparation: about 10 minutes.

0-10 explore the supplied objects; 10-15 rehearse exact rules; 15-35 investigate page 1; 35-40 move/reset; 40-55 choose a certificate or new representation; 55-60 explain one result. Pages 2-3 can be a second meeting; no requirement to complete the packet in one hour.

Launch and facilitation

Two messengers both say “red.” Must their clues tell you the same thing about the hidden bag?

Student page 1: Which messenger should change your mind?

Count equally likely outcomes; fractions; distinguish drawing from reporting.

Bag A holds R,R,B; bag B holds R,B,B. For each experiment choose one bag by a fresh fair coin. Messenger DRAWS takes one uniformly random counter. Messenger CHOOSES looks inside and deliberately shows a red counter whenever one is present. You hear “red” from one specified messenger.

The same clue, a different experiment

1 / prompt and solution

Choose a messenger. Before calculating, predict whether red favors A, favors B, or leaves them equally likely. Build the possible bag-and-counter outcomes for DRAWS. Why may you give those cards equal weight?

Solution: The six labeled outcomes each have probability $(1/2)(1/3)=1/6$. Before the draw A and B are equally likely. DRAWS reporting red leaves A1,A2,B1, so A is favored. Predictions are provisional; the explanation must use bag choice AND within-bag choice.

1 / hints to offer only after exploration

1. Label individual counters, including the two reds in A.

2 / prompt and solution

Find the chance of A after each messenger says red. Explain why the identical word did not force an identical answer.

Solution: DRAWS: two of the three surviving equal-weight outcomes are from A, giving $2/3$. CHOOSES: both bags contain red, so the specified messenger always says red regardless of the fair bag choice; the posterior is $1/2$. Do not incorrectly treat deliberate choice among counter colors as uniform random drawing.

2 / hints to offer only after exploration

1. Can CHOOSES ever give a different report with these two bags?

Student page 2: Ask the same bag twice

Count equally likely outcomes; fractions; distinguish drawing from reporting.

Choose A or B once by a fair coin. Draw uniformly, record the color, REPLACE the counter, mix, and draw again independently from that same bag. Each bag has nine labeled ordered draw pairs. The bag is not reselected between draws.

3 / prompt and solution

Use the two 3×3 grids to locate every way to hear RR. What chance of A does that clue give? Compare it with a single red draw.

Solution: The 18 bag-and-ordered-counter-pair outcomes each have probability $1/18$. A contributes four RR cells, B one: $P(A|RR)=4/5$, larger than $2/3$ after a single red. Replacement is essential to these grids; the same physical counter may appear twice.

3 / hints to offer only after exploration

1. Rows are first counter identity; columns are second counter identity.

The same clue, a different experiment

4 / prompt and solution

Choose one new report: “red then blue,” “the two colors differ,” or “at least one red.” Find its answer, then trade proofs with a group that chose another report.

Solution: RB: A contributes 2 and B contributes 2, giving $1/2$. Different colors: A contributes 4 and B contributes 4, giving $1/2$. At least one red: A contributes 8 and B contributes 5, giving $8/13$. Counts come from the same 18 equiprobable cells. RB records order, whereas the other reports pool cells; pooling is part of the reporting mechanism.

4 / hints to offer only after exploration

1. Precisely which grid cells make your chosen sentence true?

Student page 3: Can you cancel a clue?

Count equally likely outcomes; fractions; distinguish drawing from reporting.

Now select the bag by shuffling three equally likely tickets A,B,B and taking one. Then make ONE uniform counter draw. Every ticket-and-counter pair is equally likely; the two B tickets remain distinct.

5 / prompt and solution

After a red draw, are A and B equally likely? Build a card proof before using a formula.

Solution: There are 9 equal-weight micro-outcomes. Red arises from 2 A outcomes and 1 on each B ticket, hence 2 B outcomes: posterior $A=2/4=1/2$. Prior $A=1/3$; the clue still changes the prior even though its posterior happens to be $1/2$.

5 / hints to offer only after exploration

1. Keep both B tickets in the sample space.

6 / prompt and solution

Invent two equally sized bags that make a random red draw leave a fair bag choice unchanged. Make another pair where red favors A as strongly as possible while BOTH bags can produce red. State the bag size you chose.

Solution: Unchanged: both bags need the same positive red fraction, e.g. 2R and 2B in each four-counter bag. For a chosen common size $n \geq 1$ with at least one red per bag, the strongest A-favoring design has n reds in A and 1 red in B, giving $n/(n+1)$; the numerator cannot exceed n and the competitor count cannot fall below 1. If size is unrestricted, there is no single strongest finite design: $n/(n+1)$ increases toward 1. A learner may certify a fixed-size optimum without claiming a global maximum.

The same clue, a different experiment

6 / hints to offer only after exploration

1. Decide whether your bag size is fixed before saying “best.”

Worked outcome map

Facilitator key: replacement draws from one fixed bag

Bag A			Bag B				
	A1	A2	A3		B1	B2	B3
A1	RR	RR	RB	B1	RR	RB	RB
A2	RR	RR	RB	B2	BR	BB	BB
A3	BR	BR	BB	B3	BR	BB	BB

All 18 cells have probability $1/18$. RR leaves four A cells and one B cell: $4/5$. The first draw is the row; replacement permits a counter to appear twice. A different report selects a different event in this same space.

Optional facilitator extension 1

Gate: Fractions, powers and algebra; finite Bayes conditioning.

After selecting A with probability p and B with $1-p$, derive the posterior odds after k reds in the replacement experiment.

Solution: Prior odds $p/(1-p)$ multiply by $[(2/3)/(1/3)]^k = 2^k$, so posterior probability is $p \cdot 2^k / (p \cdot 2^k + 1 - p)$, for $0 < p < 1$. This conditions on exactly the specified ordered observations; stopping/report-selection changes may require reanalysis.

The same clue, a different experiment

Mathematics and prior use

Finite conditional probability and likelihood ratios. The outcome cards are the actual sample space; no continuous prior or real-world bag-selection model is silently assumed.

Prior use: Year-1 probability is related, and AP-01 in the trial uses fair paths. This new question concerns selection of evidence, not endpoint probabilities. Original atlas bag counts remain but the two-messenger and two-draw comparison are developed here.

Source: Grinstead and Snell, Introduction to Probability

§4.1 Discrete Conditional Probability, especially Bayes formula, printed pp.133-146.. The textbook supplies the conditioning framework. Messenger protocols, two-draw grids and fixed-size extremal design are independently derived finite instances. Evidence: Reopened the author-hosted PDF, checked §4.1 and Bayes discussion; all outcome counts independently enumerated in `ap1-checks.py`.

<https://math.dartmouth.edu/~prob/prob/prob.pdf>

How surprising are these labels?

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Promising with explicit statistical reasoning gates. A complement-pair representation avoids twenty rows of repetitive arithmetic. “Unusual under this no-effect model” is the payoff, never “probability training is useless.”

A complete 20-assignment reference set and a prespecified tail count, followed by an explanation of why a different assignment design gives a different answer.

Prerequisites and preparation

Core: Choose subsets systematically, add six small scores and compare sums; fractions such as $1/20$. The real reasoning gate is fixing the null assumption and actual assignment rule.

Extensions: Understand the difference between a prespecified one-sided rule and a rule selected after seeing the result; paired randomization uses three independent fair coins.

Reading and representation: Fictional experiment rules and named participant cards; no health claim or real treatment inference. Adult may read the rules.

Materials: Six cards $A=0, B=1, C=2, D=4, E=5, F=6$; three TRAINED clips; shuffled labeled assignment slips or uniformly shuffled participant cards; three fair coins for the paired variation. Preparation: about 10 minutes.

0-10 explore the supplied objects; 10-15 rehearse exact rules; 15-35 investigate page 1; 35-40 move/reset; 40-55 choose a certificate or new representation; 55-60 explain one result. Pages 2-3 can be a second meeting; no requirement to complete the packet in one hour.

Launch and facilitation

Three of six glider pilots were chosen uniformly for training. The trained pilots got the top three scores. Could assignment alone have produced that?

Student page 1: Assign the clips before explaining the result

Small subset counts and means; explicit null model and random assignment.

Six fictional pilots have scores $A=0, B=1, C=2, D=4, E=5, F=6$. Before flying, exactly three received TRAINED labels, uniformly among all three-person subsets. The observed trained group is D,E,F. For this investigation ASSUME training changed no pilot's score. Keep scores fixed and move only labels.

How surprising are these labels?

1 / prompt and solution

Try three possible assignments. Use the sum of trained scores to say which looks strongest. Why can sums replace means here?

Solution: Any three distinct participant labels form a possible assignment. Because both groups have size 3, higher trained sum S means higher trained mean $S/3$, and trained-minus-other mean is $(2S-18)/3$. Total score 18 is fixed. The observed DEF sum is 15; the other group sum is 3 and mean difference 4.

1 / hints to offer only after exploration

1. The untrained sum is the total minus the trained sum.

2 / prompt and solution

Find ALL possible assignments without listing any twice. Try putting A in one group, then pairing that assignment with the assignment that swaps the two group labels. Explain why this is complete.

Solution: Choose two partners for A from B,C,D,E,F: 10 choices. Each resulting triple and its complementary triple form a pair; exactly one contains A. This gives 20 distinct assignments, all equally likely by the stated mechanism. The ten A-containing sums are 3,5,6,7,6,7,8,9,10,11; complementary sums are 18 minus those. The sheet supplies an open paired-record area, without precomputed sums or a suggested number of pairs.

2 / hints to offer only after exploration

1. Exactly one member of each complementary pair contains A.

Student page 2: Choose the alarm rule before looking

Small subset counts and means; explicit null model and random assignment.

Before data, choose the statistic and direction. UP means larger trained sums are more extreme. EITHER WAY means larger absolute trained-minus-other mean differences are more extreme. Now the observed trained sum is 15 and the mean difference is 4; compare all assignments against these observed values.

3 / prompt and solution

Under the no-effect assumption, what fraction of assignments trigger UP? What fraction trigger EITHER WAY? Could an UP tail ever have probability smaller than $1/20$ in this exact design?

Solution: Only DEF has $S \geq 15$, so UP has $1/20$. EITHER WAY requires $S \geq 15$ or $S \leq 3$; DEF and ABC give $2/20 = 1/10$. A nonempty tail includes the observed assignment and each assignment has probability $1/20$, so no exact nonrandomized tail can be smaller than $1/20$. Ties could make the smallest tail larger.

How surprising are these labels?

3 / hints to offer only after exploration

1. Use complementary assignments to avoid recomputing the negative extreme.

4 / prompt and solution

A reporter waits to see which group did better, then advertises the one-sided 1/20 count in that direction. Under no effect, how often would this “pick the favorable direction” procedure flag either extreme? Explain what the reporter left out.

Solution: The adaptive procedure flags either DEF on top or ABC on top, two of 20 assignments, so its false-alarm probability at this extremeness is 1/10. Reporting 1/20 ignores that either tail could have been chosen after the data. This is a specified finite comparison, not a blanket statement that every data-dependent analysis doubles a p-value.

4 / hints to offer only after exploration

1. Write the full set of outcomes that could make the reporter boast.

5 / prompt and solution

What have the counts established-and what have they NOT established? Include the assumption that made scores stay fixed.

Solution: They give frequencies under the sharp no-effect null and the uniform three-of-six assignment design. 1/20 is not the probability that training has no effect; rare events still occur. No claim about a population of pilots follows without a sampling/generalization model. If treatment assignment depended on likely performance, the twenty outcomes would not have the stipulated equal probabilities.

5 / hints to offer only after exploration

1. A probability calculated assuming a model is not automatically a probability that the model is true.

Student page 3: A new assignment rule changes the answer

Small subset counts and means; explicit null model and random assignment.

Instead, form fixed pairs (A,F),(B,E),(C,D). Before the flights, use an independent fair coin in EACH pair to select exactly one trained pilot. Keep the same sharp no-effect assumption and observed trained group D,E,F.

How surprising are these labels?

6 / prompt and solution

List the possible trained groups using a three-coin tree. How often is their total at least 15 now? Why would using the old twenty-outcome map be wrong?

Solution: There are $2^3=8$ equiprobable assignments, choosing one from each pair. Their trained sums are 3,5,7,9,9,11,13,15. Only DEF reaches 15, so $1/8$. Many triples in the complete-randomization design contain both members of a fixed pair or none; they are impossible here. The reference distribution must match actual randomization.

6 / hints to offer only after exploration

1. Write one choice from each pair, never two from the same pair.

7 / prompt and solution

Design a one-sentence explanation for someone who says, “The scores stayed the same, so the probability had to stay the same.”

Solution: An acceptable explanation identifies both the fixed outcomes and the probability law over label assignments: the score cards are unchanged, but the sampling experiment now has 8 equally likely allowed assignments rather than 20. The observed extreme therefore carries a different tail proportion. Diagrams of 20 versus 8 legal assignments are a concrete certificate.

7 / hints to offer only after exploration

1. The assignment mechanism is part of the mathematical data.

Optional facilitator extension 1

Gate: Combinations with repeated values but distinct participant labels.

Replace outcomes by 0,0,1,1,2,2 but keep all six individuals labeled. What is the smallest upper tail for the top-three sum?

Solution: The maximum trained sum is 5, requiring both 2-score people and either one of the two 1-score people. Thus two of 20 assignments tie at the top and the smallest upper tail is $2/20=1/10$. Equal scores do not merge distinct assignment outcomes.

How surprising are these labels?

Mathematics and prior use

Exact randomization inference under the sharp null: each participant would have the same score with or without training. Scores are fixed only for that hypothetical calculation; treatment labels follow the actual random assignment.

Prior use: No matching inference worksheet in Weeks 1-10 or random-ten. AP-02 also conditions on a mechanism, but this asks whether randomized labels alone can explain a prespecified extremeness statistic.

Source: Allen Downey, Think Stats, third edition

Think Stats 3, §9.2 simulation/permutation mechanics and chapter 9 discussion of test statistics and null models.. The text supplies general test/permutation terminology; the sharp-null randomized-experiment argument and paired-design comparison are derived here, not attributed to observational exchangeability examples. Evidence: Reopened chapter 9, inspected the statistic/null/permutation discussion. Independently enumerated every 20 and 8 assignment and both tails.

<https://allendowney.github.io/ThinkStats/chap09.html>

A very precise wrong answer

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

First-pilot candidate when averages are familiar. The two indistinguishable hidden worlds produce a genuine impossibility argument rather than merely saying “bias is bad.” Avoid repeated draws after the mechanism is already clear.

Two different hidden populations yield exactly the same allowed observations, proving that more restricted data cannot determine the full mean; an improved sampling design is justified.

Prerequisites and preparation

Core: Means of small numbers, counting equally likely draws, and distinguishing the whole target population from the draw bag. The non-identifiability explanation needs systematic reasoning.

Extensions: Weighted averages with known group sizes; no calculus or asymptotic theorem needed.

Reading and representation: Card identities 1-8 and two draw protocols. The first reveal is staged; an adult can run the covered-card portion.

Materials: Eight numbered covered cards; values 1,1,1,1,5,5,5,5 for the first world; replacement values 9,9,9,9 for cards 5-8 in the alternate world; two draw bags of identifiers; counters; pencil. Preparation: about 10 minutes.

0-10 explore the supplied objects; 10-15 rehearse exact rules; 15-35 investigate page 1; 35-40 move/reset; 40-55 choose a certificate or new representation; 55-60 explain one result. Pages 2-3 can be a second meeting; no requirement to complete the packet in one hour.

Launch and facilitation

A sampler always reports 1. Does its perfect consistency prove the whole population’s average is 1?

Student page 1: Could two different worlds look identical?

Small population means; distinguish the sampling frame from the target.

The target is the mean of ALL eight cards. Cards 1-4 are 1,1,1,1. Cards 5-8 are hidden. The sampler chooses an identifier uniformly only from 1-4, reads its value, replaces it, and repeats independently. You may request any number of these readings.

A very precise wrong answer

1 / prompt and solution

Make two different possible sets of hidden cards 5-8 that produce exactly the same readings from this sampler but have different whole-population means. Use values from 0 to 9.

Solution: Many answers. World W has hidden 5,5,5,5 and total mean 3; World Z has hidden 9,9,9,9 and total mean 5. Both always produce 1 under the restricted rule. The same observations persist for every finite or infinite sequence of allowed draws because none accesses cards 5-8.

1 / hints to offer only after exploration

1. Can changing a card the sampler never visits change the full mean?

2 / prompt and solution

Could a million readings from this same sampler distinguish your two worlds? Propose one change to what may be observed that could help. State exactly what it would establish.

Solution: No: the observation distributions are identical. Sampling identifiers 1-8 allows unseen cards to appear; auditing one of 5-8 distinguishes the two specific homogeneous example worlds, but one audited value alone does not determine arbitrary four-card hidden contents. Reading all hidden cards would determine the exact population mean. A remedy must change information access, not just the number of identical restricted draws.

2 / hints to offer only after exploration

1. Separate “distinguish my two examples” from “identify every possible hidden population.”

Student page 2: A noisy rule with the right center

Small population means; distinguish the sampling frame from the target.

Now use this supplied population: values 1,1,1,1,5,5,5,5. New rule FULL draws an identifier uniformly from 1-8, replaces it, and draws again independently. Its estimate is the mean of the two values.

3 / prompt and solution

Try to predict all possible two-draw means. Find their exact probabilities without writing 64 individual ordered identifier pairs. Is the expected estimate the whole-population mean?

Solution: A full draw is 1 or 5 with probability $1/2$. The four value-pattern categories 11,15,51,55 each have probability $1/4$; the means are 1,3,3,5. Therefore mean 1 has probability $1/4$; 3 has $1/2$; 5 has $1/4$. The expected estimate is $(1+3+3+5)/4=3$, the population mean. Each category represents 16 of 64 identifier pairs.

A very precise wrong answer

3 / hints to offer only after exploration

1. Group equally likely identifier pairs by their two values.

4 / prompt and solution

A FULL sample happens to give 1, just like the restricted sampler. Did that one result prove the two procedures are equally trustworthy? Draw their complete distributions to explain.

Solution: No. FULL can give 1 with probability $1/4$ but is centered at 3 over repetitions; restricted always gives 1 and has bias -2 for this population. Unbiasedness does not promise every sample is correct, and repeatedly giving the same answer does not make it accurate. Draw mass $1/4, 1/2, 1/4$ at $1, 3, 5$ for FULL versus all mass at 1 for restricted.

4 / hints to offer only after exploration

1. A result is one outcome. A procedure has many possible outcomes.

Student page 3: Design a better two-card survey

Small population means; distinguish the sampling frame from the target.

Continue with the supplied population of four 1-cards and four 5-cards. Divide the identifiers into blocks 1-4 and 5-8. A BLOCK sampler draws one uniformly within each block, independently, then averages the two readings.

5 / prompt and solution

Compare BLOCK with FULL. What does BLOCK estimate on every possible draw? Explain why this certainty depends on this population rather than on a magic property of sampling one per block.

Solution: BLOCK always draws 1 and 5, so always estimates 3 here, eliminating the between-block sampling fluctuation. Certainty follows because each block is homogeneous and their sizes are equal. If block values vary, one per block can fluctuate; the design does not promise exactness for arbitrary populations.

5 / hints to offer only after exploration

1. What information about every value within each block did the reveal provide?

A very precise wrong answer

6 / prompt and solution

Change the population to TWO 1-cards in block L and SIX 5-cards in block R. A sampler still takes one card per block. Is their ordinary average the whole mean? Repair the rule using the known block sizes.

Solution: Ordinary average is 3, but the whole mean is $(2 \times 1 + 6 \times 5) / 8 = 4$. Weight the L reading by $2/8$ and the R reading by $6/8$: $1/4 \times 1 + 3/4 \times 5 = 4$. Generally these known block-size weights make one uniform independent draw per block unbiased for the full mean, even with variation inside blocks; they do not necessarily make it exact.

6 / hints to offer only after exploration

1. One observed card represents two population members; the other represents six.

Optional facilitator extension 1

Gate: Means and products of independent halves.

Let the equal-sized blocks be 0,0,2,2 and 4,4,6,6. What can BLOCK produce?

Solution: The independent within-block choices have values 0 or 2 and 4 or 6, each equally likely. Means 2,3,3,4 give probabilities $1/4, 1/2, 1/4$ and expectation 3. The certainty is gone while unbiasedness remains.

Mathematics and prior use

Sampling frames, non-identifiability, unbiasedness, and stratified estimation. Equal probability within a restricted frame is not equal coverage of the target population.

Prior use: Related to AP-02 because mechanisms matter, but distinct target: estimating a finite-population mean from a sampling frame. No corresponding trial worksheet or Week 1-10 instance.

Source: Allen Downey, Think Stats

Chapter 8, §8.1 Estimation of means; discussion of biased and unbiased estimators.. Uses the distinction between an estimator and its sampling distribution. The two-world impossibility construction and unequal-block repair are independently derived, not copied from the source examples. Evidence: Reopened estimation chapter, inspected finite-sampling and bias discussions; all 64 two-draw outcomes and block-weight calculations checked exactly.

<https://allendowney.github.io/ThinkStats/chap08.html>

Build a promise an interval can keep

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Promising with a genuine statistics game. The width optimization and intentionally unhelpful 50% procedure make coverage tangible. The coin-rule page is optional; explain its purpose before invoking general confidence-interval terminology.

A shortest 80% translation interval with a lower-bound proof, and an explanation that coverage describes the random procedure rather than a posterior distribution on the secret.

Prerequisites and preparation

Core: Signed integer addition and subtraction, closed intervals, fifths and systematic cases. Keep one unknown target fixed while the instrument error varies.

Extensions: Compare a procedure's repeated coverage with what is known from a realized report; optional randomized interval procedures use a separate independent coin.

Reading and representation: Target θ may be called "secret number"; the role distinction matters more than the symbol. Short rule cards and interval endpoints.

Materials: Number-line strips; a sealed secret integer; five equally likely error cards $-2, -1, 0, 1, 2$; transparent or paper interval strips; a fair coin for page 3. Preparation: about 10 minutes.

0-10 explore the supplied objects; 10-15 rehearse exact rules; 15-35 investigate page 1; 35-40 move/reset; 40-55 choose a certificate or new representation; 55-60 explain one result. Pages 2-3 can be a second meeting; no requirement to complete the packet in one hour.

Launch and facilitation

The instrument gives a slightly wrong reading. Design the shortest interval rule that catches the fixed secret at least four times out of five.

Student page 1: What does the strip catch?

Integer intervals and fractions; track one fixed secret across random errors.

A secret integer θ stays fixed. The instrument reports $X = \theta + E$, where E is freshly chosen uniformly from $-2, -1, 0, 1, 2$. Try the rule $[X-1, X+1]$, including both endpoints. The interval moves with X ; the secret does not move.

Build a promise an interval can keep

1 / prompt and solution

Choose a practice secret and run all five possible errors once, as an exact audit rather than a random sample. Place each reported interval. Which catch your secret?

Solution: For any chosen integer θ , the interval $[\theta+E-1, \theta+E+1]$ contains θ exactly when $-1 \leq E \leq 1$. Errors $-1, 0, 1$ succeed; $-2, 2$ fail. Every error has probability $1/5$ in the actual instrument, so coverage is $3/5$. Auditing each error once enumerates the model rather than estimating coverage from five random trials.

1 / hints to offer only after exploration

1. Check the secret against both interval endpoints.

2 / prompt and solution

Without redoing five calculations, explain why moving the secret to another integer leaves the success fraction unchanged. What assumption about the errors did your explanation use?

Solution: Translating θ and all readings/intervals by the same amount preserves containment. The success condition depends only on E , not θ . Its $3/5$ probability relies on the same uniform error distribution at every target. Translation geometry alone would not determine probabilities under a different error mechanism.

2 / hints to offer only after exploration

1. Draw the original and translated pictures on two aligned strips.

Student page 2: The shortest four-out-of-five promise

Integer intervals and fractions; track one fixed secret across random errors.

Restrict to translation rules $[X+a, X+b]$, where $a \leq b$ are fixed integers chosen BEFORE seeing X . Width means $b-a$, not the number of integer dots. The same offsets must be used for every reading.

3 / prompt and solution

Design a rule with coverage at least $4/5$ for EVERY integer secret. Make its width as small as possible. You may put the reading off-center. Find every shortest rule.

Solution: Minimum width is 3. The two width 3 rules with coverage $4/5$ are $[X-2, X+1]$ and $[X-1, X+2]$. A rule succeeds for errors E with $-b \leq E \leq -a$. To cover four of the five consecutive possible errors, the reflected interval must contain either $-2, -1, 0, 1$ or $-1, 0, 1, 2$. These yield the two stated offsets. Both are translation-invariant, so the proof holds for every θ .

3 / hints to offer only after exploration

1. Which four error values will your rule protect against?

Build a promise an interval can keep

4 / prompt and solution

Prove that a narrower rule cannot meet the promise. Then design a 100% rule and explain exactly which model assumption makes its guarantee possible.

Solution: An interval of integer width L contains at most $L+1$ integer errors. Coverage $\geq 4/5$ requires at least four distinct equally likely errors, hence $L \geq 3$. For 100%, $[X-2, X+2]$ has width 4 and covers all possible errors. The guarantee uses the known hard error bound $|E| \leq 2$; real instruments without that bound would need another model. The optimality claim is for fixed integer-offset closed intervals, not arbitrary confidence sets or randomized widths.

4 / hints to offer only after exploration

1. Count protected error values, not places you happened to measure.

Student page 3: An honest percentage can hide a bad rule

Optional: independent randomization and distinction between coverage and posterior belief.

Optional: after measuring X , flip a fresh independent fair coin. On HEADS report $[X-2, X+2]$. On TAILS report the one-point interval $[X+10, X+10]$. Always reveal which branch you used. Keep the same bounded-error model.

5 / prompt and solution

What is this procedure's coverage for each fixed secret? Prove your answer using all ten error-and-coin cases.

Solution: All five HEADS cases cover because the full error range is included. All five TAILS cases miss, since $\theta = X - E$ and $|E| \leq 2$ cannot equal $X+10$. The 10 cases are equally likely, so coverage is $1/2$ for every fixed θ . This is an exact randomized confidence procedure, although a poor practical one.

5 / hints to offer only after exploration

1. Which branch guarantees a catch? Which makes a catch impossible?

6 / prompt and solution

You see $X=7$ and the TAILS report $[17, 17]$. Is it reasonable to call 17 a 50% plausible secret just because the procedure has 50% coverage? Explain what the percentage actually certifies.

Solution: No. The measurement model forces $\theta \in \{5, 6, 7, 8, 9\}$, so 17 is impossible for this report. The $1/2$ describes averaging over the future random error-and-coin procedure with θ fixed; it is not a posterior probability for each realized interval. A distribution on θ after a report requires an additional prior model. No such prior has been specified.

Build a promise an interval can keep

6 / hints to offer only after exploration

1. Use the possible errors to list targets consistent with the actual reading.

7 / prompt and solution

Would you recommend this 50% rule over your shortest 80% rule? Give a reason beyond simply choosing the larger percentage.

Solution: A useful comparison mentions width, informative reports and guaranteed failure on one branch. The coin rule sometimes gives a knowingly impossible interval and otherwise width 4; the 80% rule always has width 3 and a clearly specified miss direction. Coverage alone does not rank every procedure or make every realized report useful. Preference must identify its criterion; no universal statistical-optimality claim is required.

7 / hints to offer only after exploration

1. What does the user receive on each possible branch?

Optional facilitator extension 1

Gate: Same core; reinforces model dependence.

What happens to the original narrow rule if error is -2 or 2 with probability $1/2$ each?

Solution: It never covers: neither possible error lies in $[-1, 1]$. The printed interval formula is unchanged while its coverage falls from $3/5$ to 0 . A coverage statement always includes the error law.

Mathematics and prior use

Exact finite confidence coverage $P(\theta \in I(X))$. The instrument model is $X = \theta + E$ with fixed unknown integer θ and a fresh uniform five-card error. No prior on θ is assumed.

Prior use: No matching trial activity; AP-02 has probabilities for a randomly selected bag, whereas this holds an unknown target fixed. Keeping those different probability models explicit is part of the investigation.

Source: NIST/SEMATECH e-Handbook of Statistical Methods

NIST/SEMATECH §1.3.5.2, introductory interpretation paragraphs before the t-interval formula.. Uses only the repeated-sampling meaning of coverage. The five-error model, shortest-offset proof and independent-coin counterexample are separately constructed; no t-distribution calculation is borrowed.

Evidence: Directly reread the interpretation distinguishing procedure coverage from the probability assigned to a realized interval. Enumerated offsets/errors and randomized branches exactly.

<https://www.itl.nist.gov/div898/handbook/eda/section3/eda352.htm>

When the tangent robot comes back

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Advanced first-pilot candidate for learners taking calculus. The safeguard gives more than a familiar failure example. Do not relabel tangent motion as a lower-prerequisite guessing game.

An exact non-root 2 cycle despite available roots, plus a modified method whose sign bracket shrinks by a proved factor.

Prerequisites and preparation

Core: CALCULUS: derivative as tangent slope, evaluating cubics, rational arithmetic and a tangent-line zero. The loop is a Newton-method investigation only with this gate retained.

Extensions: Continuity/Intermediate Value Theorem and algebraic interval-width bounds for a safeguarded method; a supplied geometric-decay fact is enough for finite error certification.

Reading and representation: Symbolic f, f' and a short iterative algorithm. Calculator optional; exact fractions establish the cycle.

Materials: Supplied graph of $f(x)=x^3-5x$ on $-3 \leq x \leq 3$, with axes and no tangent answers; ruler; blank tangent workspace; calculator; bracket strips for $[1,3]$. Preparation: about 10 minutes.

0-10 explore the supplied objects; 10-15 rehearse exact rules; 15-35 investigate page 1; 35-40 move/reset; 40-55 choose a certificate or new representation; 55-60 explain one result. Pages 2-3 can be a second meeting; no requirement to complete the packet in one hour.

Launch and facilitation

A tangent robot wants a zero of x^3-5x . Choose a start, draw its first step, and look for a start that makes the robot revisit a guess.

Student page 1: Follow a tangent, not the curve

Calculus: derivatives and tangent updates; continuity for the safeguard.

Use $f(x)=x^3-5x$ and $f'(x)=3x^2-5$. From guess x , draw the tangent at $(x, f(x))$; its horizontal-axis crossing is the next guess. Then move vertically to the curve and repeat. The initial guess is x_0 .

When the tangent robot comes back

1 / prompt and solution

Choose two starts from $-2, -1, 0, 1, 2$. Draw or calculate two steps for each. Predict what a third step might do. Keep your exact values when they are simple.

Solution: For 0 the method stays at the true root 0. For 1 the next guesses are $-1, 1$; for -1 they are $1, -1$. For 2 they are $16/7$ and $8192/3661$; for -2 the values are the negatives because the Newton map is odd. Approximate $-2/2$ runs head toward $-\sqrt{5}/\sqrt{5}$; two numerical steps alone are not a convergence proof. Every chosen pair is acceptable if the tangent construction is correct.

1 / hints to offer only after exploration

1. Find the zero of $y=f(x)+f'(x)(t-x)$.

2 / prompt and solution

Derive a formula for the next guess. For a start at 1, predict x_{1000} and x_{1001} without making a long table. Prove the prediction.

Solution: Setting the tangent equation to zero gives $N(x)=x-f(x)/f'(x)=2x^3/(3x^2-5)$, where the denominator is nonzero. $N(1)=-1$ and $N(-1)=1$, so induction gives $x_{\text{even}}=1$ and $x_{\text{odd}}=-1$. Hence $x_{1000}=1, x_{1001}=-1$. The exact two-arrow cycle is the certificate; rounded repeats alone would not prove an infinite loop.

2 / hints to offer only after exploration

1. Once a deterministic rule returns to an earlier exact state, its future repeats.

Student page 2: A failed run is not a failed equation

Calculus: derivatives and tangent updates; continuity for the safeguard.

Keep the same cubic. A solver must distinguish a true zero, a repeating guess and an undefined tangent step.

3 / prompt and solution

Find every real root. Find the starts where the Newton formula is undefined. Explain why “the robot failed” does not imply “there is no root.”

Solution: Factor $f(x)=x(x^2-5)$, so the roots are $0, \pm\sqrt{5}$. The formula is undefined at $x=\pm\sqrt{5/3}$, where $f'=0$ and $f\neq 0$. Starts ± 1 cycle through nonroots despite three actual roots. A horizontal tangent can miss the axis while the curve has zeros elsewhere.

3 / hints to offer only after exploration

1. Solve the equation and the denominator condition separately.

When the tangent robot comes back

4 / prompt and solution

Design a warning that catches the exact two-cycle. Could a warning alone force the solver to reach a root? Give a separate certificate that a root lies in $[1,3]$.

Solution: Storing earlier exact iterates and signaling on a repeated nonroot detects the cycle; checking $x_{(n+1)}=x_{(n-1)}$, with distinct consecutive iterates and $f(x_n) \neq 0$, also detects this two-cycle. Detection alone supplies no new successful step. $f(1)=-4, f(3)=12$ and continuity imply a root in $[1,3]$, specifically $\sqrt{5}$. The warning and the existence bracket are distinct guarantees.

4 / hints to offer only after exploration

1. A diagnostic tells you a run failed; it is not automatically a repair.

Student page 3: Give the tangent robot a guardrail

Calculus plus continuity and interval inequalities.

Keep a bracket $[L,R]$ with opposite-sign endpoint values and a current guess x . Propose its Newton step. ACCEPT the proposal only if it is defined and lies in the middle half $[L+(R-L)/4, R-(R-L)/4]$. Otherwise use the midpoint. If the chosen point is a root, stop; otherwise retain the subinterval whose endpoints have opposite signs. Use the chosen point as the next current guess.

5 / prompt and solution

Start with $[L,R]=[1,3]$ and $x=1$. Carry out two guarded updates. Record which proposal was accepted and the new bracket each time.

Solution: First Newton proposes -1 , outside $[3/2, 5/2]$, so choose midpoint 2. $f(2)=-2$, hence bracket $[2,3]$. Next Newton from 2 proposes $16/7$, inside the new middle half $[9/4, 11/4]$, so accept it. $f(16/7)=176/343 > 0$, so bracket $[2, 16/7]$. Widths change $2 \rightarrow 1 \rightarrow 2/7$. Exact fractions, not plotting accuracy, decide signs and acceptance.

5 / hints to offer only after exploration

1. The allowed middle half changes when the bracket changes.

When the tangent robot comes back

6 / prompt and solution

Prove that the bracket always retains a root and its width is at most $3/4$ of the old width after an update. Give a bound after n updates starting from width 2. How many updates suffice to make the width less than 0.01?

Solution: Continuity and opposite signs retain a root after either sign-based endpoint replacement; an exact zero stops successfully. Every selected point lies in the middle half, including a fallback midpoint, so neither resulting subinterval can exceed $3/4$ of the old width. Induction gives $\text{width} \leq 2(3/4)^n$. At $n=19$ this is $<1/100$; $n=18$ does not make that worst-case bound $<1/100$. Thus 19 updates suffice for the bracket-width guarantee, not necessarily the exact number actually required. Midpoint error is then at most half the width. The argument does not claim ungarded Newton converges from every start.

6 / hints to offer only after exploration

1. Track an invariant and a shrinking quantity separately.

Optional facilitator extension 1

Gate: Algebraic inequalities and geometric powers.

Replace the safeguard's middle half by $[L+\delta(R-L), R-\delta(R-L)]$, with fixed $0 < \delta \leq 1/2$. Prove the new width bound.

Solution: A chosen point in this central interval leaves either side of width $\leq (1-\delta)(R-L)$. A midpoint fallback also qualifies. Repetition gives $\text{width} \leq (1-\delta)^n$ times the initial width. $\delta=1/2$ forces pure midpoint steps; smaller δ permits more Newton proposals but weakens the guaranteed contraction.

Mathematics and prior use

Newton iteration as numerical dynamics; a sign-bracket invariant plus a contraction bound gives a global certificate in this specific interval. The new cubic differs from the atlas's shared AP-06/GA-28 counterexample.

Prior use: AP-06 and GA-28 in the original atlas share x^3-2x+2 . This investigation deliberately changes the example to x^3-5x and develops a bracket-preserving safeguard. Delivered AP-07 studies explicit-Euler stability, accuracy and an optimization certificate: both diagnose numerical procedures, but the failure mechanism and certificate differ here.

When the tangent robot comes back

Source: Yano, Penn, Konidakis and Patera, Math, Numerics, and Programming

Yano et al., Unit VI, §29.2.1 tangent construction; §29.2.3 failure checks and bisection discussion, PDF pp.5-8.. Retains the Newton derivation but changes the cubic to avoid merely repeating GA-28. The middle-half safeguard and its $3/4$ width bound are independently designed/proved. Evidence: Reopened the MIT primary textbook PDF and inspected the formula, derivative-zero issue and bracketing discussion. Exact rational iterates and safeguard decisions independently checked.

https://ocw.mit.edu/courses/2-086-numerical-computation-for-mechanical-engineers-spring-2013/60af88c658ba3e83cdf7c43289aeb5b4_MIT2_086S13_Unit6_Textbook.pdf

How many rooms must a machine remember?

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

First-pilot candidate with low arithmetic requirements and substantial proof. Learners get construction and adversarial testing before state-distinguishability is introduced. Do not confuse passing several short tests with a universal proof.

A working three-room machine and a proof that every two-room machine fails for some finite string; optional six-room extension has its own lower bound.

Prerequisites and preparation

Core: Follow directed arrows labeled R/B, count red cards in triples, and reason that the current room is the machine's only memory. Lower bounds require comparing histories followed by the same suffix.

Extensions: Two simultaneous remainder conditions and a six-state construction; a pairwise distinguishability argument proves minimality.

Reading and representation: R and B symbol strings, including the empty string; rules can be spoken. No hidden memory in the operator's head.

Materials: Blank state circles; one movable token; R/B arrow cards; start arrow and accepting-star stickers; short input slips; optional floor circles. Preparation: about 8 minutes.

0-10 explore the supplied objects; 10-15 rehearse exact rules; 15-35 investigate page 1; 35-40 move/reset; 40-55 choose a certificate or new representation; 55-60 explain one result. Pages 2-3 can be a second meeting; no requirement to complete the packet in one hour.

Launch and facilitation

Build a machine that accepts exactly strings with a multiple of three red cards, including zero. Another group will try to break it.

Student page 1: Build it. Try to break it.

Follow labeled arrows; remainders; compare histories using common futures.

A machine starts in its marked room, reads one symbol at a time, and follows exactly one arrow for that symbol. Each room needs one outgoing R arrow and one outgoing B arrow. It answers YES only if it ends in a starred room. Its room is its entire memory. The target: number of R cards is a multiple of 3, including 0; B cards do not count.

How many rooms must a machine remember?

1 / prompt and solution

Try a two-room design. Mark start and YES rooms and both arrows from every room. Ask a partner for short strings that might break it. Start with the empty string, B, R, RR, RBR and RRR.

Solution: No two-room design solves all strings, but attempts are valuable. Target answers in the given order are YES, YES, NO, NO, NO, YES. Empty input leaves the machine in its start room, so the start must accept. A partner's failed test must trace the machine correctly, not secretly count in the operator's memory.

1 / hints to offer only after exploration

1. How does a machine answer the empty string?

2 / prompt and solution

A machine passes EVERY string of length at most 2. Its rooms are START (starred) and OTHER (not starred); B stays put, and every R goes to OTHER. Find its shortest failing string. What did the short tests fail to prove?

Solution: This machine accepts exactly strings containing no R. It agrees with the target for lengths 0,1,2, but fails on RRR at length 3 because the target accepts three reds while the machine remains in OTHER. No shorter failure exists by the stated agreement. Exhaustion confirms this is the unique fixed-start two-state design passing all $\text{length} \leq 2$ tests, but the worksheet needs only this explicit counterexample.

2 / hints to offer only after exploration

1. A long enough accepted input must contain some reds.

Student page 2: Build a certificate about memory

Follow labeled arrows; remainders; compare histories using common futures.

You may now use three rooms. A history is the part of a string already read. A suffix is the SAME future string attached to each of two histories.

3 / prompt and solution

Build a correct three-room machine. Explain what each room remembers and prove your explanation survives the next R or B.

Solution: Rooms q_0, q_1, q_2 record the red count remainder 0,1,2. Start and accept at q_0 . R cycles $q_0 \rightarrow q_1 \rightarrow q_2 \rightarrow q_0$; B loops at each room. Initially remainder 0 is correct; each R adds 1 mod 3 and each B leaves it unchanged, proving by induction that every finite string is classified correctly.

3 / hints to offer only after exploration

1. Discard complete triples of red cards; what unfinished remainder is left?

How many rooms must a machine remember?

4 / prompt and solution

Consider the histories empty, R and RR. For EACH pair, find one common suffix that makes exactly one completed string acceptable. Why does that rule out every two-room machine?

Solution: Empty versus R: empty suffix distinguishes YES from NO. Empty versus RR: empty suffix likewise. R versus RR: suffix R produces RR (NO) and RRR (YES). If any pair of histories ended in the same machine room, identical future symbols would lead to the same final answer, contradicting its distinguishing suffix. Three pairwise distinguishable histories therefore require at least three states. This excludes every deterministic two-state design, not only the one children tried.

4 / hints to offer only after exploration

1. Different pasts can share a room only if every possible common future treats them the same.

Student page 3: Remember two things at once

Optional: remainder pairs and a systematic lower-bound proof.

New target: the number of R cards is a multiple of 3 AND the number of B cards is even. Zero counts satisfy both requirements. Every finite input length is allowed.

5 / prompt and solution

Design a machine for this new target. Choose your room layout. Test RRR, BBB, RBRBRBB and the empty string. Explain what a room needs to remember.

Solution: Use six rooms (r,b) with $r \in \{0,1,2\}, b \in \{0,1\}$; R adds 1 to $r \bmod 3$ and B leaves r unchanged while toggling b . Start/accept at $(0,0)$. Target answers: RRR YES; BBB NO; RBRBRBB has 3R and 4B, YES; empty YES. The pair is a sufficient state because updates depend only on it. Layout may be a 2×3 array, two linked cycles, or another unambiguous drawing.

5 / hints to offer only after exploration

1. A room might record two small pieces of information rather than a long history.

6 / prompt and solution

Can fewer rooms work? Use the six histories empty, R, RR, B, RB, RRB. Explain how to choose a common suffix that separates ANY chosen pair.

Solution: Each history represents one distinct remainder pair (r,b) . Given a first pair (r,b) , append enough R cards to make its red remainder 0 and enough B cards to make its parity 0: $R^{\{(-r \bmod 3)\}} B^{\{(-b \bmod 2)\}}$. It accepts. For any different second pair, at least one difference survives addition of the same suffix, so that completed string rejects. Thus all six histories are pairwise distinguishable and six rooms are necessary. This is a proof for every pair, with 15 pair instances exhaustively checked in the script.

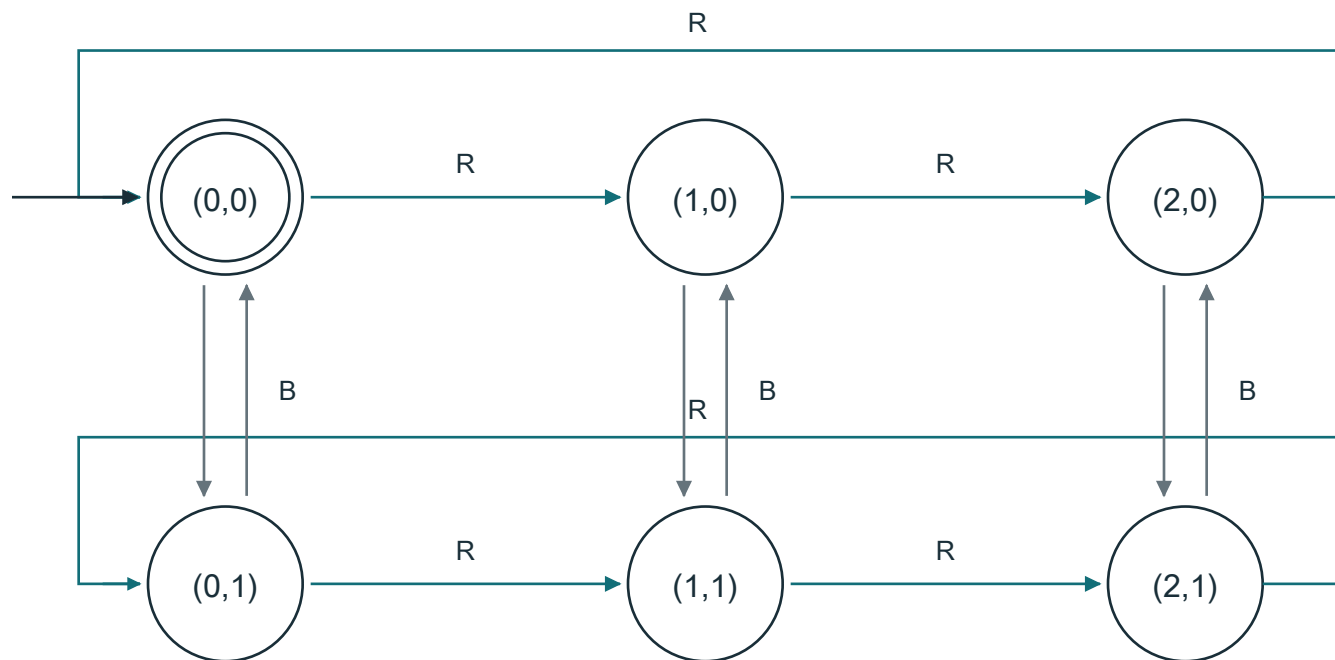
How many rooms must a machine remember?

6 / hints to offer only after exploration

1. Complete one selected history to an accepted state; see what the same completion does to the other.

A machine with a reason

Facilitator key: states record red count modulo 3 and blue count modulo 2



The double ring marks the only YES room. R advances the red remainder around a row; B changes rows. The start is (0,0). Each history pair is distinguished by completing the first to (0,0): a different remainder pair cannot also become (0,0) after the same suffix.

Optional facilitator extension 1

Gate: Modular arithmetic and a parameterized distinguishing-suffix argument.

Replace multiples of 3 by multiples of m . Prove that m rooms suffice and are necessary.

Solution: Remainder states $0, \dots, m-1$ with R cycle and B self-loops suffice. Histories R^r for $0 \leq r < m$ are pairwise distinguishable: append $R^{\{(m-r) \bmod m\}}$ to accept one chosen remainder; any other remains nonzero. Thus at least m states are needed. $m=1$ has a single accepting state.

How many rooms must a machine remember?

Mathematics and prior use

Deterministic finite automata and the concrete distinguishable-prefix principle underlying Myhill-Nerode minimality. The finite lower bound is proved directly without invoking the full theorem.

Prior use: Delivered AP-26 studies state memory and recurrence; Weeks 1-10 also use state diagrams. This investigation adds a new payoff: distinguishing common future strings gives a lower-bound certificate for every smaller deterministic machine, alongside explicit correct constructions.

Source: Michael Sipser, MIT Theory of Computation

Sipser MIT18.404J Lecture 1, PDF pp.6-9, finite automata definition, computation and even-count language example.. Uses the formal room/transition/acceptance model. The three- and six-state constructions, short-test counterexample and lower-bound certificates are independently proved. Evidence: Reopened the MIT lecture PDF and inspected state/alphabet/transition/start/accept definitions and computation.

Enumerated all 64 fixed-start two-state machines plus all short strings and all 15 six-state history pairs.

https://ocw.mit.edu/courses/18-404j-theory-of-computation-fall-2020/b4d9bf1573dccea21bee82cfba4224d4_MIT_18_404f20_lec1.pdf

Two motions hiding in one machine

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Promising with algebra; specialist continuation with calculus. The classification and sum/difference coordinates give a strong mathematical payoff. Keep temporal frequencies and recurrence claims behind the calculus gate.

Classify the two force patterns and uniquely decompose any initial displacement. With calculus, verify the motion and explain why two periodic modes can combine into a motion with no exact common repeat.

Prerequisites and preparation

Core: ALGEBRA: signed displacement coordinates, linear force formulas, proportional pairs and factorization. The complete static force-pattern classification is a real stop; paper movement alone is not the whole result.

Extensions: CALCULUS: differentiate cosine twice and verify a two-component ODE. Exact nonrepetition additionally uses the irrationality of $\sqrt{3}$ and cosine's period.

Reading and representation: Two coordinates measured from separate equilibrium marks; read short formulas for force. A facilitator should demonstrate that both positive displacements point right.

Materials: Paper mass markers on two parallel displacement axes; supplied wall-spring-mass-spring-mass-spring-wall diagram; pencil/ruler; calculator optional. No physical spring apparatus required. Preparation: about 10 minutes.

0-10 explore the supplied objects; 10-15 rehearse exact rules; 15-35 investigate page 1; 35-40 move/reset; 40-55 choose a certificate or new representation; 55-60 explain one result. Pages 2-3 can be a second meeting; no requirement to complete the packet in one hour.

Launch and facilitation

Choose a pair of displacements. When does the restoring-force pair point exactly opposite the displacement pair, without changing their relative shape?

Student page 1: Which shapes pull straight back?

Algebra for force modes; calculus for oscillation and exact nonrepetition.

Two equal unit masses move horizontally between fixed walls. All three springs have stiffness 1. Displacements x_1, x_2 are measured from each mass's own rest mark; positive is right for BOTH. The ideal force law is $F_1 = -2x_1 + x_2$, $F_2 = x_1 - 2x_2$. Ignore friction and assume the linear spring law.

Two motions hiding in one machine

1 / prompt and solution

Choose three nonzero displacement pairs. Draw each pair and its force pair. Try to find a pair whose force is a NEGATIVE multiple of the whole displacement pair. Find a second essentially different shape.

Solution: Many trials. Equal pairs (a,a) have force $(-a,-a)=-1$ times the pair. Opposite pairs $(a,-a)$ have force $(-3a,3a)=-3$ times the pair. Scaling or sign-reversing a pair does not create a new direction. A contrasting trial $(1,0)$ gives $(-2,1)$, not proportional. Require both coordinates to share the same multiplier.

1 / hints to offer only after exploration

1. Equal displacement leaves the middle spring unchanged; opposite displacement changes it.

2 / prompt and solution

Can a third shape work? Give a complete algebraic reason, not just more tested pairs. Exclude the zero pair when discussing a shape.

Solution: For $x=(u,v)$, proportionality requires $u F_2 - v F_1 = 0$. Substitution gives $u(u-2v) - v(-2u+v) = u^2 - v^2 = (u-v)(u+v)$. Thus $u=v$ or $u=-v$. For nonzero pairs these indeed give negative factors -1 and -3 , so there are exactly two eigendirections. The zero pair has no distinguished direction and is explicitly excluded.

2 / hints to offer only after exploration

1. Two vectors in the plane are parallel exactly when their 2×2 determinant is zero.

Student page 2: A different pair of coordinates

Algebra for force modes; calculus for oscillation and exact nonrepetition.

Now use the two shapes you found as building blocks. Add displacement pairs coordinate by coordinate; this is a mathematical decomposition, not two separate physical copies of the masses.

3 / prompt and solution

Build the displacement $(3,1)$ from one equal-motion piece and one opposite-motion piece. Then do it for a pair (u,v) of your choosing. Why is the decomposition unique?

Solution: $(3,1)=2(1,1)+1(1,-1)$. Generally $u=a+b, v=a-b$, hence $a=(u+v)/2, b=(u-v)/2$. These formulas give existence and uniqueness for every real pair. A learner's chosen numerical pair is checked by adding the two contributions back.

3 / hints to offer only after exploration

1. Add and subtract the two coordinate equations.

Two motions hiding in one machine

4 / prompt and solution

Define $S=x_1+x_2$ and $D=x_1-x_2$. Express F_1+F_2 and F_1-F_2 using S,D . What interaction disappeared in these coordinates?

Solution: $F_1+F_2=-(x_1+x_2)=-S$; $F_1-F_2=-3(x_1-x_2)=-3D$. The force sum depends only on S , and the force difference only on D ; each new coordinate has its own scalar restoring factor. Without calculus this is an exact algebraic decoupling, not yet a statement about time.

4 / hints to offer only after exploration

1. Collect the coefficients of x_1 and x_2 before naming the result.

Student page 3: Can periodic pieces fail to repeat together?

Calculus, cosine periods, irrationality; ODE uniqueness may be supplied.

CALCULUS continuation. Unit mass means acceleration equals force. Release from $x(0)=(3,1)$ with both initial velocities 0. A scalar motion $A \cos(\omega t)$ has second derivative $-\omega^2 A \cos(\omega t)$. Use radians.

5 / prompt and solution

Use your decomposition to propose the two mass positions $x_1(t), x_2(t)$. Verify both force equations and all four initial conditions by differentiation.

Solution: $x(t)=2 \cos(t)(1,1)+\cos(\sqrt{3}t)(1,-1)$, so $x_1=2 \cos t+\cos(\sqrt{3}t), x_2=2 \cos t-\cos(\sqrt{3}t)$. Differentiating twice gives $x_1''=-2 \cos t-3 \cos(\sqrt{3}t), x_2''=-2 \cos t+3 \cos(\sqrt{3}t)$, exactly $-2x_1+x_2$ and x_1-2x_2 . At 0 the positions are 3,1 and both sine-based velocities vanish. Standard uniqueness for a linear ODE may be supplied to identify this verified solution as the actual ideal-model motion.

5 / hints to offer only after exploration

1. Each shape keeps its own coefficient but acquires the frequency whose square is its restoring factor.

6 / prompt and solution

Each mode is periodic. Can the combined system return EXACTLY to its starting positions at a positive time? Distinguish an exact proof from a plot that looks close.

Solution: Starting positions require $S=4$ and $D=2$ again, so $\cos T=1$ and $\cos(\sqrt{3}T)=1$. Thus $T=2\pi m$ and $\sqrt{3}T=2\pi n$ for positive integers m,n , implying $\sqrt{3}=n/m$, impossible. Therefore there is no positive exact return to the initial positions (and hence no full-state period). Approximate near-returns can occur; no finite plot proves or disproves the exact statement. Irrationality of $\sqrt{3}$ may be supplied or proved by the usual divisibility contradiction.

Two motions hiding in one machine

6 / hints to offer only after exploration

1. What must BOTH mode coefficients equal for the original pair to return?

7 / prompt and solution

Change only the middle spring stiffness to $k > 0$. Choose k so that the opposite mode completes exactly TWO cycles while the equal mode completes ONE. Show that the combined motion then repeats.

Solution: The force matrix has equal-mode factor 1 and opposite-mode factor $1+2k$. Frequency ratio 2 requires $\sqrt{1+2k}=2$, hence $k=3/2$. Both modal cosines repeat after 2π because their frequencies are 1 and 2; derivatives do too. This changes the ideal model, not merely a plotted scale.

7 / hints to offer only after exploration

1. Recompute the force of the middle spring; equal motion still leaves it unstretched.

Optional facilitator extension 1

Gate: Calculus/trigonometry and rational versus irrational frequency ratios.

For which positive k does the opposite mode finish an integer number m of cycles in one equal-mode cycle?

Solution: $\sqrt{1+2k}=m$, so $k=(m^2-1)/2$ with integer $m \geq 2$. More generally any rational frequency ratio greater than 1 gives a common period, while an irrational ratio and two nonzero amplitudes prevent exact joint recurrence to this released state. The ratio criterion assumes the specified ideal linear system.

Mathematics and prior use

Normal modes diagonalize a coupled linear system. With unit masses and three unit Hookean springs, $\mathbf{x}'' = -\mathbf{K}\mathbf{x}$ for $\mathbf{K} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$. Algebra finds eigendirections; differentiation verifies their oscillations.

Prior use: AP-10 also uses spring laws, but studies static effective stiffness. Trial AP-07 studies a numerical recurrence, not exact coupled motion. No earlier coupled-mode sheet is documented.

Source: David Tong, Classical Dynamics

Tong, Classical Dynamics §2.6 Small Oscillations and Stability; neighboring examples of normal-mode diagonalization.. Uses the general normal-mode framework. The two-unit-mass worksheet, full shape classification, nonrepeat proof and retuned middle spring are independently derived for the stated matrix. Evidence: Reopened Tong's author-hosted section, inspected the small-oscillation/eigenmode setup. Matrix identities and decompositions checked exactly; time formulas checked analytically in the key, not inferred from numerical plots.

<https://davidtong.org/teaching/classical-dynamics/dynhtml/S2>

Add a spring: softer or stiffer?

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

First-pilot candidate for an algebra-ready group. The complete four-response classification and series/parallel duality make this more than substituting into two formulas. Younger exploration of hardware alone is not advertised as the full mathematical result.

Explain the series/parallel reversal from compatibility and force balance; classify all three-unit-spring responses and certify an impossible target, then construct it with four.

Prerequisites and preparation

Core: Fractions and algebra with $F=kx$; distinguish shared force from shared extension. Proving completeness for three springs requires structural cases, not calculus.

Extensions: Series-parallel recursive constructions and reciprocal duality. Continuum elasticity, material failure and non-series-parallel networks remain outside this core.

Reading and representation: Spring diagrams with fixed walls, massless connectors and rigid guided bars; labels F, k, x . Explain the physical connection constraints before arithmetic.

Materials: Printed spring pieces labeled $k=2, k=6$ and three pieces $k=1$; blank two-terminal network frames; force and extension tags; pencil. A physical elastic demonstration is optional and does not certify Hooke's law. Preparation: about 10 minutes.

0-10 explore the supplied objects; 10-15 rehearse exact rules; 15-35 investigate page 1; 35-40 move/reset; 40-55 choose a certificate or new representation; 55-60 explain one result. Pages 2-3 can be a second meeting; no requirement to complete the packet in one hour.

Launch and facilitation

Two springs can be joined end-to-end or side-by-side. Predict which arrangement stretches more under the same total load, then design a connection that reaches a target response.

Student page 1: What is shared?

Algebra: force/extension equations, fractions and recursive networks.

Ideal spring A has $F=2x$; spring B has $F=6x$. Compare END-TO-END with SIDE-BY-SIDE between common rigid guided bars. Connectors are massless, motion is along the springs, and the load is static. Apply total force 6. The bars in the side-by-side drawing translate without rotating.

Add a spring: softer or stiffer?

1 / prompt and solution

Predict which arrangement extends farther. Draw force and extension labels on EACH spring, then solve. Which quantity must be equal in the two springs for each arrangement?

Solution: End-to-end shares force 6: extensions 3 and 1 sum to 4. Side-by-side shares extension x : forces $2x$ and $6x$ sum to 6, giving $x=3/4$; branch forces are $3/2$ and $9/2$. Equal extension follows from common translating bars; equal force in series follows from balance at the massless internal connector. The diagram's constraints, not visual resemblance alone, decide what is shared.

1 / hints to offer only after exploration

1. Separate each spring as a free body; at a connector the forces must balance.

2 / prompt and solution

Someone gives each side-by-side spring half the total force. Show exactly what fails. Then explain how adding a spring can make a system either softer or stiffer.

Solution: Force 3 on A requires extension $3/2$; force 3 on B requires $1/2$. Those unequal extensions contradict the rigid common guided bars. For general positive k_1, k_2 , parallel stiffness k_1+k_2 exceeds either, whereas series stiffness $k_1k_2/(k_1+k_2)$ is less than either. Under a fixed total force, larger stiffness means smaller extension. These are ideal static statements, not universal rubber-band behavior.

2 / hints to offer only after exploration

1. Check compatibility as well as the total force.

Student page 2: How many three-spring responses exist?

Algebra: force/extension equations, fractions and recursive networks.

Now use EXACTLY THREE identical springs, each $k=1$. A permitted network starts with single springs and repeatedly joins two disjoint smaller networks either in SERIES (end-to-end) or PARALLEL (between common guided bars). Use every spring once; no other connections are allowed. Mirrored drawings count as the same construction type.

3 / prompt and solution

Build as many different responses as you can. For each, find the total extension under force 6. Compare different-looking drawings that give the same result.

Solution: The four possible effective stiffnesses are 3 (all parallel), $1/3$ (all series), $3/2$ (one spring parallel to a two-spring series branch), $2/3$ (one spring in series with a two-spring parallel pair). Their force 6 extensions are 2, 18, 4, 9 respectively. Permuting identical springs or reparenthesizing all-series/all-parallel does not create a new response. The key provides four worked AST/network drawings after learner attempts.

Add a spring: softer or stiffer?

3 / hints to offer only after exploration

1. At your final join, one subnetwork has one spring and the other has two.

4 / prompt and solution

Prove your list of possible responses is complete. Is total extension 6 under force 6 possible with exactly these three springs? If not, give a certificate covering EVERY permitted network.

Solution: The final join is S or P, splitting three leaves into 1 and 2. The two-spring part is $S(1,1)$ or $P(1,1)$, with stiffness $1/2$ or 2 . Root S gives $1/3$ or $2/3$; root P gives $3/2$ or 3 . These exhaust the two final-join choices times two child choices, after branch exchange. None equals 1 , which would be needed for extension 6 under force 6; the target is impossible within the explicitly restricted class. No claim is made about networks with extra rigid paths or other constitutive laws.

4 / hints to offer only after exploration

1. Classify the last connection, not every possible drawing separately.

Student page 3: Turn a design into its reciprocal

Algebra: force/extension equations, fractions and recursive networks.

A dual design swaps EVERY series join for a parallel join, and every parallel join for a series join; each unit spring stays a unit spring. Use the four verified three-spring designs as your first experiments.

5 / prompt and solution

Pair each three-spring design with its dual. What happens to stiffness and to extension under unit force? Explain from the two combination rules.

Solution: All-series $1/3$ pairs with all-parallel 3 . $S(1,P(1,1))$ has $2/3$ and pairs with $P(1,S(1,1))$ of $3/2$. In every case dual stiffness is reciprocal. Inductive proof: if dual subnetwork stiffnesses are $1/a, 1/b$, a series parent has $ab/(a+b)$, and the dual parallel parent has $1/a+1/b=(a+b)/(ab)$; a parallel parent $a+b$ duals to series $(1/a)(1/b)/(1/a+1/b)=1/(a+b)$. The single unit spring is the base case. Under unit load extension is $1/k$, so original and dual extension values also reciprocate.

5 / hints to offer only after exploration

1. Apply the claim to the two pieces before the final join.

Add a spring: softer or stiffer?

6 / prompt and solution

Use EXACTLY FOUR unit springs to achieve extension 6 under force 6. Find two visibly different arrangements and certify both. Explain why this does not contradict your three-spring impossibility proof.

Solution: $S(P(1,1),P(1,1))$ has two stiffness 2 blocks in series, giving 1. $P(S(1,1),S(1,1))$ has two stiffness $1/2$ branches in parallel, also giving 1. Thus both extend 6 under force 6 and are dual to each other. The earlier exhaustive proof assumed exactly three leaves; changing that resource count changes the permitted construction set.

6 / hints to offer only after exploration

1. Try making two identical two-spring blocks, then join the blocks the other way.

All three-unit-spring responses

Facilitator key: every leaf is an ideal unit spring



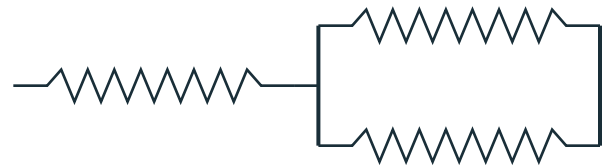
$k = 3$; extension = 2



$k = 1/3$; extension = 18

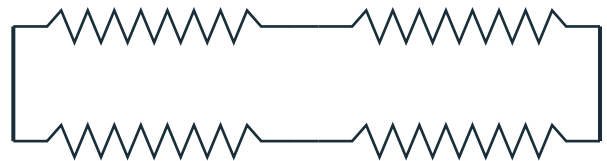


$k = 3/2$; extension = 4



$k = 2/3$; extension = 9

Extensions shown use total force 6. A final series or parallel join splits three leaves into one plus two. The two-leaf part is itself series or parallel; these four cases exhaust the permitted networks.



Four springs: two parallel pairs in series, or two series pairs in parallel. Both have $k=1$ and extension 6 under force 6. The drawings above suppress rest lengths; common vertical connectors are rigid bars constrained to translate.

All three-unit-spring responses

Optional facilitator extension 1

Gate: Algebraic induction on a recursively defined network.

Can the reciprocal proof work if individual springs have arbitrary positive stiffness?

Solution: Yes if dualization ALSO replaces each leaf stiffness k by $1/k$. The same inductive combination identities apply. Swapping joins alone while leaving arbitrary nonunit leaf stiffnesses unchanged does not generally reciprocate the response. This distinguishes a theorem's base case from its recursive step.

Mathematics and prior use

Finite ideal elastic networks: constitutive equations plus connection constraints determine effective stiffness. Series-parallel composition has an exact reciprocal symmetry for unit springs.

Prior use: AP-09 shares the linear spring model but studies coupled motion. Electrical resistor networks have related algebra; cross-link the representation instead of counting a relabeled network as unrelated content.

Source: Allan Bower, Applied Mechanics of Solids

Bower, Applied Mechanics of Solids §3.2, linear elastic constitutive assumptions; finite spring laws derived in the worksheet.. Uses linear-elastic model discipline. Series/parallel compatibility, all three-spring responses and reciprocal duality are derived directly, not claimed as a theorem printed in this continuum section. Evidence: Reopened the author-hosted linear-elastic section. Independently solved force/extension equations and exhaustively generated all recursive three- and four-spring stiffness values with exact fractions.

https://solidmechanics.org/Text/Chapter3_2/Chapter3_2.php

What can a flow map actually tell you?

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Promising with fractions and units. The finite section/branch argument is a real conservation special case; the vector-calculus extension remains advanced. Physical model assumptions need explicit adult reading.

A conservation certificate rejects a map, while explicit distinct profiles and branch splits show what the same certificate leaves undetermined.

Prerequisites and preparation

Core: Area, speed and volume units; multiplication, division and weighted means. Algebra describes every branch split. A storage ledger is the core conservation argument.

Extensions: Vector calculus and the divergence theorem for the local continuity equation. Density and mass-rate arithmetic for the compressible continuation.

Reading and representation: Short labeled flow diagrams; a facilitator may read rules aloud. Distinguish a point speed, section mean normal speed, volume rate and mass rate.

Materials: Pencil, ruler, optional equal-volume blocks; supplied tube, split-section and branch diagrams. All arrows label mean normal speed unless a patch is explicitly uniform. No running-water demonstration is needed. Preparation: about 8 minutes.

0-8 move equal-volume packets through two gates; 8-20 design and reject section labels; 20-35 compare velocity profiles; 35-50 classify branch choices; later 20 minutes for changed storage/density models. Hand out one page at a time.

Launch and facilitation

A wide gate passes 12 volume units each second. Can a narrow gate carry that same flow? Make a plausible-looking label that the stored-water count exposes.

Student page 1: Catch an impossible label

Core: area, speed and volume per time

A rigid full tube carries constant-density liquid steadily, with no leaks. At a cross-section, $Q = \text{area} \times \text{mean forward normal speed}$. Q is volume per second. If speeds vary, use the area-weighted mean.

What can a flow map actually tell you?

1 / prompt and solution

The wide section has area 6 and mean speed 2. The narrow section has area 3. Invent one possible and one impossible narrow mean-speed label. For the impossible label, explain where the volume count fails after one second.

Solution: Incoming $Q=12$. The only conserving narrow mean is 4. Any nonnegative speed $w \neq 4$ fails: the intervening inventory changes at rate $12-3w$. For example $w=3$ gives a gain of 3 volume units per second; $w=5$ requires a loss of 3. Both contradict a steady rigid full tube with constant-density liquid and no leaks. Equal Q passes only this necessary conservation test, not all fluid equations.

1 / hints to offer only after exploration

1. Count the volume crossing each gate in one second.
2. Keep a ledger for the region between the gates.

2 / prompt and solution

Split the narrow section into patches of areas 1 and 2, each with a uniform forward speed. Design two different nonnegative speed pairs that carry the required flow. Then test patch speeds 2 and 5. Is the ordinary average of those two readings the section mean?

Solution: Patch speeds u, v satisfy $u+2v=12$. Examples $(0,6)$, $(8,2)$, and $(4,4)$ all work; the complete nonnegative family has $0 \leq v \leq 6$ and $u=12-2v$. For $(2,5)$, $Q=1 \times 2 + 2 \times 5 = 12$, so the section mean is $12/3=4$, while the unweighted average is $7/2$. A local fast reading does not determine Q . These piecewise-uniform profiles are section data for the conservation audit, not asserted viscous solutions.

2 / hints to offer only after exploration

1. The larger patch contributes twice as much area.
2. Try keeping the weighted sum fixed while changing the two speeds.

Student page 2: A branch does not choose its own flow

Core continuation: classify a nonnegative family

The same incoming volume rate 12 now splits into two steady outlets with areas 2 and 1. Their mean forward speeds are u and v . There are no other outlets or storage changes.

What can a flow map actually tell you?

3 / prompt and solution

Design two conserving splits: in one, the smaller outlet is faster; in the other, it is slower. Describe EVERY nonnegative pair (u,v) that passes the conservation test. What are the largest possible u and v ?

Solution: Conservation is $2u+v=12$, with $u,v \geq 0$. For example $(u,v)=(2,8)$ has the smaller outlet faster; $(5,2)$ has it slower. Every solution is $v=12-2u$ with $0 \leq u \leq 6$. Thus $\max u=6$ at $v=0$ and $\max v=12$ at $u=0$. Zero flow is permitted by the stated nonnegative convention. Narrower-implies-faster required equal Q in a single stream tube; separate branches need not have equal Q .

3 / hints to offer only after exploration

1. Add the two outgoing volume rates.
2. Describe the allowed portion of a line, including its endpoints.

4 / prompt and solution

A designer says, "The areas alone determine both speeds." Refute that claim with a certificate. Then a sensor reports $u=4$. What does this additional observation determine? Does it determine the pressure?

Solution: The two distinct pairs in prompt 3 are an explicit nonuniqueness certificate for the same incoming rate and areas. With $u=4$, conservation gives $v=4$, and outgoing rates are 8 and 4. It still does not determine pressure: no force, energy, viscosity or boundary-pressure law has been supplied. Distinct pressure/velocity-profile details are not excluded by this ledger.

4 / hints to offer only after exploration

1. Which unknowns occur in the conservation equation?
2. A new measured speed is different from a new physical law.

Student page 3: Which assumption changed?

Continuation: storage and mass density

Keep an observed inlet volume rate 12 and outlet volume rate 9. Change the model explicitly in each task. Rates and dimensions use consistent units.

What can a flow map actually tell you?

5 / prompt and solution

Replace the rigid full tube by a tank with horizontal area 4, initial volume 8 and capacity 20. The two rates remain constant; there is no leak. Predict the water-level rise per second and the first overflow time. Could the same two flow readings alone prove that a pipe has a leak?

Solution: Before overflow, volume is $V(t)=8+(12-9)t=8+3t$. Height is $V/4$, so its rate is $3/4$ length unit per second; initial height is 2 and rim height is 5. Capacity is reached at $t=4$ seconds; overflow begins thereafter. The readings alone do not identify a leak: changing stored volume is another explanation. Under the original steady, full, constant-density assumptions, a third outlet of rate 3 would balance the ledger. The two explanations rely on different stated models.

5 / hints to offer only after exploration

1. A volume increase divided by horizontal area is a height increase.
2. Compare the inventory change with the unaccounted boundary flow.

6 / prompt and solution

Now use a steady compressible gas in a rigid tube, with uniform density at each section. Inlet density is 1; its area is 6 and mean speed is 2. Outlet area is 3 and mean speed is 3. Choose an outlet density that conserves MASS. Explain why conserving volume would give the wrong test here.

Solution: Mass rate is density $\times$ area $\times$ mean normal speed when section density is uniform. Inlet mass rate is 12, outlet is 9ρ , so $\rho=4/3$ is required. Volume rates 12 and 9 differ without steady mass accumulation because the gas is denser downstream. This is a continuity-compatible data set, not a proof of a full gas-dynamical solution. Constant-density incompressibility was deliberately removed.

6 / hints to offer only after exploration

1. Track mass carried by each unit of volume.
2. Do not assume the inlet and outlet densities agree.

Optional facilitator extension 1

Gate: Partial derivatives, integration and the divergence theorem.

Derive the control-volume mass law from $\partial\rho/\partial t + \text{div}(\rho\mathbf{u})=0$. For constant density, test $\mathbf{u}=(ax,by)$ on the rectangle $0\leq x\leq 2$, $0\leq y\leq 1$ per unit depth. Which a,b pass local incompressibility?

Solution: For a fixed region V , differentiation under the integral and the divergence theorem give $d/dt \int_V \rho dV = - \int_V \rho \text{div} \mathbf{u} dV$. With spatially and temporally constant positive ρ , $\text{div} \mathbf{u}=0$. Here $\text{div} \mathbf{u}=a+b$; hence $b=-a$. Right-edge outward volume flux is $(2a)\times 1=2a$, top-edge flux $b\times 2=2b$, and the $x=0/y=0$ edge fluxes are zero, so total outward flux is $2(a+b)$. This rectangle is a control region with permeable boundaries, not an impermeable-wall tube. The family $(x,-y)$ passes; (x,y) fails. This supplies no momentum solution.

What can a flow map actually tell you?

Mathematics and prior use

Integral conservation, weighted means and an affine family of admissible fluxes. Passing continuity is a necessary test, not a solution of momentum, pressure or viscosity equations.

Prior use: Week 10 and trial GA-25 concern graph traversal or trees. This is material flux and underdetermination, not another route puzzle. AP-30 later uses an analogous conservation law plus constitutive information; cross-link that distinction.

Source: David Tong, Fluid Mechanics

§1.1.3, printed pp. 9-10 / PDF pp. 13-14; narrowing tube, printed p. 20 / PDF p. 24.. Use mass continuity and constant-density volume balance. Patch profiles, branch classification and changed-model examples are new exact instances. Evidence: Reopened 2026-09-25: read continuity (1.2), its integral/divergence form, incompressibility (1.4), and $UA=ua$ in the narrowing-pipe example. No pressure consequence is borrowed.

<https://davidtong.org/pdfs/teaching/fluid-mechanics/fluids1.pdf>

Choose the fastest crossing

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Advanced/specialist calculus investigation. The trial stage is accessible with roots and ratios, but alone is not marketed as a proof of the real optimization.

A unique global minimizing crossing, a correct boundary optimum when access is restricted, and an inverse construction making a chosen crossing optimal.

Prerequisites and preparation

Core: Pythagoras and distance/speed for the launch; derivatives and a global monotonicity or convexity argument are required for the satisfying optimum.

Extensions: Implicit or monotonic comparative reasoning for changing a medium speed; no wave-boundary or diffraction theorem is claimed.

Reading and representation: Coordinate pairs, square roots, derivatives and angles measured from the normal. Calculators may support experiments but not replace the continuum proof.

Materials: Supplied coordinate diagrams, ruler and a movable crossing marker; optional calculator. The upper and lower regions have constant positive speeds. No actual water is needed. Preparation: about 8 minutes.

0-12 choose crossings and compare time with distance; 12-30 derive and certify the optimum; 30-45 handle a restricted crossing gate; 45-60 design a new speed. The calculus proof is core, not an optional decoration.

Launch and facilitation

Travel from $A=(0,4)$ to $B=(7,-3)$. Above the shoreline speed is 3; below it speed is 4. Choose where to cross. A shorter-looking route may take longer.

Student page 1: Race time, not distance

Calculus core; Pythagoras starts the experiment

$A=(0,4)$, $B=(7,-3)$. A permitted route is a straight segment from A to $C=(x,0)$, then a straight segment to B . Any real x is allowed. Speeds are constant: 3 above $y=0$ and 4 below. Crossing takes no extra time.

Choose the fastest crossing

1 / prompt and solution

Choose two crossing points and compare both total distance and total time. Find where the direct straight line from A to B crosses. Must that shortest route also be fastest?

Solution: For any x , lengths are $L1=\sqrt{(x^2+16)}$, $L2=\sqrt{((7-x)^2+9)}$, distance $L1+L2$ and time $T=L1/3+L2/4$. Any correctly calculated chosen trials are acceptable. The direct line crosses at $x=4$ because its vertical drop to $y=0$ is $4/7$ of the total drop 7. At $x=4$, lengths are $4\sqrt{2}$ and $3\sqrt{2}$, so distance $7\sqrt{2}$ and time $25\sqrt{2}/12$. At $x=3$, lengths 5,5, distance 10 and time $35/12$. Since $35 < 25\sqrt{2}$ ($1225 < 1250$), $x=3$ is faster despite $10 > 7\sqrt{2}$. This comparison disproves fastest=shortest here, before the full optimum is proved.

1 / hints to offer only after exploration

1. Use distance divided by the speed on that side.
2. Compare a convenient pair of right triangles.

2 / prompt and solution

Build $T(x)$ from the picture. Find a candidate crossing that cannot be improved by a tiny move either way. Give an exact time, not just a graph estimate.

Solution: $T(x)=\sqrt{(x^2+16)}/3+\sqrt{((7-x)^2+9)}/4$. Its derivative is $x/[3\sqrt{(x^2+16)}]-(7-x)/[4\sqrt{((7-x)^2+9)}]$. At $x=3$ the two terms are $3/15$ and $4/20$, both $1/5$, so $T'(3)=0$ and $T(3)=35/12$. A stationary calculation alone is not yet a global certificate; prompt 3 supplies that.

2 / hints to offer only after exploration

1. Differentiate both distance terms, keeping the minus sign in $7-x$.
2. Look for two 3-4-5 triangles.

Student page 2: Certify the best route

Calculus: a global guarantee and endpoint constraints

A stationary value is a candidate. Your certificate must cover every allowed crossing, including points outside the segment from $x=0$ to $x=7$.

Choose the fastest crossing

3 / prompt and solution

Prove that your best crossing is the ONLY global minimizer over all real x . Then express its derivative balance using the two angles from the vertical normal.

Solution: $T''(x) = 16/[3(x^2+16)^{3/2}] + 9/[4((7-x)^2+9)^{3/2}] > 0$ for every real x . Hence T' is strictly increasing, negative for $x < 3$ and positive for $x > 3$, since $T'(3) = 0$. By the mean value theorem T decreases then increases, proving the unique global minimum. At the optimum $0 < x < 7$, $\sin \theta_1 = x/L_1$ and $\sin \theta_2 = (7-x)/L_2$, with both acute angles measured from the normal. The equation is $\sin \theta_1/3 = \sin \theta_2/4$; here θ_1 has sine $3/5$ and θ_2 has sine $4/5$. No total-internal-reflection claim follows from this endpoint problem.

3 / hints to offer only after exploration

1. Can you show the derivative changes in only one direction?
2. Mark the horizontal component of each unit direction.

4 / prompt and solution

A fence permits crossing only on the closed interval $4 \leq x \leq 6$. Choose the fastest allowed crossing and prove it. Does the derivative have to vanish at this new winner?

Solution: $T' > 0$ throughout $[4, 6]$, so the unique constrained minimum is $x = 4$, with $T = 25\sqrt{2}/12$. Its derivative does not vanish: $T'(4) = 1/(3\sqrt{2}) - 1/(4\sqrt{2}) = 1/(12\sqrt{2}) > 0$. A boundary minimum need not have zero derivative because the improving leftward direction is forbidden. The certificate uses the allowed interval, not a new physical refraction law.

4 / hints to offer only after exploration

1. Where does the unconstrained optimum lie relative to the gate?
2. Which direction could lower time, and is it permitted?

Student page 3: Design the medium to fit your crossing

Calculus and algebra: inverse design

Remove the fence. Keep A , B and the upper speed 3. You may choose any finite positive lower speed w .

Choose the fastest crossing

5 / prompt and solution

Choose a target crossing s strictly between 0 and 7. Find a lower speed w that makes it the unique fastest crossing. Verify your own choice; also recover the original example.

Solution: Stationarity requires $s/[3\sqrt{(s^2+16)}] = (7-s)/[w\sqrt{((7-s)^2+9)}]$. Thus $w = 3(7-s)\sqrt{(s^2+16)}/[s\sqrt{((7-s)^2+9)}]$, positive and finite for $0 < s < 7$. The same positive-second-derivative proof works for every $w > 0$, so this is the unique global minimizer. At $s=3$ it gives $w=4$; at $s=4$ it gives $w=3$. Arbitrary learner targets are verified by substitution into this formula and the positivity argument.

5 / hints to offer only after exploration

1. Solve the derivative balance for the speed you control.
2. A stationary design still needs the same global certificate.

6 / prompt and solution

Could a finite positive w make $x=0$ or $x=7$ the unconstrained winner? If w increases from 4, predict which way the winner moves and justify the direction.

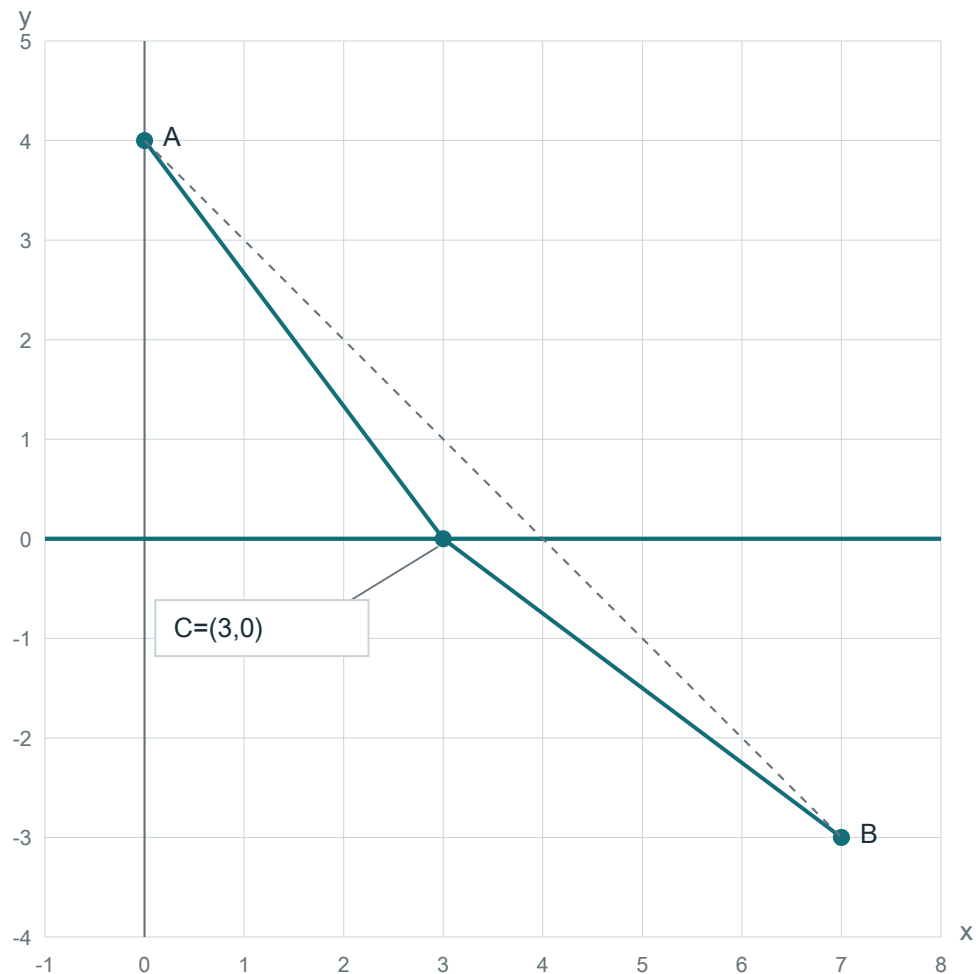
Solution: $T'(0) = -7/[w\sqrt{58}] < 0$ and $T'(7) = 7/[3\sqrt{65}] > 0$, so neither endpoint can be optimal. For every positive finite w , strict increase of T' and these endpoint signs give one root in $(0,7)$. Increasing w makes the negative second term smaller in magnitude, so T' at the old root becomes positive. The new unique root must lie to its left. Faster travel below the boundary shifts the crossing toward A, spending less distance in the slower upper medium. Limits $w \rightarrow \infty$ and $w \rightarrow 0^+$ approach $x=0$ and $x=7$ respectively, without attaining either at finite positive w .

6 / hints to offer only after exploration

1. Check the derivative at the endpoints directly.
2. Evaluate the new derivative at the old optimal point.

Worked route certificate

Facilitator diagram / reveals the solution



Solid: the fastest route crosses at $C=(3,0)$, with two length-5 legs and time $5/3+5/4=35/12$. Dashed: the direct line crosses at $x=4$; its time is $25\sqrt{2}/12$, which is larger.

The diagram supports the comparison. Strict positivity of T'' on all real x , together with $T'(3)=0$, is the global uniqueness certificate. Equal horizontal and vertical coordinate units are used.

Worked route certificate

Optional facilitator extension 1

Gate: Calculus, positive parameters and strict convexity.

Replace the heights by $a, b > 0$, horizontal separation by $d > 0$ and speeds by $v_1, v_2 > 0$. Prove the optimizer lies strictly between 0 and d and state the constrained answer for any closed gate $[L, U]$, $L \leq U$.

Solution: $T = \sqrt{(x^2 + a^2)}/v_1 + \sqrt{((d-x)^2 + b^2)}/v_2$ has $T'' = a^2/[v_1(x^2 + a^2)^{3/2}] + b^2/[v_2((d-x)^2 + b^2)^{3/2}] > 0$. $T'(0) < 0$ and $T'(d) > 0$, so there is a unique root $x^* \in (0, d)$, with T decreasing before it and increasing after it. The constrained minimum is x^* when it lies in $[L, U]$, L when $x^* < L$, and U when $x^* > U$, including the single-point gate. This proves a global minimum for the stipulated route class; it does not establish all of wave optics.

Mathematics and prior use

Strict convexity of a sum of weighted distances, Snell's sine law from a stationary time, and constrained versus unconstrained optima.

Prior use: Week 9 reflections and GA-21 mirror optimization use equal-speed reflected distance. This is unequal-speed refraction, with an actual derivative law and a different inverse-design problem.

Source: MIT 12.201, Essentials of Geophysics, Chapter 4: Seismology

§4.14, printed pp. 163-164 / PDF pp. 27-28, equations (4.91)-(4.94).. Use the two constant-speed straight-segment time model and angles from the normal. The 3-4-5 optimum, global proof, gate and inverse-design tasks are derived here. Evidence: Reopened 2026-09-25 and read travel-time differentiation and the distinction between stationary and minimum time. Strict global minimality is established for this worksheet, not inferred for all rays.

https://ocw.mit.edu/courses/12-201-essentials-of-geophysics-fall-2004/b78f19037066b01c0977aa9dda43b225_ch4.pdf

A dimmer signal, a hidden phase

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Promising with explicit trigonometry. Pointwise cancellation is a useful earlier stop, but the phase-identification proof is the full core. Scalar same-frequency coherence must remain visible.

A complete brightness range and a deliberately chosen extra reference signal that removes a genuine two-phase ambiguity.

Prerequisites and preparation

Core: Signed waveform addition, sine/cosine and squares. The full investigation needs trigonometric addition and a two-coordinate representation; period averages are supplied explicitly.

Extensions: Integration proves the supplied period averages; averaging over an explicitly uniform drifting phase gives an incoherent comparison.

Reading and representation: Phase is an angle in radians, not a time delay in arbitrary units. Follow a shared time coordinate and distinguish the total signed field from its squared average.

Materials: Supplied sine/cosine templates, tracing paper or pencil, unit-circle diagram and blank resultant plots. A ruler suffices; no sound or laser apparatus is needed. Preparation: about 12 minutes.

0-12 add in-phase and opposite-phase strips; 12-25 compare detector readings; 25-45 derive the phase formula and its attainable range; later 25 minutes for the inverse phase experiment.

Launch and facilitation

Two nonzero signals arrive at one detector. You can shift the second one. Can its arrival make the detector less bright? Can brightness alone reveal the shift?

Student page 1: Add a signal and reduce the reading

Core launch: signed heights and squared amplitude

At one detector, two coherent scalar waves have the same fixed frequency and polarization. Their signed readings add. $A(t)=2 \sin t$. You may add $B(t)=\sin(t+\phi)$. Start with $\phi=0$ or π . For $E(t)=C \sin t$, define normalized brightness $I=C^2$.

A dimmer signal, a hidden phase

1 / prompt and solution

Choose the alignment that you think will dim the detector. Draw the summed waveform over one whole period. Compare A alone with BOTH permitted alignments. Explain using the signed heights, before using brightness.

Solution: For $\phi=0$, $B=\sin t$ and $E=3 \sin t$, so $I=9$. For $\phi=\pi$, $B=-\sin t$ and $E=\sin t$, so $I=1$. A alone has $I=4$. The opposite-phase signal therefore reduces the local reading although its own amplitude is nonzero. Each equality holds at every time, not only at plotted sample points. Waveform templates share the same scale and time axis; sum corresponding signed heights.

1 / hints to offer only after exploration

1. A negative height subtracts from a positive height.
2. Check a peak, a trough and an ordinary intermediate time.

2 / prompt and solution

You may now change B's amplitude as well as its phase. Design a nonzero B that makes the total signal zero for the whole period. Does a zero reading here tell you that energy vanished everywhere?

Solution: Choose $B(t)=-2 \sin t$, equivalently amplitude 2 and phase π , so $E(t)=0$ pointwise. Among $B=b \sin(t+\phi)$ with $b>0$, exact cancellation requires $b \cos \phi=-2$ and $b \sin \phi=0$, hence $b=2$ and $\phi=\pi$ modulo 2π . No conclusion about energy everywhere follows from this one-detector scalar model. The spatial field, sources and energy transport have not been specified; local cancellation is the stated result.

2 / hints to offer only after exploration

1. Match both size and opposite sign, not just a single zero crossing.
2. Which locations does the supplied model describe?

Student page 2: Which brightness values are possible?

Core: sine addition and a complete range

Return to amplitude 1 for B, and allow any real phase ϕ . Define $I=2$ times the average of $E(t)^2$ over a full period. You may use these exact averages: $\text{average}(\sin^2 t)=\text{average}(\cos^2 t)=1/2$; $\text{average}(\sin t \cos t)=0$.

A dimmer signal, a hidden phase

3 / prompt and solution

Write $E(t) = P \sin t + Q \cos t$. Use the supplied averages to derive I as a function of ϕ . Find the least and greatest possible brightness, and show that every value between them can occur.

Solution: Sine addition gives $P = 2 + \cos \phi$ and $Q = \sin \phi$. Expanding the square and applying the stated averages gives $I = P^2 + Q^2 = (2 + \cos \phi)^2 + \sin^2 \phi = 5 + 4 \cos \phi$. Since $\cos \phi$ takes every value in $[-1, 1]$, the exact range is $[1, 9]$. The minimum 1 occurs at $\phi = \pi$ modulo 2π and maximum 9 at $\phi = 0$ modulo 2π . For any target $i \in [1, 9]$, choose a phase with $\cos \phi = (i - 5)/4$; such a phase exists. In particular, amplitude-1 B can never cancel amplitude-2 A completely.

3 / hints to offer only after exploration

1. Keep the cross term until you average it.
2. On the unit circle, what range can the horizontal coordinate take?

4 / prompt and solution

Choose a target brightness strictly between 1 and 9 and find two different phases in $[0, 2\pi)$ that produce it. Describe every phase that makes the detector dimmer than A alone. What information did squaring and averaging lose?

Solution: For a chosen target $i \in (1, 9)$, set $c = (i - 5)/4 \in (-1, 1)$. The two phases are $\arccos c$ and $2\pi - \arccos c$, distinct in $[0, 2\pi)$; they have opposite sine but equal cosine. A convenient choice $i = 5$ gives $\pi/2$ and $3\pi/2$. Dimming means $5 + 4 \cos \phi < 4$, or $\cos \phi < -1/4$: on $[0, 2\pi)$ the open interval $(\arccos(-1/4), 2\pi - \arccos(-1/4))$. For all real phases, take this interval and all its translates by $2\pi k$, $k \in \mathbb{Z}$. Equality at its endpoints gives the original brightness 4. The first observation recovers cosine but loses the sign of sine, and hence generally does not identify phase.

4 / hints to offer only after exploration

1. Compare phases reflected across the horizontal axis.
2. Translate “dimmer than 4” into an inequality for cosine.

Student page 3: Design a second observation

Core continuation: recover what the first measurement lost

The unknown phase stays fixed while you repeat the experiment on identically prepared signals. You may add a known coherent reference $R(t)$. All intensities use the same full-period normalization and unchanged amplitude scale.

A dimmer signal, a hidden phase

5 / prompt and solution

A first reading gives $I=5$. Choose ONE reference, $R(t)=\sin t$ or $R(t)=\cos t$, that distinguishes the compatible phases modulo 2π . Predict the second reading in each case. Explain why the other reference fails.

Solution: $I=5$ gives $\cos \varphi=0$, so φ is $\pi/2$ or $3\pi/2$ modulo 2π and $E=2 \sin t \pm \cos t$. Adding $R=\cos t$ yields $E+R=2 \sin t+2 \cos t$ in the plus case (brightness 8), and $2 \sin t$ in the minus case (brightness 4). Adding $R=\sin t$ yields $3 \sin t \pm \cos t$, brightness 10 in both cases, so it fails. This is a controlled repeat with stable relative phase, not a claim that unrelated experimental runs share an unknown phase.

5 / hints to offer only after exploration

1. Which coordinate changed sign between the two hidden candidates?
2. A useful reference must interact with that coordinate.

6 / prompt and solution

Let J be the second reading after adding $R(t)=\cos t$. Derive a rule that recovers both $\cos \varphi$ and $\sin \varphi$ from I, J . Decode $I=37/5$, $J=10$. Could $I=5$, $J=6$ come from this exact model? Give a certificate.

Solution: Adding the reference changes Q to $Q+1$, so $J=P^2+(Q+1)^2=I+1+2 \sin \varphi$. Therefore $\cos \varphi=(I-5)/4$ and $\sin \varphi=(J-I-1)/2$. For $I=37/5, J=10$ these are $3/5$ and $4/5$, locating a unique phase modulo 2π in the first quadrant ($\tan \varphi=4/3$). A pair of exact readings is compatible precisely when $((I-5)/4)^2+((J-I-1)/2)^2=1$; each point of this unit circle defines a phase and reproduces both readings. For $I=5, J=6$ both coordinates would be 0, violating the identity. That rejects the joint assumptions/data; it does not identify which physical assumption failed.

6 / hints to offer only after exploration

1. Compare J with I before expanding the unchanged coordinate again.
2. Any decoded sine/cosine pair must lie on the unit circle.

Optional facilitator extension 1

Gate: Integration and, for the last comparison, a specified phase probability law.

Prove the supplied period averages by integration over $[0, 2\pi]$. Then suppose the relative phase is uniformly distributed on $[0, 2\pi)$ across long observation blocks, each with a stable phase and a full-period average. What is the mean brightness for $A=2$ and $B=1$?

Solution: The identities $\sin^2 t=(1-\cos 2t)/2$, $\cos^2 t=(1+\cos 2t)/2$ and $\sin t \cos t=\sin 2t/2$ integrate to averages $1/2, 1/2, 0$ on a full period. At each fixed phase the already-derived brightness is $5+4 \cos \varphi$. Uniform phase averaging gives $(1/(2\pi)) \int_0^{2\pi} (5+4 \cos \varphi) d\varphi=5$. This is a precisely specified block-average model for loss of coherence, not the same as assigning brightness 5 to every coherent phase.

A dimmer signal, a hidden phase

Mathematics and prior use

A coherent amplitude pair is a plane vector; squared detector output loses direction. Two controlled intensity observations recover both phase coordinates, with a consistency circle.

Prior use: GA-14 later examines temporal aliasing and GA-15 orthogonal signal coordinates. This family asks what coherent intensity loses and how a phase-controlled reference restores it, not just how to add a waveform.

Source: David Tong, Electromagnetism

§4.3.1-4.3.2, printed pp. 84-88 / PDF pp. 18-22; §4.4, printed pp. 93-94 / PDF pp. 27-28.. Use linear same-frequency wave superposition and the quadratic, period-averaged field signal. The normalization and reference-wave inverse problem are defined and proved here. Evidence: Reopened 2026-09-25: inspected sine-wave solutions, phase/polarization representation, superposition and average $\sin^2=1/2$ in the Poynting calculation. The worksheet fixes one scalar component at one location; no whole-field energy inference is made.

<https://davidtong.org/pdfs/teaching/electromagnetism/electro3.pdf>

A sales pitch that balances energy and still fails

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Promising with algebra and careful model reading. Tokens support energy bookkeeping only; entropy and Kelvin temperatures remain actual supplied physics, not a “messiness” analogy.

A sharp bound, a reversible cascade achieving the same ideal bound, and an exact formula converting entropy generation into lost work for the specified reservoirs.

Prerequisites and preparation

Core: Fractions, algebraic inequalities and a clear heat/work sign convention. The second-law entropy inequality is a supplied physical law, not derived from token accounting.

Extensions: Reverse-cycle bookkeeping for a refrigerator. No microscopic derivation of thermodynamics is asserted.

Reading and representation: Reservoir temperatures are positive kelvin values. Q_h and Q_c denote positive heat magnitudes in the stated directions; W is positive work output.

Materials: Reservoir and engine diagram cards, pencil, optional twelve equal heat tokens. This is a paper model of complete cycles; no hot substances or physical machine is needed. Preparation: about 8 minutes.

0-10 sort chosen heat/work claims; 10-25 prove the upper bound; 25-40 compare two-stage designs; later 20 minutes for entropy generation and refrigerator extension. Keep the supplied physical law beside every ledger.

Launch and facilitation

A machine returns to its original state after every cycle. Can a claim balance every unit of energy and still violate a physical constraint?

Student page 1: Which claims can survive both tests?

Core: supplied physical laws and algebra

A cyclic engine takes $Q_h=12$ heat units from a 600 K reservoir, rejects $Q_c \geq 0$ to a 300 K reservoir, and outputs $W \geq 0$. It returns to its starting internal state. Supplied laws: $W+Q_c=12$ and $S_{\text{gen}}=Q_c/300-12/600 \geq 0$. Equality in the entropy law is the reversible ideal case.

A sales pitch that balances energy and still fails

1 / prompt and solution

Invent two energy-balanced claims: one that fails the entropy test and one that passes. Label all three energy transfers and calculate S_{gen} . Then audit the advertised claim $W=8$, $Q_c=4$.

Solution: Any nonnegative split $W+Q_c=12$ is energy-balanced. Entropy balance requires $Q_c \geq 6$. Thus a choice with $Q_c < 6$ fails, such as $(W, Q_c)=(8, 4)$ with $S_{\text{gen}}=4/300-12/600=-1/150$. A passing example $(4, 8)$ has $S_{\text{gen}}=1/150$; $(6, 6)$ has $S_{\text{gen}}=0$. Diagrams have heat arrows hot→engine and engine→cold, and work engine→load. Passing these necessary laws is not a physical construction.

1 / hints to offer only after exploration

1. The engine's own stored energy and entropy return to their starting values.
2. Which reservoir loses entropy, and which gains it?

2 / prompt and solution

Find the largest work output permitted by both laws, and prove that every larger claim fails. For general $Q_h > 0$ and $T_h > T_c > 0$, derive the corresponding bound. What happens when the reservoir temperatures are equal?

Solution: The inequality gives $Q_c \geq Q_h T_c/T_h$, hence $W=Q_h-Q_c \leq Q_h(1-T_c/T_h)$. For the supplied values $Q_c \geq 6$ and $W \leq 6$. Reversible ideal equality is compatible with both constraints. If $T_h=T_c > 0$, $Q_c \geq Q_h$ and $W \leq 0$; with the stipulated $W \geq 0$ the only energy-balanced passing engine has $W=0, Q_c=Q_h$. Ratios use kelvin, not Celsius. The second law is an input; this algebra does not prove it from energy conservation.

2 / hints to offer only after exploration

1. Turn the entropy inequality into a lower bound on discarded heat.
2. Substitute that lower bound into the energy equation.

Student page 2: Can a middle reservoir beat the bound?

Core continuation: reversible cascades

Use two reversible ideal engines. The first works from 600 K down to T_m ; the second works from T_m down to 300 K. All heat rejected by the first is absorbed by the second, so the middle reservoir has no net heat change over the paired cycles. Initial heat remains 12.

A sales pitch that balances energy and still fails

3 / prompt and solution

Choose $T_m=400$ K or 450 K. Compute each engine's work and every transferred heat. Compare with a single reversible engine from 600 K to 300 K.

Solution: For reversible stages, $Q_m=12 T_m/600$. At $T_m=400$, $Q_m=8$, $W_1=4$, $Q_c=8 \times 300/400=6$ and $W_2=2$. At $T_m=450$, $Q_m=9$, $W_1=3$, $Q_c=9 \times 300/450=6$ and $W_2=3$. Both total work outputs are 6. The middle reservoir receives and gives the same Q_m , and its entropy changes cancel. A single reversible engine also gives $W=6, Q_c=6$.

3 / hints to offer only after exploration

1. Reversible equality fixes the heat ratio in each stage.
2. Use the first stage's rejected heat as the second stage's input.

4 / prompt and solution

Prove the total work is independent of T_m for every $300 < T_m < 600$. Could adding more reversible middle stages improve it? Give an argument covering any finite number of stages.

Solution: $Q_m=12 T_m/600$ and $Q_c=Q_m \times 300/T_m=6$, so $W_1+W_2=(12-Q_m)+(Q_m-6)=6$. For a descending sequence $T_0=600 > T_1 > \dots > T_n=300$, reversible equality gives $Q_i=Q_{i-1} T_i/T_{i-1}$. Multiplication telescopes to $Q_n=12 T_n/T_0=6$. Summing work differences gives $12-6=6$. This covers every finite reversible cascade with no net middle-reservoir storage; adding stages cannot exceed the same external-reservoir bound.

4 / hints to offer only after exploration

1. Multiply consecutive temperature ratios.
2. Add the work ledgers and cancel internal heat transfers.

Student page 3: Locate the lost work

Core continuation: entropy generation as a certificate

Use $T_m=400$ K. A changed first engine takes 12 from 600 K but rejects 9 to 400 K. The second engine is reversible and accepts all 9, then rejects heat to 300 K. Both engines are cyclic; the middle reservoir has no net heat change.

A sales pitch that balances energy and still fails

5 / prompt and solution

Audit each stage and compute total work, final rejected heat and total reservoir entropy change. Compare with the reversible limit. Where did the shortfall arise?

Solution: First-stage work is 3 and entropy generation is $9/400 - 12/600 = 1/400$. The reversible second stage rejects $Q_c = 9 \times 300/400 = 27/4$ and outputs $W_2 = 9/4$, with zero entropy generation. Total work is $3 + 9/4 = 21/4$; the reversible limit is 6, so the shortfall is $3/4$. External reservoir entropy change is $(27/4)/300 - 12/600 = 1/400$, agreeing with the sum of stage values; middle terms cancel. The first stage creates the positive entropy generation.

5 / hints to offer only after exploration

1. First audit the two stages separately, then cancel the middle transfers.
2. The second engine sees an input of 9, not 12.

6 / prompt and solution

For any cyclic device or cascade exchanging heat only with fixed T_h and T_c , keep the same Q_h and define $S_{\text{gen}} = Q_c/T_c - Q_h/T_h$. Derive an exact relation between S_{gen} and the work shortfall from the reversible bound. Can adding a reversible stage erase already positive total entropy generation?

Solution: Let $W_{\text{rev}} = Q_h(1 - T_c/T_h)$ and $W = Q_h - Q_c$. Subtracting gives $W_{\text{rev}} - W = Q_c - Q_h T_c/T_h = T_c S_{\text{gen}}$. With fixed Q_h, T_h, T_c , positive total entropy generation forces a strictly positive shortfall, and equality of work with the reversible bound requires $S_{\text{gen}} = 0$. A reversible added stage has zero entropy generation, so it cannot cancel a positive amount; all allowed stage generations are nonnegative. Changing external reservoirs or heat budget changes the comparison and must be stated. In the example $300 \times (1/400) = 3/4$.

6 / hints to offer only after exploration

1. Subtract the two work expressions before replacing anything.
2. Keep the external reservoirs fixed when comparing designs.

Optional facilitator extension 1

Gate: Algebra and a reversed heat/work convention.

A refrigerator extracts $Q_c > 0$ from a cold reservoir T_c and deposits Q_h into a hot reservoir $T_h > T_c$, using positive work input W_{in} . Derive the least permitted input and the greatest coefficient of performance Q_c/W_{in} . Evaluate $Q_c = 6$, $T_c = 300$ K, $T_h = 600$ K.

Solution: Energy gives $Q_h = Q_c + W_{\text{in}}$. Reservoir entropy is $Q_h/T_h - Q_c/T_c \geq 0$, giving $W_{\text{in}} \geq Q_c(T_h/T_c - 1) = Q_c(T_h - T_c)/T_c$. Thus $Q_c/W_{\text{in}} \leq T_c/(T_h - T_c)$, with reversible equality. Here $W_{\text{in}} \geq 6$ and the coefficient is at most 1. Directions differ from the engine: cold $\rightarrow$ refrigerator, work $\rightarrow$ refrigerator, refrigerator $\rightarrow$ hot. The positive-input premise excludes division by zero, since $T_h > T_c$ and $Q_c > 0$.

A sales pitch that balances energy and still fails

Mathematics and prior use

Intersecting a conservation affine line with an inequality gives a sharp certificate. Reversible intermediate exchanges telescope; irreversible entropy generation has a quantified work cost in this two-reservoir comparison.

Prior use: AP-11 also uses conservation, but here a second independent thermodynamic inequality rejects energy-balanced proposals. No prior circle thermodynamic cycle is documented.

Source: David Tong, Statistical Physics: Classical Thermodynamics

§4.3.2 Carnot's theorem and ideal-gas cycle, equation (4.162); §4.3.3 entropy and Clausius inequality.. Use the two-reservoir efficiency limit and reservoir entropy balance. The chosen heat ledger, intermediate-temperature designs and loss identity are worked self-contained. Evidence: Reopened 2026-09-25: read Carnot's theorem, $\eta=1-T_c/T_h$, the heat-sign convention and the irreversible-cycle inequality. The law is explicitly supplied on the student launch; passing it is not a constructed engine.

<https://davidtong.org/teaching/statistical-physics/statmechhtml/S4>

The question between two identical questions

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Advanced/specialist vector/probability investigation, with a compact four-state entry. Keep the Born rule and update visible; replacing them with an unexplained coin game would lose the actual mathematics.

Exact recorded-outcome statistics show an order effect, and an algebraic bound identifies the best single unrecorded intermediate basis for changing a formerly certain answer.

Prerequisites and preparation

Core: Normalized two-component vectors, dot products, squared probabilities and branch trees. The Born probability and projection-update rules are supplied as actual quantum model assumptions.

Extensions: Matrix multiplication for projectors; trigonometry for the optimal rotated basis and additional intermediate measurements. No complex-state or entanglement claim.

Reading and representation: State vectors represent real quantum amplitudes, not hidden physical arrow directions. Outcomes and state changes must both be recorded.

Materials: Printed definitions of four unit states $e_0=(1,0)$, $e_1=(0,1)$, $+=(1,1)/\sqrt{2}$, $-=(1,-1)/\sqrt{2}$; blank probability trees and two-component coordinate diagrams. No physical quantum apparatus is implied. Preparation: about 12 minutes.

0-10 check unit vectors and a sample projection; 10-25 compare chosen two-question orders; 25-40 distinguish repeated measurement from an intervening question; later 25 minutes for projector and optimal-basis proofs.

Launch and facilitation

Start in e_0 . A Z question has outcomes e_0, e_1 ; an X question has outcomes $+, -$. Choose an order, then build the exact tree. The next question acts on the state you actually reached.

Student page 1: Choose the order, keep the state

Core: dot products and branch probabilities

Start with the normalized real state $\psi=e_0=(1,0)$. Define $e_1=(0,1)$. Z measures in $\{e_0, e_1\}$; X measures in $\{+, -\}$, where $\pm=(1, \pm 1)/\sqrt{2}$. Supplied rule: outcome b has probability $(b \cdot \psi)^2$, then the state becomes b . Branch probabilities multiply; disjoint branches add. No evolution occurs between questions.

The question between two identical questions

1 / prompt and solution

Choose X then Z or Z then X . Build its full outcome tree, labeling every branch probability and every new state. Then build the other order. In each experiment, record the Z answer wherever it occurs. How often is that answer e_1 ?

Solution: For $X \rightarrow Z$, the first outcomes $+$ and $-$ each have probability $1/2$. From either, $Z=e_0$ or e_1 each has probability $1/2$. All four joint branches have probability $1/4$, so recorded $Z=e_1$ has probability $1/2$. For $Z \rightarrow X$, the first Z is e_0 with probability 1 and e_1 with probability 0; from e_0 , X is $+$ or $-$ with probability $1/2$. Hence its recorded $Z=e_1$ probability is 0. Zero-probability branches need not be continued. These are different recorded statistics; no claim is made that the final unconditional mixed states differ for these particular two bases.

1 / hints to offer only after exploration

1. Square the dot product before using it as a probability.
2. The second question uses the branch's new state.

2 / prompt and solution

Check that probabilities sum to 1 at every live branch. A proposed shortcut computes both questions from the original e_0 and never updates it. Find a specific branch whose probability that shortcut gets wrong.

Solution: Each orthonormal basis has squared projections summing to $||\psi||^2=1$. Explicitly $1/2+1/2=1$ for the cross-basis transitions, and $1+0=1$ for Z from e_0 . In $X \rightarrow Z$, after the $+$ branch the correct conditional probability of $Z=e_1$ is $1/2$; the shortcut uses e_0 again and gets 0. Its joint $(+,e_1)$ probability 0 contradicts the correct $1/4$. This isolates the missed update rather than a generic failure of classical probability arithmetic.

2 / hints to offer only after exploration

1. Look at the state immediately after the first X outcome.
2. Which part of the supplied rule did the shortcut discard?

Student page 2: A certain answer can become uncertain

Core continuation; matrices in prompt 4

Every run again starts in e_0 . Compare a repeated question with a question inserted between repeats. An outcome may be unreported, but the projection update still occurs.

The question between two identical questions

3 / prompt and solution

Choose $Z \rightarrow Z$ or $Z \rightarrow X \rightarrow Z$. Predict whether the final Z can differ from the first, then compute the exact probability. If the middle X result is erased from the record, does the final Z distribution return to certainty?

Solution: $Z \rightarrow Z$ always gives e_0 then e_0 , so change probability 0. $Z \rightarrow X \rightarrow Z$ starts with e_0 certainly; X gives $\pm$ with probability $1/2$ and each then gives e_1 with probability $1/2$. The final answer differs with probability $1/2$. Erasing the classical X label means summing the two branches, not undoing their physical state updates, so it remains $1/2$. Repeating one basis immediately is stable; inserting another basis changes the subsequent statistics.

3 / hints to offer only after exploration

1. Deleting a label and deleting an interaction are different operations.
2. Sum the unreported branches only after updating their states.

4 / prompt and solution

Matrix gate: let $P_0 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $P_+ = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}/2$. Compute $P_0 P_+$ and $P_+ P_0$. Applied to e_0 , the squared norm of each unnormalized vector is the probability of its ordered pair of selected outcomes. Match the two results with your trees.

Solution: $P_0 P_+ = \begin{bmatrix} 1/2 & 1/2 \\ 0 & 0 \end{bmatrix}$ while $P_+ P_0 = \begin{bmatrix} 1/2 & 0 \\ 1/2 & 0 \end{bmatrix}$, so they do not commute. Acting on e_0 gives $(1/2, 0)$ and $(1/2, 1/2)$, with squared norms $1/4$ and $1/2$. Since the rightmost matrix acts first, these are respectively $X=+$ then $Z=e_0$, and $Z=e_0$ then $X=+$. They agree with the corresponding tree branches. Normalizing the intermediate state and multiplying conditional probabilities gives the same result because the intermediate norm factors cancel. A zero intermediate vector represents a zero-probability branch.

4 / hints to offer only after exploration

1. Read matrix products from right to left when they act on a state.
2. Keep vectors unnormalized only for the stated branch-probability calculation.

Student page 3: Choose the most disruptive single question

Continuation: trigonometry and a sharp bound

Start in e_0 , ask ONE intermediate question Y , do not retain its outcome, then ask Z . Choose Y 's orthonormal basis $b_0 = (\cos \theta, \sin \theta)$, $b_1 = (-\sin \theta, \cos \theta)$, with $0 \leq \theta \leq \pi/2$. The target is final $Z=e_1$.

The question between two identical questions

5 / prompt and solution

First test the rational basis $b_0=(3/5,4/5)$, $b_1=(-4/5,3/5)$. Find the target probability using both intermediate branches. Is it larger or smaller than inserting X?

Solution: The first probabilities are $(b_0 \cdot e_0)^2=9/25$ and $(b_1 \cdot e_0)^2=16/25$. From b_0 , final e_1 has probability $16/25$; from b_1 , $9/25$. Total target probability is $2 \times 9 \times 16/625=288/625$. It is less than $1/2$ because $576 < 625$. Inserting X corresponds to $\theta=\pi/4$ and gives $1/2$. Both basis vectors are unit and perpendicular: $(3/5)(-4/5)+(4/5)(3/5)=0$.

5 / hints to offer only after exploration

1. Calculate the two joint target branches separately.
2. Compare the fraction exactly, without rounding.

6 / prompt and solution

Design the angle θ that makes final $Z=e_1$ as likely as possible. Prove that no SINGLE unreported intermediate projective question in this real model can do better. Can the target become certain?

Solution: The probability is $\cos^2 \theta \sin^2 \theta + \sin^2 \theta \cos^2 \theta = 2 \sin^2 \theta \cos^2 \theta$. Put $p=\sin^2 \theta \in [0,1]$. Then $2p(1-p)=1/2-2(p-1/2)^2 \leq 1/2$, with equality exactly $p=1/2$. On the given angle interval this is $\theta=\pi/4$, the X basis up to outcome signs/labels. It cannot make the target certain. The bound is only for one unreported intermediate basis with no feedback or additional evolution; prompt 1's order effect is not a universal $1/2$ bound on all quantum procedures.

6 / hints to offer only after exploration

1. A square is nonnegative.
2. State exactly how many intermediate questions your argument covers.

The question between two identical questions

Optional facilitator extension 1

Gate: Trigonometric dot products and complete probability-tree summation.

Allow TWO unreported intermediate questions, with basis angles $\pi/6$ then $\pi/3$, followed by Z. Starting at e_0 , can the probability of final e_1 exceed the single-question bound? Derive it without selecting only favorable records.

Solution: Let $b(\alpha) = (\cos \alpha, \sin \alpha)$ and its perpendicular partner define each basis. Transition probabilities between angles α and β are $\cos^2(\beta - \alpha)$ for keeping the basis label and $\sin^2(\beta - \alpha)$ for switching it. At $\alpha = \pi/6$, plus-label probability is $3/4$. At $\beta = \pi/3$ the angle difference is $\pi/6$, so plus-label probability becomes $(3/4)(3/4) + (1/4)(1/4) = 5/8$; minus is $3/8$. Final $Z = e_1$ has probability $(5/8)\sin^2(\pi/3) + (3/8)\cos^2(\pi/3) = 15/32 + 3/32 = 9/16 > 1/2$. All intermediate records were summed; no postselection or feedback was used. This does not contradict a bound explicitly limited to one intermediate question.

Mathematics and prior use

Noncommuting projectors, conditional branch composition and an optimal disturbance in a real two-dimensional projective-measurement model.

Prior use: AP-02 uses classical conditional evidence. Here the stipulated projection update changes the state used for the next measurement. No prior finite quantum-measurement investigation is documented.

Source: David Tong, Quantum Mechanics: The Formalism of Quantum Mechanics

§3.3 Measurement, equations (3.71)-(3.72), “First One Thing, Then The Other”; §3.4 commutation relations.. Restrict the stated Born/update rules to real two-dimensional orthonormal bases. All four-state probabilities, matrices and optimal-basis examples are derived here. Evidence: Reopened 2026-09-25: read normalization, outcome probabilities, post-measurement eigenstate and repeated versus intervening measurements. Final unobserved mixed states are not claimed different for the X/Z example.

<https://www.davidtong.org/teaching/quantum-mechanics/qmhtml/S3>

Build a sampler for a tiny magnet

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

First-pilot candidate for confident counters, with fractions needed for the sampler and a later probability gate for rejection sampling. It is a genuine finite ensemble; the energy interpretation is supplied explicitly in the guide.

A first-spin/link encoding proves the weighted distribution and yields a small exact sampler; closing the ring reveals a new parity constraint and a corrected conditional sampler.

Prerequisites and preparation

Core: Four binary choices, complete counting and weighted probabilities. Distinguish the weight of one configuration from total weight of a class; arithmetic reaches 82.

Extensions: Exponentials connect the stipulated weights to Ising energy; binomial/parity algebra gives general finite chains and rings. No finite-system phase transition is claimed.

Reading and representation: An open row has exactly three interaction links. A ring adds a fourth link. Spin labels are + and –, not magnetic field directions in a laboratory setup.

Materials: Four two-sided counters, a three-link strip and a four-link ring diagram, open configuration record and three equal tickets for an exact sampler. Draw independently with replacement when a ticket model is used. Preparation: about 8 minutes.

0-12 make and compare configurations; 12-30 organize a complete catalog; 30-45 design and prove the sampler; later 25 minutes for the ring and conditional sampling. Do not require drawing sixteen near-identical pictures if a bijective encoding replaces them.

Launch and facilitation

A perfectly aligned row gets more weight than any one mixed row. Does it beat all mixed rows taken together? Can you generate exactly the right distribution without making a large weighted ticket bag?

Student page 1: A favorite arrangement or a favorite class?

Core: four spins and weighted counts

Four labeled sites 1,2,3,4 form an OPEN row, with links 12,23,34 only. Each spin is + or –. If a configuration has a agreeing links, assign weight 2^a . Probability equals its weight divided by the total weight of all configurations.

Build a sampler for a tiny magnet

1 / prompt and solution

Choose several configurations, including two with the same number of + spins but different neighbor patterns. Find their agreeing links and weights. Does counting + spins alone determine the weight?

Solution: Every chosen pattern is checked along the three specified links. Examples $++--$ has $a=2$ and weight 4, while $+-+-$ has $a=0$ and weight 1; both have two + spins. Thus number of + spins alone does not determine weight. $++++$ and $----$ have $a=3$ and weight 8; $+++-$ has $a=2$ and weight 4. Global sign reversal preserves every agreement and hence the weight.

1 / hints to offer only after exploration

1. Mark each link as agree or disagree.
2. Keep the number of + spins fixed while rearranging their positions.

2 / prompt and solution

Predict which agreement class has the greatest TOTAL weight. Organize all configurations by $a=0,1,2,3$, without counting any twice. Compare the probability of perfect alignment in this model with uniform choice of the sixteen configurations.

Solution: Class multiplicities for $a=0,1,2,3$ are 2,6,6,2. Total class weights are 2,12,24,16, summing to 54. The $a=2$ class has greatest total weight 24, although each aligned configuration individually has the greatest weight 8. Perfect alignment has probability $16/54=8/27$, compared with $2/16=1/8$ under uniform sampling. The remaining mixed configurations together have weight $38>16$. Any complete nonduplicated catalog or the encoding proof developed on page 2 is accepted.

2 / hints to offer only after exploration

1. Pair each pattern with its global sign reversal.
2. A class's total weight is its size times one member's weight.

Student page 2: Replace a large raffle by tiny choices

Core continuation: a bijection and an exact probability law

Record a configuration by its first spin and, for each link from left to right, A for agree or D for disagree. The later spins must be reconstructed from that record.

Build a sampler for a tiny magnet

3 / prompt and solution

Prove that this record determines exactly one configuration and that every configuration has one record. Use it to certify your class counts and total weight without checking sixteen drawings individually.

Solution: Given the first spin, A copies the current sign and D flips it, so the three symbols determine all later spins uniquely. Conversely, every spin row has one first sign and one link record; these constructions invert each other. For a agreements, choose their a positions among 3 links and choose the first spin in 2 ways, giving 2 times $C(3,a)$: 2,6,6,2. The weighted total is $2(1+2)^3=54$, because each independent record position contributes weight 1 for D or 2 for A. This is a complete bijection, not only a count guessed from examples.

3 / hints to offer only after exploration

1. Can two rows have the same first sign and link record?
2. Sum weights at each record position before multiplying.

4 / prompt and solution

Design an exact random sampler: choose the first spin fairly, then choose each link independently, with the same chance p of A. What p makes each row's probability proportional to 2^a ? Give a small-ticket implementation and prove the probability of EVERY row.

Solution: For $p=2/3$, use a fresh independent draw with replacement from three equally likely tickets A1,A2,D at each link. Choose the first spin with a separate fair coin. A row with a agreements then has probability $(1/2)(2/3)^a(1/3)^{3-a}=2^a/54$, exactly its weight divided by the partition sum. To derive p , compare two records differing at one bond: the desired weight ratio is 2, so $p/(1-p)=2$ and $p=2/3$. Independent fair spins instead make all 16 configurations equally likely and do not produce these weights.

4 / hints to offer only after exploration

1. Compare the chance of an A with the chance of a D.
2. Multiply the first-spin probability and all three link probabilities.

Student page 3: Close the last link

Continuation: parity and conditional sampling

Now add link 41, making a RING of four labeled sites. Weight is still 2^a , but a now counts agreements on all FOUR links. Two configurations are identical exactly when every labeled site has the same spin in both.

Build a sampler for a tiny magnet

5 / prompt and solution

Try to prescribe the four A/D link symbols freely and reconstruct a spin ring. Which records fail to close? Use that obstruction to find all agreement-class sizes, the total ring weight and the new alignment probability.

Solution: Walking around the ring flips the spin once for each D. Returning to the first site requires an even number of D symbols; this condition is also sufficient. With $d=0,2,4$ disagreements, counts are 2 $C(4,d)=2,12,2$, and $a=4,2,0$ respectively. Odd a classes are empty. Class weights are 32,48,2, summing to 82. Perfect alignment has probability $32/82=16/41$. The $a=2$ class still outweighs the aligned class. The first spin supplies the factor 2; labeling the sites means rotations are not identified.

5 / hints to offer only after exploration

1. What does an odd number of flips do to the first sign?
2. The final link is a consistency test, not a free extra spin.

6 / prompt and solution

Try this repair: choose a fair first spin and FOUR independent links with $P(A)=2/3$. Reject a record if it fails to close; then redraw all choices independently until one closes. Prove that the accepted ring has the desired weighted distribution. Find the acceptance probability.

Solution: A particular first-spin/four-link proposal with a agreements has probability $(1/2)(2/3)^a(1/3)^{4-a}=2^a/162$. The accepted proposals correspond bijectively to the 16 valid labeled rings. Their weights sum to 82, so acceptance probability is $82/162=41/81$. Conditioning on acceptance divides each valid proposal probability by $41/81$, giving $2^a/82$, the desired ring law. Repeated independent retries yield this same conditional law and terminate with probability 1 because the failure probability after n attempts is $(40/81)^n \rightarrow 0$. Forcing the last bond after only three draws would keep the open-row distribution instead, so rejection is doing real work.

6 / hints to offer only after exploration

1. Add the probabilities of every valid proposal before normalizing.
2. The sampler must redraw rejected choices, not silently change their last bond.

Worked link-count audit

Facilitator table / reveals the enumerations

Four labeled spins; open links 12,23,34, or ring links 12,23,34,41. The first spin has two choices. The link record determines all later spins.

a	Open count	Open weight	Ring count	Ring weight
0	2	2	2	2
1	6	12	0	0
2	6	24	12	48
3	2	16	0	0
4	-	-	2	32

Open total: 54. Ring total: 82. The ring accepts exactly the records with an even number of D links; the accepted probability is $82/162=41/81$.

The ring proposal gives each valid configuration weight $2^a/162$ before conditioning. Dividing by $41/81$ gives exactly $2^a/82$. Redrawing all choices after rejection matters.

Worked link-count audit

Optional facilitator extension 1

Gate: Exponentials for the physics connection; binomial expansion for general finite size.

Explain the Ising energy behind b^a weights. For $n \geq 3$ labeled spins derive total weights for an open row and a ring at any $b > 0$, and state why these formulas do not prove a finite-temperature phase transition.

Solution: For an open row, $E = -J \sum_i s_i s_{i+1} = -J(2a - (n-1))$. The Boltzmann weight $\exp(-\beta E)$ has common factor $\exp(-\beta J(n-1))$ times b^a with $b = \exp(2\beta J)$; for ferromagnetic $J, \beta > 0$, $b > 1$. The four-spin $b=2$ choice means $\beta J = (\ln 2)/2$. For any positive abstract b , the open total is $2(1+b)^{(n-1)}$. On a ring, the number d of disagreements is even, giving $2 \sum_{(d \text{ even})} C(n,d) b^{(n-d)} = (b+1)^n + (b-1)^n$. The finite sums are positive for $b > 0$ and analytic there, so no singular finite-size transition is established. A thermodynamic-limit statement would require a separate limit analysis.

Mathematics and prior use

Finite Ising partition functions, a bijection from spin configurations to first spin plus bond agreements, product measures on a path and the parity constraint on a cycle.

Prior use: Week 6 binary coding and AD-02 bit strings provide familiar representations. The new substance is interaction energy, multiplicity and a weighted sampler whose independence changes when a loop closes.

Source: David Tong, Statistical Physics: Phase Transitions

§5.2 Ising model, equations (5.192)-(5.193); §5.3.1 one-dimensional chain.. Use zero-field nearest-neighbor Ising energy and canonical weights. Four-spin open/ring enumeration and the exact samplers are independently derived here. Evidence: Reopened 2026-09-25: read $E = -J \sum_i s_i s_j - B \sum_i s_i$ and $Z = \sum \exp(-\beta E)$, ferromagnetic sign and finite-chain setting. Here $B=0$ and $\exp(2\beta J)=2$; a common factor is canceled from all weights.

<https://davidtong.org/teaching/statistical-physics/statmechhtml/S5>

Choose a route for a clock

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Promising with algebra and careful diagrams; the boosted-frame page is advanced. The main clock comparison preserves actual special relativity under a supplied metric, not a rubber-sheet analogy.

Design a specified elapsed time, prove the stationary route gives the largest possible proper time in the stated class, and verify the answer survives a change of inertial coordinates.

Prerequisites and preparation

Core: Coordinates, slopes, squares and square roots. The flat-spacetime proper-time rule is supplied; algebra proves the finite piecewise-inertial comparison.

Extensions: Calculus and strict concavity for optimizing the turnaround time at a fixed excursion distance. Curved-spacetime gravity is outside this model.

Reading and representation: Event pairs are always (t,x) , although x is the horizontal drawing axis and t is vertical. Units make light speed $c=1$. Coordinate time and a clock's proper time are different quantities.

Materials: Supplied spacetime grids with equal coordinate scales, ruler and two colors. Printed formulas for inertial-segment duration and one Lorentz coordinate change. No Euclidean-length measurement is used as clock time. Preparation: about 8 minutes.

0-12 choose and reject turn events using the speed limit; 12-25 compute reunited clocks; 25-40 inverse-design a duration and prove the maximum; later 25 minutes for the boosted frame. Treat instantaneous turns explicitly as idealized limits.

Launch and facilitation

Both clocks depart at $(0,0)$ and reunite at $(10,0)$. One stays at $x=0$. You choose where the other turns. A longer line on the picture need not record more time.

Student page 1: Which turn points are legal?

Core: a supplied spacetime rule

Events are written (t,x) . The drawing has x horizontal and t vertical. Use units with light speed 1. A massive clock's straight segment must have $\Delta t > 0$ and $|\Delta x| < \Delta t$. Its elapsed proper time is $\sqrt{(\Delta t)^2 - (\Delta x)^2}$. Add proper times over segments. An instantaneous turn is an idealization, not an inertial segment.

Choose a route for a clock

1 / prompt and solution

Departure is $O=(0,0)$, reunion $R=(10,0)$. Choose a turn point $T=(t,d)$ with $d \neq 0$ and draw $O \rightarrow T \rightarrow R$ with both legs legal. Describe the whole region of legal turn points. Test $T=(5,3)$, $(2,3)$ and $(5,5)$.

Solution: For $T=(t,d)$, legality requires $0 < t < 10$, $|d| < t$ and $|d| < 10-t$. Thus $|d| < \min(t, 10-t)$, the open diamond between the departure's future light rays and the reunion's past light rays. Nonzero excursion means $d \neq 0$. $T=(5,3)$ is legal; $(2,3)$ has an outward speed $3/2 > 1$ and fails; $(5,5)$ has both legs at light speed and is excluded for a massive clock. Any chosen interior point with $d \neq 0$ works. The boundary is not included.

1 / hints to offer only after exploration

1. Check each leg separately, with forward time.
2. A light-speed boundary is not a legal massive-clock route.

2 / prompt and solution

Clock A stays at $x=0$. Clock B uses $T=(5,3)$. Compute each clock's total reading at reunion. Then compute the reading for your own legal turn point. Why should measuring the drawn line with a ruler give a different result?

Solution: A has $\Delta t=10, \Delta x=0$, so $\tau_A=10$. Each B leg has $\Delta t=5$ and $|\Delta x|=3$, giving $\sqrt{(25-9)}=4$; hence $\tau_B=8$. For a learner's $T=(t,d)$, $\tau = \sqrt{(t^2-d^2)} + \sqrt{((10-t)^2-d^2)}$. The Euclidean drawing length would use a plus sign and is not the supplied Lorentzian proper-time rule. Both clocks share departure and reunion events but follow different spacetime paths.

2 / hints to offer only after exploration

1. The return leg has negative Δx , but it is squared.
2. Coordinate time 10 is shared; proper time is accumulated along the route.

Student page 2: Design a clock reading, then bound every route

Core continuation: algebra and a global certificate

First restrict to symmetric excursions $O \rightarrow (5,d) \rightarrow R$ with $|d| < 5$. Then allow any finite number of forward-time straight segments, all with speed strictly less than 1, from O to R .

Choose a route for a clock

3 / prompt and solution

Design a symmetric route on which the traveler records exactly 6 time units. Which target readings can symmetric legal routes attain? Can the clock record exactly 0?

Solution: The symmetric duration is $\tau = 2\sqrt{(25-d^2)}$. Setting $\tau=6$ gives $d^2=16$, so $d=4$ or -4 ; each leg has speed $4/5 < 1$. For any $\tau \in (0, 10]$, choose $d = \pm\sqrt{(25-\tau^2/4)}$, with $d=0$ for $\tau=10$. If a nonzero excursion is demanded, the range is $(0, 10)$. Duration 0 would require $|d|=5$, a lightlike boundary excluded for a massive clock. Values can approach 0 as $|d|$ approaches 5 from below, but no legal route attains 0.

3 / hints to offer only after exploration

1. Solve for the excursion distance rather than guessing turn points.
2. Check whether an endpoint value obeys the strict speed condition.

4 / prompt and solution

Could a route with many turns make the traveler's clock read more than A's 10? Give a proof for EVERY permitted finite route, including the equality case.

Solution: For every leg with $\Delta t > 0$, $\sqrt{(\Delta t)^2 - (\Delta x)^2} \leq \Delta t$, with equality exactly $\Delta x = 0$. Summing over the finitely many legs gives $\tau \leq \sum \Delta t = 10$. Equality requires every $\Delta x = 0$, so the whole route stays at $x=0$ (subdivision is allowed). Any genuine movement on at least one leg makes the inequality strict. This proves a maximum in A's rest frame for the stated finite piecewise-inertial class; it does not compare Euclidean lengths or rely on symmetric turn times.

4 / hints to offer only after exploration

1. Bound each leg by its coordinate-time interval.
2. When can every term attain equality simultaneously?

Student page 3: Change the coordinates, keep the events

Advanced algebra: one inertial boost

Return to $T=(5,3)$. An observer moving at speed $3/5$ uses $t'=(5t-3x)/4$ and $x'=(5x-3t)/4$. This single inertial coordinate system covers the WHOLE trip. Event pairs below are (t', x') ; use the same proper-time rule.

Choose a route for a clock

5 / prompt and solution

Transform O, T and R. Replot BOTH worldlines in the primed grid. Recalculate A's duration and both B-leg durations from the transformed differences. Do the clock readings agree with the first calculation?

Solution: $O'=(0,0)$, $T'=((25-9)/4,(15-15)/4)=(4,0)$, $R'=(50/4,-30/4)=(25/2,-15/2)$. A is the straight line $O' \rightarrow R'$, with squared interval $(25/2)^2 - (15/2)^2 = 100$ and duration 10. B first goes $O' \rightarrow T'$ with duration 4. Its return difference is $(17/2, -15/2)$, giving squared interval $(289-225)/4=16$ and duration 4. Thus B still records 8. T' is B's turn event, not an event on A's worldline. The two return segments do not share equal coordinate durations in this frame.

5 / hints to offer only after exploration

1. Transform the turnaround event as well as departure and reunion.
2. Subtract primed endpoint coordinates for each segment.

6 / prompt and solution

Expand $(\Delta t')^2 - (\Delta x')^2$ and prove it equals $(\Delta t)^2 - (\Delta x)^2$ for any segment. In the primed frame, which clock changes velocity? Explain why "each sees the other moving" does not make these two complete trips interchangeable.

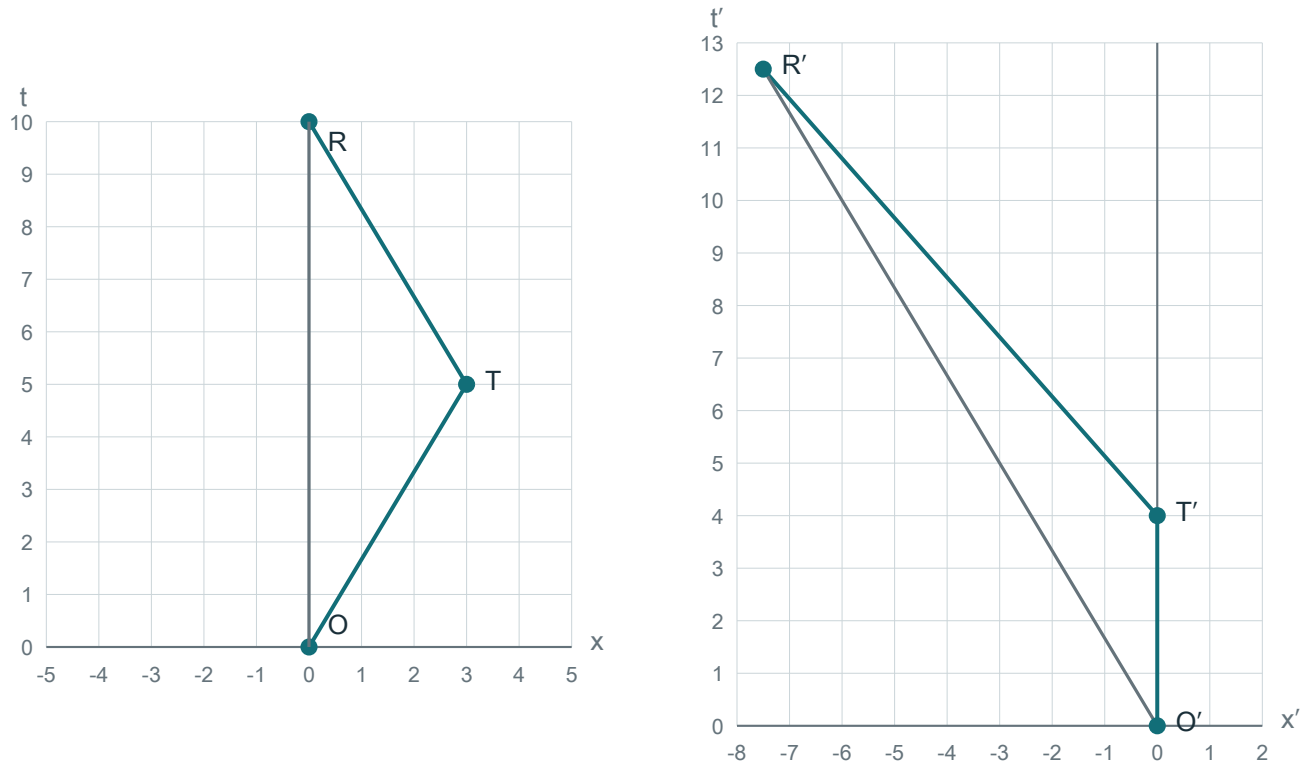
Solution: Expansion gives $[(5\Delta t - 3\Delta x)^2 - (5\Delta x - 3\Delta t)^2]/16 = (16\Delta t^2 - 16\Delta x^2)/16$. Cross terms cancel, so the squared interval and positive proper duration are unchanged. Future timelike legs remain future because $\Delta t' = (5\Delta t - 3\Delta x)/4 > (5-3)\Delta t/4 > 0$. A has constant primed velocity $(-15/2)/(25/2) = -3/5$. B has velocity 0 on its outward leg and $(-15/2)/(17/2) = -15/17$ on return, so B changes velocity at T in both frames. One inertial clock versus a clock changing inertial segments is not a symmetric exchange of entire histories. The ideal turn is the acceleration limit; its duration is omitted explicitly, not mistaken for another rest frame valid for the whole trip.

6 / hints to offer only after exploration

1. Expand both squares before simplifying.
2. A frame moving with the outward traveler does not move with its return leg.

Worked spacetime comparison

Coordinates are events; drawn Euclidean lengths are not clock readings



Original (t, x) : $O = (0, 0)$, $T = (5, 3)$, $R = (10, 0)$. Primed (t', x') : $O' = (0, 0)$, $T' = (4, 0)$, $R' = (25/2, -15/2)$. Both diagrams use equal units on their own two axes.

Primed A: $(25/2)^2 - (15/2)^2 = 100$, so proper time 10. Primed B: first leg 4; return has differences $(17/2, -15/2)$, interval square 16, and proper time 4. Total 8.

Worked spacetime comparison

Optional facilitator extension 1

Gate: Calculus and strict concavity; square-root endpoint limits.

Keep the excursion distance $d=3$ but choose turnaround coordinate time t . Find the legal interval and maximize the traveler's total proper time. Is its least value attained?

Solution: Legality is $3 < t < 7$. The duration $F(t) = \sqrt{t^2 - 9} + \sqrt{(10 - t)^2 - 9}$ is symmetric about 5. Its derivative is $t/\sqrt{t^2 - 9} - (10 - t)/\sqrt{(10 - t)^2 - 9}$, zero at 5. Its second derivative is $-9/(t^2 - 9)^{3/2} - 9/((10 - t)^2 - 9)^{3/2} < 0$, so the unique maximum is $F(5) = 8$. The function rises then falls, and its endpoint limits are $\sqrt{40}$; the infimum $\sqrt{40}$ is not attained because an endpoint would make one leg lightlike. This compares two straight-leg excursions at fixed distance, not all possible accelerations.

Mathematics and prior use

Lorentzian interval, a global proper-time bound for finite polygonal worldlines and exact invariance under a rational Lorentz boost.

Prior use: Week 9 paths and AP-12 time optimization concern ordinary spatial distances. Here the quadratic interval has a minus sign, time is an axis, and the invariant is the clock reading along a worldline.

Source: David Tong, Dynamics and Relativity

§7.1, equations (7.2), (7.4), (7.5), printed pp. 109-110 / PDF pp. 115-116; §7.2.3 twin comparison, pp. 120-121; §7.4.1, pp. 133-134 / PDF pp. 139-140.. Use the Lorentz change of coordinates and proper-time sum. Chosen event coordinates, inverse route design and exact boosted-frame audit are new worked instances. Evidence: Reopened 2026-09-25: read both coordinate transformations, the turnaround asymmetry, invariant interval and additive proper-time integral. Each printed numerical interval is independently recomputed.

<https://davidtong.org/pdfs/teaching/dynamics-and-relativity/dynrel.pdf>

What can an orbit reveal about an unseen center?

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Promising with algebra. The observation assumptions do substantial work and must stay visible. The exact/noisy/projected cases are separate pages, not silently combined into one misleading astronomy story.

A parameter is inferred with an exact consistency certificate; an explicit ambiguity is exhibited and a new measurement is proved sufficient to remove it.

Prerequisites and preparation

Core: Algebra, squares, positive square roots and units. Circular centripetal acceleration and inverse-square gravity are supplied physical assumptions for the main inverse problem.

Extensions: Trigonometric differentiation derives centripetal acceleration. A common unknown projection factor is a separately stated simplified observation model, not an inference from a picture.

Reading and representation: r is true orbital radius from the center, not diameter. v is true speed on the first two pages; w is the projected measured quantity only on the last page. Distinguish a bounded-error interval from a probability distribution.

Materials: Supplied concentric-circle diagram, exact radius/speed cards, number-line interval workspace and a later observation-model diagram. Pencil and optional calculator; all astronomical data are invented. Preparation: about 8 minutes.

0-10 interpret radii and model assumptions; 10-25 infer and falsify a common mass; 25-45 design a discriminating bounded-error measurement; later 25 minutes for projection ambiguity and an added period measurement.

Launch and facilitation

Three tiny orbiters may reveal the same unseen mass. Choose a first observation, make a prediction and try to catch a card that cannot fit the same exact model.

Student page 1: One center, several witnesses

Core: circular-orbit model and algebra

Assume tiny test particles in circular Newtonian orbits about one fixed dominant point mass M . Gravity is the only force. Radii r and speeds v are true three-dimensional values. Supplied circular balance: $v^2/r = GM/r^2$. Use consistent units with $G=1$. All data on this page are exact.

What can an orbit reveal about an unseen center?

1 / prompt and solution

Choose any two cards: $(r,v)=(1,6),(4,3),(9,2)$. Infer the central mass from each, then test the third. Explain why the orbiting particle's mass is absent and why a slower outer orbiter need not imply a different center.

Solution: Rearranging gives $M=v^2r$ in these units. The cards yield $6^2 \times 1 = 36, 3^2 \times 4 = 36, 2^2 \times 9 = 36$, so all three are compatible with $M=36$. In full force balance $GmM/r^2 = mv^2/r$, the positive test mass m cancels. At fixed M , $v=\sqrt{M/r}$, so outer circular orbiters move more slowly while indicating the same mass. Agreement supports consistency with this model; it does not establish that all its assumptions are true.

1 / hints to offer only after exploration

1. Solve the balance equation for the common unknown.
2. Keep radius and speed paired on each card.

2 / prompt and solution

Predict the speed at radius 16. Invent an exact radius/speed card that cannot fit this same M , and give a numerical certificate. Also audit the card $(r,v)=(4,4)$. What if a labeled diameter were mistaken for a radius?

Solution: At $r=16$, $v=\sqrt{(36/16)}=3/2$. Any positive r,v with $v^2r \neq 36$ is incompatible with the same exact circular point-mass model; for example $(1,5)$ gives 25. The supplied $(4,4)$ gives 64, so it fails. Treating a diameter $2r$ as radius would double the inferred mass v^2r/G . An inconsistent card identifies a failure of the joint data/model claims, not specifically a faulty gravity law, a bad instrument or a noncircular orbit.

2 / hints to offer only after exploration

1. A certificate should exhibit two unequal estimates of the same parameter.
2. The center-to-orbiter segment is a radius, not a full crossing of the circle.

Student page 2: How precise must the next observation be?

Core continuation: guaranteed error bounds, not probabilities

For this page radii remain exact, but speeds may lie in stated closed intervals. No distribution or independence is assumed. A common mass is compatible if some speed in each interval satisfies the same circular model.

What can an orbit reveal about an unseen center?

3 / prompt and solution

At radius 4 the speed lies in $[2.9, 3.1]$; at radius 9 it lies in $[1.9, 2.1]$. Find EVERY compatible mass. Prove both that masses outside your answer fail and that each mass inside it can fit both observations.

Solution: Since $v > 0$ and $M = rv^2$ increases with v , the first interval gives $M \in [4(2.9)^2, 4(3.1)^2] = [33.64, 38.44] = [841/25, 961/25]$. The second gives $[9(1.9)^2, 9(2.1)^2] = [32.49, 39.69] = [3249/100, 3969/100]$. Their intersection is $[33.64, 38.44]$. Necessity follows from the two bounds. For any M in the intersection, choose $v_1 = \sqrt{M/4}$, $v_2 = \sqrt{M/9}$; monotonicity puts each in its allowed interval, proving sufficiency. This is a compatibility set, not a confidence interval or posterior probability.

3 / hints to offer only after exploration

1. Map both interval endpoints through the increasing function rv^2 .
2. Intersection proves necessity; construct the two speeds for sufficiency.

4 / prompt and solution

A new experiment must distinguish two exact hypotheses, $M=36$ and $M=64$. You can measure speed at radius 16 with guaranteed absolute error at most ϵ . Find the largest error threshold below which the possible readings are disjoint. What happens at the threshold itself?

Solution: Predicted true speeds are $3/2$ and 2 . Possible measured readings are $[3/2 - \epsilon, 3/2 + \epsilon]$ and $[2 - \epsilon, 2 + \epsilon]$. They are disjoint exactly when $3/2 + \epsilon < 2 - \epsilon$, or $\epsilon < 1/4$, for $\epsilon \geq 0$. There is no largest permissible ϵ because the inequality is strict; the supremum threshold is $1/4$. At $\epsilon = 1/4$ the intervals meet at $7/4$, so that reading cannot distinguish the hypotheses with certainty. Greater precision means a smaller guaranteed bound, not a favorable random draw.

4 / hints to offer only after exploration

1. Compare the upper end of the lower-speed interval with the lower end of the other.
2. The intervals are closed, so a touching endpoint matters.

Student page 3: A hidden projection can imitate a different mass

Continuation: identifiability and experiment design

Change the observation model explicitly. Radii are still true and exact, but the instrument reports $w = qv$, with the SAME unknown factor $0 < q \leq 1$ for all orbiters. The exact observed (r, w) cards are $(1, 6), (4, 3), (9, 2)$. This common-factor model is stipulated; it is not implied by a drawing.

What can an orbit reveal about an unseen center?

5 / prompt and solution

Can these three observations determine M ? Construct two different pairs (M, q) that fit all three. Describe EVERY allowed pair and give a lower bound on M .

Solution: The measured products are $w^2r = q^2v^2r = q^2M = 36$. Thus every pair with $M = 36/q^2$ and $0 < q \leq 1$ fits all three cards; equivalently $M \geq 36$ and $q = 6/\sqrt{M}$. For example $(M, q) = (36, 1)$ and $(144, 1/2)$, or $(64, 3/4)$, are distinct compatible models. More cards obeying the same product cannot separate M from q . The lower bound is 36; no finite upper bound follows because q can approach zero.

5 / hints to offer only after exploration

1. Identify the parameter combination that actually enters the observations.
2. Scale true speeds and compensate with the same projection factor.

6 / prompt and solution

You can also measure the TRUE orbital period T of the radius-4 orbiter, independently of the speed projection. Its circumference is $2\pi r$. Show that one exact period determines M and q . Decode $T = 4\pi/3$, using the existing observed $w = 3$.

Solution: Circular true speed is $v = 2\pi r/T$, so $M = v^2r = 4\pi^2r^3/T^2$ and $q = w/v = wT/(2\pi r)$. At $r = 4, T = 4\pi/3$, $v = 8\pi/(4\pi/3) = 6$, $M = 6^2 \times 4 = 144$ and $q = 3/6 = 1/2$. The positive period gives a unique positive v and M , then unique q ; checking $0 < q \leq 1$ audits the stipulated observation model. For these observed cards, a period producing $q > 1$ would reject the combined assumptions. The period must be independently known in the same time units, not reconstructed from the projected speed.

6 / hints to offer only after exploration

1. A period measures the time for one complete circumference.
2. Use it first to find true speed, then compare with the reported speed.

What can an orbit reveal about an unseen center?

Optional facilitator extension 1

Gate: Trigonometric differentiation and dimensional reasoning.

Derive the supplied circular acceleration from $x(t)=r \cos(\omega t)$, $y(t)=r \sin(\omega t)$. Restore G in both mass formulas and check dimensions. Explain which parameter is identifiable if G were also unknown.

Solution: Differentiating twice gives acceleration $-\omega^2(x,y)$, inward with magnitude $r\omega^2$. Speed is $r|\omega|$, so acceleration magnitude is v^2/r . Balancing with GM/r^2 yields $M=v^2r/G=4\pi^2r^3/(GT^2)$. Since $[G]=L^3/(\text{mass}\cdot\text{time}^2)$, both expressions have mass units. If G were unknown, circular kinematics determines only $\mu=GM$, even with true speed or period. Separating G and M would require independent information; this is a distinct ambiguity from the projection factor.

Mathematics and prior use

Inverse parameter inference, interval consistency, experimental separation and a one-parameter family of observationally equivalent models.

Prior use: AP-04 distinguishes a population from a restricted sampler. This is a dynamical inverse problem with projection nonidentifiability. AP-19 and AP-27 later revisit designing an extra observation under different evolution laws.

Source: OpenStax, College Physics for AP Courses 2e: Satellites and Kepler's Laws

§6.6, "Derivation of Kepler's Third Law for Circular Orbits," equations (6.60)-(6.67).. Use small test mass, dominant central mass, circular force balance and $v=2\pi r/T$. Numerical cards, error-bound design and the common-factor observation ambiguity are derived here. Evidence: Reopened 2026-09-25: inspected gravity=centripetal force, cancellation of the test mass, circular-speed relation and period-based parent-mass inference. No virial or elliptical-orbit approximation is imported.

<https://openstax.org/books/college-physics-ap-courses-2e/pages/6-6-satellites-and-keplers-laws-an-argument-for-simplicity>

Design a survey that can identify the depth

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Promising for algebra-ready learners. The receiver-design payoff is substantial; this is not a physical tsunami model or an elementary substitute for wave equations.

A complete ambiguity family, followed by a chosen extra measurement that uniquely identifies both positive depths; an optimal receiver placement under an explicit error model.

Prerequisites and preparation

Core: Distance $\div$ speed, positive square roots, fractions and two linear equations. The noise-design continuation uses exact interval bounds. The wave-speed law is supplied, not derived.

Extensions: Inequalities and weighted means for the effective-depth extension.

Reading and representation: Read a two-region channel and distinguish measured time, known length, unknown depth and supplied model.

Materials: Pencil, ruler and supplied channel/receiver diagram. Optional calculator checks only; no water experiment is needed. Preparation: about 8 minutes.

0-10 try chosen depths; 10-25 build indistinguishable models; 25-45 place a second receiver and invert its readings; later 25 minutes for placement under bounded timing error. Give one page at a time.

Launch and facilitation

The end receiver reports five time units. Choose a first-region depth and construct a second region that makes the same report. Can a receiver placed inside the channel settle the disagreement?

Student page 1: One reading, many hidden channels

Core: square roots and positive travel times

Use a supplied one-dimensional travel-time model for small-amplitude long waves, no current or rotation. Each region has constant positive finite depth H and speed $\sqrt{gH}$; segment times add. Ignore reflections. Choose units with $g=1$. The channel runs from $x=0$ to 12; its depth changes at $x=6$. The end travel time is $T=5$.

1 / prompt and solution

Choose H_1 from 4, 9 or 16. Find H_2 that gives the measured total time. Compare with a partner who chose differently. Can equal total times certify equal channels?

Solution: The first region takes $6/\sqrt{H_1}$, leaving $5-6/\sqrt{H_1}$ for region 2. Thus $H_2=[6/(5-6/\sqrt{H_1})]^2$. Choices 4, 9, 16 give $H_2=9, 4, 144/49$ respectively. The positive remaining times are 2, 3, $7/2$. Distinct pairs give the same T , so the end reading does not uniquely identify depths.

Design a survey that can identify the depth

1 / hints to offer only after exploration

1. Allocate the measured time between the two known lengths.

2 / prompt and solution

Describe EVERY pair of positive finite depths with $T=5$. Include an equal-depth channel and explain which proposed first depths are impossible.

Solution: Put $u=1/\sqrt{H_1}$ and $v=1/\sqrt{H_2}$. Exactly the positive pairs with $u+v=5/6$ work. In depth coordinates $H_1 > 36/25$ and $H_2 = [6/(5-6/\sqrt{H_1})]^2$; this is both necessary and sufficient. $H_1=36/25$ leaves zero time for a positive-length region and would require infinite H_2 ; smaller H_1 already takes too long. Equal depths have $u=v=5/12$, hence $H_1=H_2=144/25$. Neither endpoint of the open slowness segment is permitted.

2 / hints to offer only after exploration

1. Use reciprocal speeds; both must be strictly positive.

Student page 2: Where should the second receiver go?

Core continuation: design and invert a measurement

Place one receiver at a chosen x between 0 and 12. It measures S , the first arrival time from the same launch at 0. The end reading remains $T=5$. Let $u=1/\sqrt{H_1}$ and $v=1/\sqrt{H_2}$.

3 / prompt and solution

Choose $x=3, 6$ or 9 . A test channel gives $S=3/2, 3$ or 4 respectively. Recover both depths from your chosen reading and T . Explain why the new equation supplies information the end reading lacks.

Solution: At $x=3$, $S=3u$, so $u=1/2$ and $v=(5-6u)/6=1/3$. At $x=6$, $S=6u$ gives the same. At $x=9$, $S=6u+3v=4$ and $T=6u+6v=5$, so $v=1/3$ and $u=1/2$. Each gives $(H_1, H_2)=(4, 9)$. The new length row is not proportional to $(6, 6)$, so the independent equations determine both slownesses. A direct substitution proves sufficiency.

3 / hints to offer only after exploration

1. A receiver records only the lengths already crossed.

Design a survey that can identify the depth

4 / prompt and solution

For an arbitrary receiver position x , decide exactly which positions identify both depths. Give recovery formulas and the restrictions on S for positive finite depths. Include $x=0$ and $x=12$.

Solution: For $0 < x \leq 6$, $S = xu$: $u = S/x$ and $v = T/6 - S/x$. Positive depths require $0 < S < xT/6$. For $6 \leq x < 12$, $v = (T - S)/(12 - x)$ and $u = T/6 - v$; positivity requires $(x - 6)T/6 < S < T$. Both formulas agree at $x = 6$, where $0 < S < T$. Every interior position uniquely determines positive depths when these inequalities hold, with $H_1 = u^{-2}$ and $H_2 = v^{-2}$. At $x = 0$ the reading is always 0; at $x = 12$ it merely repeats T . Those endpoint positions cannot distinguish the prompt-2 family.

4 / hints to offer only after exploration

1. Split the calculation according to which region contains the receiver.

Student page 3: A unique answer can still be sensitive

Continuation: deterministic error bounds

The end time $T = 5$ is exact. The inside receiver reports $\hat{S}$ with unknown error $e = \hat{S} - S$ satisfying $|e| \leq 1/20$. The same absolute error bound applies at every position. Recover using $\hat{S}$ as though it were exact. Slowness means reciprocal speed, not depth.

5 / prompt and solution

Compare placing the receiver at $x = 3$ with report $\hat{S} = 3/2$, or at $x = 6$ with report $\hat{S} = 3$. Find every positive depth pair consistent with each report and error bound. Which placement gives the tighter slowness uncertainty?

Solution: At $x = 3$, $u \in [29/60, 31/60]$ and $v = 5/6 - u \in [19/60, 21/60]$. Thus $H_1 \in [3600/961, 3600/841]$ and $H_2 \in [400/49, 3600/361]$, with the two depths linked by $u + v = 5/6$, not independent choices in a rectangle. At $x = 6$, $u \in [59/120, 61/120]$, $v \in [39/120, 41/120]$, so $H_1 \in [14400/3721, 14400/3481]$ and $H_2 \in [14400/1681, 1600/169]$, again linked. All endpoint slownesses are positive. The slowness half-width is $1/60$ at $x = 3$ and $1/120$ at $x = 6$, so the interface receiver is twice as precise under this error model.

5 / hints to offer only after exploration

1. Recover slowness intervals first; reciprocal squares reverse endpoint order.

Design a survey that can identify the depth

6 / prompt and solution

Choose ANY interior x . Derive the largest absolute error in either recovered slowness caused by $|e| \leq \epsilon$, with T still exact. Prove which receiver position minimizes this error amplification. Does a good position remove the wave-model assumptions?

Solution: For $0 < x \leq 6$, the errors are $\Delta u = e/x$ and $\Delta v = -e/x$. For $6 \leq x < 12$, $\Delta v = -e/(12-x)$ and $\Delta u = e/(12-x)$. The sharp algebraic bound in either coordinate is $\epsilon/\min(x, 12-x)$, attained at $e = \pm\epsilon$. For physical states restrict to errors that keep recovered slownesses positive; the bound remains valid and is attained at interior states with enough margin. The denominator is at most 6, uniquely maximized at $x=6$, so that position uniquely minimizes amplification. Better conditioning does not validate the supplied shallow-water model, the depth segmentation, or the assumed error bound.

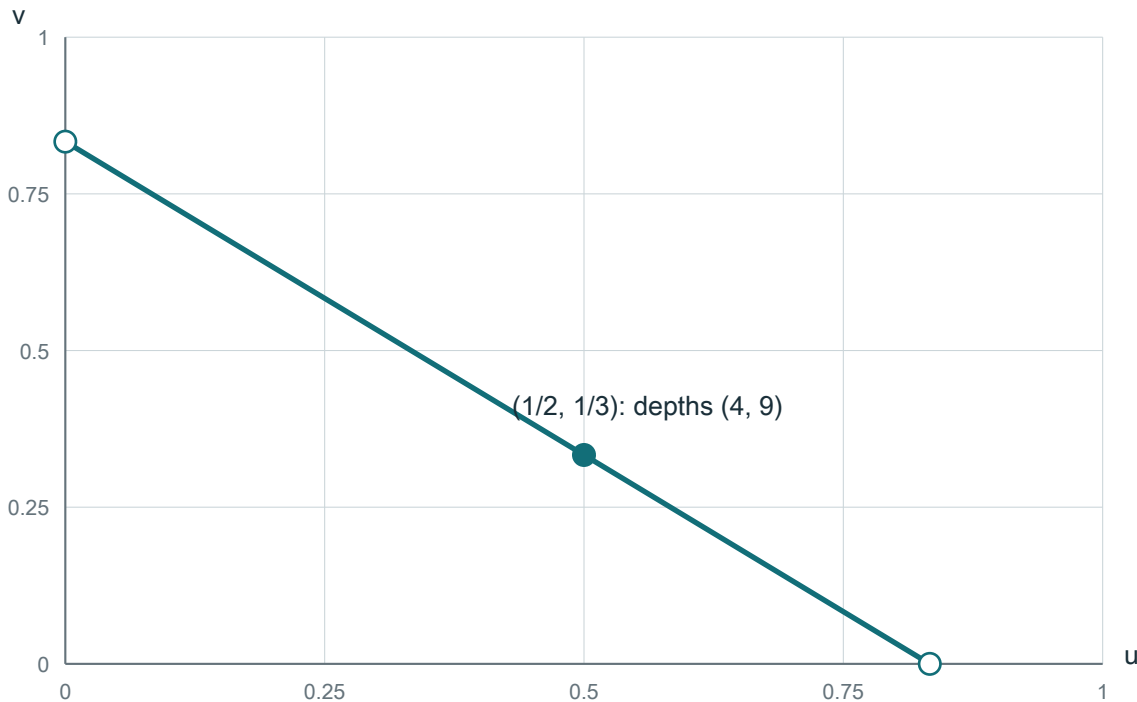
6 / hints to offer only after exploration

1. Track the error through the recovery formulas, not through a numerical experiment.

Design a survey that can identify the depth

Worked visual / facilitator only

A single end reading leaves a whole positive line of slowness pairs. The endpoints are excluded: they would require an infinite depth.



For $x=6$, the interior reading $S=3$ fixes $u=S/6=1/2$ and $v=(5-S)/6=1/3$. Under $|e|\leq 1/20$, the two recovered slowness errors have equal magnitude and opposite signs; they do not independently fill a rectangle.

Optional facilitator extension 1

Gate: Positive weighted means and reciprocal inequalities.

A uniform-depth interpretation assigns $H_{\text{eff}}=(12/T)^2$. Prove that it lies between the two actual depths, and test (4,9). Is it the arithmetic average of the depths?

Solution: The effective speed is $12/T=2/(u+v)$, the harmonic mean of the two speeds. Since $(u+v)/2$ lies between u and v , its reciprocal square lies between H_1 and H_2 , strictly if they differ. For (4,9), $T=5$ and $H_{\text{eff}}=144/25$, whereas the arithmetic depth average is $13/2$. The uniform interpretation fits the same single time but is a different channel. If g were unknown, only products gH_i could be recovered: replacing g by kg and H_i by H_i/k for $k>0$ preserves every travel time.

Design a survey that can identify the depth

Mathematics and prior use

A nonlinear inverse depth problem becomes linear in slowness. Identifiability, positivity and conditioning are separate questions.

Prior use: AP-18 separates orbital inference from model assumptions. AP-27 studies dynamic observability and close decay rates. Here the learner physically places a receiver, and the inverse parameters are wave slownesses and depths.

Source: Geoffrey Vallis, *Atmospheric and Oceanic Fluid Dynamics*, supplementary material for the second edition

§7.1.3, printed pp. 288-289 / PDF pp. 75-76, shallow-water limit following (7.19).. Retain $c=\sqrt{gH}$ as a supplied long-wave approximation. The two-region inverse data, receiver placement and error analysis are new. Evidence: Reopened 2026-09-26: inspected the limit $KH \ll 1$ giving equal phase/group speed $\sqrt{gH}$. Piecewise travel-time addition is an explicitly idealized model here; interface reflections and dispersion are omitted.

https://empslocal.ex.ac.uk/people/staff/gv219/aofd/aofd2_r2.pdf

Sell products, then price a proof

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

First-pilot candidate for fraction-ready learners. The counter launch works earlier, but the complete universal certificate needs weighted inequalities.

A feasible plan and matching universal resource-price upper bound, followed by an integer gap and a decision about which extra resource to buy.

Prerequisites and preparation

Core: Counting resources and revenue; fractions and linear inequalities for a price certificate. Whole products and divisible production are explicitly separate models.

Extensions: Piecewise affine formulas for arbitrary real resource capacities.

Reading and representation: Read two recipe cards. Adult may read the challenge aloud; algebra is needed for the universal price proof.

Materials: At least 6 red and 6 blue counters, recipe cards, pencil and open planning grid. Recipes are supplied; no price numbers are preprinted. Preparation: about 6 minutes.

0-10 build products; 10-25 challenge a best plan; 25-45 invent resource prices and prove optimality; 45-60 change the stock and distinguish fractional from whole production. General capacity extension can be a later meeting.

Launch and facilitation

Make the highest-value bundle you can from this stock. A search record is useful; can you also invent a price for each resource that proves nobody can earn more?

Student page 1: Build a plan worth challenging

Core: whole products and resource accounting

Product A uses 2 red and 1 blue counter and sells for 3 points. Product B uses 1 red and 2 blue and sells for 4 points. You have 5 red and 4 blue. Counters cannot be reused. On this page make whole products only.

Sell products, then price a proof

1 / prompt and solution

Choose a production plan, trade recipes with a partner, and try to improve the revenue. Record the counters used and left over. Give your best plan and a challenge another plan must beat.

Solution: For nonnegative integers a, b , feasibility is $2a + b \leq 5$ and $a + 2b \leq 4$, with revenue $3a + 4b$. The best plan is $a=2, b=1$, using all stock and earning 10. The complete feasible list is $(0,0), (1,0), (2,0), (0,1), (1,1), (2,1), (0,2)$; respective revenues are 0,3,6,4,7,10,8. Any learner proposal is audited by the two resource counts. Page 2 supplies a universal proof without depending on this enumeration.

1 / hints to offer only after exploration

1. One product can consume a resource needed by a more valuable combination.

2 / prompt and solution

Someone says, “B sells for more, so making as many B products as possible must be best.” Give a counterexample using this stock. What would a proof against ALL competing plans need to control?

Solution: Two B products use (2 red, 4 blue) and earn 8, leaving three red unusable alone. The plan $2A+B$ earns 10. Sale value per product ignores the scarce ingredients and how recipes fit together. A proof must bound every feasible combination, not merely compare pure-A and pure-B plans.

2 / hints to offer only after exploration

1. Track unused counters as well as completed products.

Student page 2: Invent prices that certify your plan

Core proof: fractions and inequalities

Assign nonnegative audit prices r to one red counter and b to one blue counter. These are proof tools, not extra income. Require the ingredients of each recipe to be worth at least its sale value. For this page fractional quantities of products are also allowed.

3 / prompt and solution

Design audit prices that make the total stock worth exactly your 10-point plan. Check both recipe inequalities. Prove these prices bound every feasible plan, including fractional ones.

Solution: Use $r=2/3$ and $b=5/3$. Then $2r+b=3$ and $r+2b=4$, and stock value $5r+4b=10$. Let x, y be the nonnegative real quantities of A, B. Every feasible plan satisfies $3x+4y \leq r(2x+y)+b(x+2y) \leq 5r+4b=10$. The first inequality follows from recipe prices and nonnegative quantities; the second from resource limits and nonnegative prices. The feasible plan $(x,y)=(2,1)$ attains this bound. This proves optimality even when production may be fractional.

Sell products, then price a proof

3 / hints to offer only after exploration

1. Try making each recipe inequality an equality, then check the stock value.

4 / prompt and solution

Prove whether another fractional plan can also earn 10. Explain which equality conditions in your price proof matter.

Solution: Both prices are strictly positive, so equality in the stock inequality requires using all 5 red and 4 blue: $2x+y=5$ and $x+2y=4$. These equations uniquely give $x=2, y=1$. Both recipe inequalities are already equalities. Therefore this is the unique fractional optimum and also the unique whole-product optimum. Merely finding one plan that reaches the bound would prove optimality but not uniqueness.

4 / hints to offer only after exploration

1. A leftover counter with a positive audit price makes the upper bound strict.

Student page 3: When a fractional certificate leaves a gap

Continuation: whole-number restrictions

Change the stock to 4 red and 4 blue. Keep the same recipes and sale values. First allow fractional production; then restore the whole-product rule.

5 / prompt and solution

Find the best fractional plan and certify it. Then find the best whole-product plan and prove that no whole plan does better. Explain why the fractional certificate alone does not finish the second task.

Solution: The same prices give upper bound $4(2/3)+4(5/3)=28/3$, attained uniquely by $x=y=4/3$. Whole products each use three counters, so at most $\text{floor}(8/3)=2$ can be made; each earns at most 4, giving $\text{revenue} \leq 8$. Two B products are feasible and attain 8. The fractional bound $28/3$ exceeds 8 and even its floor 9 does not exclude a hypothetical 9-point whole plan. The product-count argument uses the extra integer rule.

5 / hints to offer only after exploration

1. How many total counters does every whole product consume?

Sell products, then price a proof

6 / prompt and solution

You may add ONE red or ONE blue counter to the 4-and-4 stock. Choose which addition gives more whole-product revenue. Certify both resulting optima. Compare the gain with what the audit price of that counter predicts for fractional production.

Solution: Adding red gives stock (5,4), optimum 10 at (2,1). Adding blue gives (4,5), optimum 11 at (1,2), using all resources. The same price pair certifies bounds 10 and 11 respectively, so blue is better for whole products: gains 2 and 3 from the original whole optimum 8. Fractional revenue starts at $28/3$, so its gains are exactly $2/3$ and $5/3$, the audit prices. Whole-product gains differ because the original stock had an unattained fractional bound. No universal interpretation of prices as integer marginal gains follows.

6 / hints to offer only after exploration

1. Keep the fractional baseline and the whole-product baseline separate.

Optional facilitator extension 1

Gate: Linear inequalities and a piecewise formula with real quantities.

Keep blue stock 4 and let red stock R be any nonnegative real number. Allow fractional production. Find the exact maximum revenue for every R and supply a price certificate in every range.

Solution: For $0 \leq R \leq 2$, make $B=R$ and $A=0$, earning $4R$; audit prices (red,blue)=(4,0) bound revenue by $4R$. For $2 \leq R \leq 8$, solve both resource equalities: $A=(2R-4)/3$, $B=(8-R)/3$, giving $(2R+20)/3$ and prices $(2/3, 5/3)$. For $R \geq 8$, make $A=4, B=0$, earning 12; prices (0,3) certify 12. The formulas agree at $R=2$ and 8. All constructions are nonnegative in their stated ranges, and every stated price pair satisfies both recipe inequalities. This proves the complete piecewise optimum directly.

Mathematics and prior use

Weak duality proved directly by pricing resources, with equality conditions and a distinction between linear and integer optimization.

Prior use: Week 1 uses packing bounds and AP-07 diagnoses numerical optimization. This is an exact primal/dual resource certificate; the integer gap is not settled by a fractional graph alone.

Source: Stephen Boyd and Lieven Vandenbergh, Convex Optimization

§5.1.3, printed p. 216 / PDF p. 230, lower bounds from dual feasibility.. Use the weak-duality idea in a maximization problem with nonnegative resource prices. All recipes, capacity changes and equality conditions are proved directly. Evidence: Reopened 2026-09-26: inspected the feasible-point inequality underlying (5.2). No strong-duality theorem is needed or inferred.

https://www.seas.ucla.edu/~vandenbe/cvxbook/bv_cvxbook.pdf

Build a randomizer your opponent cannot exploit

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Promising with fractions and expected value. Playing is a launch, but short-run score averages do not verify the guarantee. The lower/upper certificate is the satisfying core.

Two independently sampled three-ticket randomizers with matching universal payoff bounds; an exact account of how deviations or revealed actions change the guarantee.

Prerequisites and preparation

Core: Fractions, independent probabilities and expected payoff; linear inequalities for worst-case guarantees. No general minimax theorem is required.

Extensions: Positive real parameters and solving two linear equalities.

Reading and representation: Read a two-by-two payment table; distinguish knowing a distribution from seeing its actual sampled action.

Materials: Two opaque cups or envelopes, up to six equal-sized slips per player, row U/D cards and column L/R cards, counters only for scores. Replace and mix slips independently before each round. Preparation: about 8 minutes.

0-10 try private randomizers; 10-25 challenge against pure opponents; 25-45 construct matching lower/upper certificates; later 20 minutes for near-optimal ranges and changed information rules.

Launch and facilitation

Choose a secret action using your own bag. Your opponent knows the bag recipe, but cannot see the draw. Can a deliberately unequal bag protect you better than a fair coin?

Student page 1: Choose a bag; let a rival challenge it

Core: independent hidden actions and averages

Row chooses U or D; Column chooses L or R. Choices are simultaneous and privately sampled, using independent randomizers. Column pays Row the table entry: U/L pays 2, D/R pays 1, the other entries pay 0. Row wants more; Column wants less. Both know each bag recipe, never the other draw.

Build a randomizer your opponent cannot exploit

1 / prompt and solution

Build a Row bag with at most six equally likely slips, each marked U or D. Test its EXPECTED payment against a Column player who always chooses L, then one who always chooses R. A rival chooses the worse opponent for your bag. Revise your bag if you can.

Solution: If a proportion p of slips are U, expected payment against L is $2p$ and against R is $1-p$. The worse guarantee is $\min(2p, 1-p)$. Any chosen bag is evaluated by these two exact numbers; trials need not equal their expectations. One U and two D slips gives $p=1/3$ and both payments $2/3$; two U and four D is equivalent. A fair bag only guarantees $1/2$. Do not announce the best recipe before the learner has explored.

1 / hints to offer only after exploration

1. An opponent can exploit the less profitable column, even while unable to see your draw.

2 / prompt and solution

Now allow any real p between 0 and 1 as the probability of U. Find the largest payment Row can guarantee against EVERY independent Column mixture. Prove which p attain it.

Solution: For Column probability q of L, payoff is $q(2p) + (1-q)(1-p)$, a weighted average of the two pure-column payoffs. Its minimum over q is $\min(2p, 1-p)$. If $p \leq 1/3$ this is $2p \leq 2/3$; if $p \geq 1/3$ it is $1-p \leq 2/3$. Equality occurs only at $p=1/3$. Thus Row uniquely maximizes its worst-case expected payoff at $2/3$. This is an expected guarantee, not a positive payment on every round.

2 / hints to offer only after exploration

1. The opponent may use either pure column; mixtures cannot go below both endpoint values.

Student page 2: The opponent needs a certificate too

Core proof: matching lower and upper bounds

Let q be Column's probability of L, independently of Row's action. A Column guarantee is an upper bound on what Row can earn against that fixed distribution.

Build a randomizer your opponent cannot exploit

3 / prompt and solution

Design a Column randomizer that keeps EVERY Row mixture at or below the guarantee you found. Prove it works and decide whether its probability q is unique. What do the two certificates establish together?

Solution: Against pure U the payoff is $2q$, and against pure D it is $1-q$. Every Row mixture is a weighted average. The smallest possible upper bound $\max(2q, 1-q)$ is $2/3$, uniquely at $q=1/3$: for $q < 1/3$, $1-q > 2/3$, and for $q > 1/3$, $2q > 2/3$. One L and two R slips works. Matching Row lower and Column upper bounds prove the value exactly $2/3$, and the product of these independent distributions is an equilibrium. No general minimax theorem is invoked.

3 / hints to offer only after exploration

1. Find an upper bound by testing both pure rows.

4 / prompt and solution

Suppose Column advertises $q=3/4$ and really samples independently from it. Choose Row's best response and compute its payoff. Why can a strategy with a larger payoff here have a worse worst-case guarantee?

Solution: Pure U earns $2q=3/2$, while pure D earns $1-q=1/4$. Since mixtures average these, U is the unique best response and earns $3/2$. But if Column changes to pure R, U earns 0. The minimax mixture protects against every Column distribution and earns exactly $2/3$ against any fixed q ; exploiting one known opponent is a different objective.

4 / hints to offer only after exploration

1. A best response compares against this opponent; a guarantee compares against all opponents.

Student page 3: How much imperfection can the bag tolerate?

Continuation: inequalities and information rules

Keep independent private sampling for the first task. Then change what the opponent can see, and state that change explicitly.

5 / prompt and solution

Choose $0 \leq \epsilon \leq 1/3$. Find EVERY p that guarantees Row at least $2/3 - \epsilon$, and EVERY q that keeps Row at most $2/3 + \epsilon$. Test $\epsilon = 1/15$. Prove both necessity and sufficiency.

Solution: Row needs both $2p \geq 2/3 - \epsilon$ and $1-p \geq 2/3 - \epsilon$, giving $1/3 - \epsilon/2 \leq p \leq 1/3 + \epsilon$. Column needs $2q \leq 2/3 + \epsilon$ and $1-q \leq 2/3 + \epsilon$, giving $1/3 - \epsilon \leq q \leq 1/3 + \epsilon/2$. These ranges stay in $[0, 1]$ for the stated ϵ . At $\epsilon = 1/15$, Row has $3/10 \leq p \leq 2/5$ and Column has $4/15 \leq q \leq 11/30$. Pure opponents prove necessity; weighted-average payoffs prove sufficiency.

Build a randomizer your opponent cannot exploit

5 / hints to offer only after exploration

1. Demand the promised bound against each pure action separately.

6 / prompt and solution

Change the rules: Column sees Row's actual action before choosing. Design Column's response and determine Row's best guarantee. Then suppose both instead use the SAME uniform ticket from $\{1,2,3\}$: Row uses U only on 1 and Column uses L only on 1. Compute the payoff and explain why the independent-mixture formula cannot be used.

Solution: With the revealed action, Column chooses R after U and L after D, giving zero on every round. All entries are nonnegative, so the value under this information rule is exactly 0. With the shared ticket, (U,L) occurs with probability $1/3$ and (D,R) with probability $2/3$, giving expected payoff $2/3 + 2/3 = 4/3$. Both marginal probabilities are $1/3$ but the choices are correlated; multiplying their marginals would incorrectly give $2/3$. The earlier certificates apply to independent private sampling, not these changed rules.

6 / hints to offer only after exploration

1. Write the actual joint outcomes before calculating an expectation.

Optional facilitator extension 1

Gate: Algebra with positive real parameters.

Replace the two positive diagonal payments by $A > 0$ and $B > 0$, keeping the off-diagonal payments zero. Find both optimal mixtures and the value; prove the full lower/upper certificate.

Solution: The pure-column payoffs for Row are Ap and $B(1-p)$, so the unique maximizer of their minimum is $p = B/(A+B)$, with value $AB/(A+B)$. The pure-row payoffs against Column are Aq and $B(1-q)$, so $q = B/(A+B)$ uniquely minimizes their maximum at the same value. Independent mixed payoffs are weighted averages, proving both bounds. Positivity ensures the intersection is interior and uniqueness; zero payments would need separate boundary treatment.

Mathematics and prior use

Finite zero-sum minimax, best responses, and exploitability proved by affine payoff inequalities.

Prior use: Weeks 7-8 use perfect-information games. AP-21 studies backward induction. Here a strategy is a private distribution and the certificate is a worst-case expected payoff, not a sure win in each round.

Build a randomizer your opponent cannot exploit

Source: Giacomo Bonanno, Game Theory

§§5.2-5.3, printed pp. 190-198 / PDF pp. 197-205; mixed strategies and best replies.. Use private independent mixtures and indifference as a candidate-finding tool. Prove both minimax bounds directly for the chosen matrix. Evidence: Reopened 2026-09-26: inspected the mixed-strategy definition and support-indifference discussion, including the warning that indifference alone is not a sufficient equilibrium test.

<https://arxiv.org/pdf/1512.06808>

A fair copying rule can erase a color

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Promising with probability-ready learners. Copying has an accessible physical launch; expectation versus realized history is the essential conceptual distinction. The limiting proof is a genuine prerequisite gate.

Exact finite-time probabilities and a proof that eventual absorption has probability one, even while the expected count stays unchanged; an asymmetric three-parent fixation result.

Prerequisites and preparation

Core: Independent sampling with replacement, fractions and finite outcome trees. Geometric tails and conditional averages for eventual-absorption proofs.

Extensions: Finite conditional expectations and a uniform absorption bound for arbitrary population size.

Reading and representation: Differentiate the current parent population from the new offspring population. Colors represent neutral inherited types, not human groups.

Materials: Three numbered parent cards, two colors of counters, opaque cup, and separate parent/offspring mats. Draw labels uniformly with replacement; make every child from the unchanged old generation before replacing all parents. Preparation: about 8 minutes.

0-10 practice copying without replacing parents early; 10-25 contrast chosen legal histories; 25-40 compute two-parent transitions and tails; later 25 minutes for a three-parent population and an asymmetric fixation probability.

Launch and facilitation

Each child copies a randomly chosen parent. No color has an advantage. Invent a legal family history that loses one color, and another that stays mixed for several generations.

Student page 1: Build two possible family histories

Core: copy from the OLD generation

Start with two parents: one red and one blue. To make each of two children, independently choose one parent uniformly, with replacement, and copy its color. Both children choose from the unchanged old generation. Only then replace the parents by the two children. Use fresh independent draws each generation. No mutation or color advantage occurs.

A fair copying rule can erase a color

1 / prompt and solution

Choose and draw a legal three-generation history that loses a color, and another that stays mixed throughout. Name the selected parent for each child. Does either history contradict the fair copying rule?

Solution: For early loss, choose the red parent twice in generation 1, giving RR; all later offspring remain RR. For persistent mixing, choose each color once in each generation, giving one red and one blue each time. Parent labels can be chosen in either child order. Both histories have positive probability under independent with-replacement draws. Fairness means the sampling law treats parent labels equally, not that every realized generation preserves both colors.

1 / hints to offer only after exploration

1. One parent can be chosen by both children.

2 / prompt and solution

From one red and one blue, list the ordered parent choices for one generation and their probabilities. Group them by the new red count. Find the expected red count. Then give the transition rule from an all-red or all-blue population.

Solution: The ordered choices RR, RB, BR, BB each have probability $1/4$. Thus next red counts 2, 1, 0 have probabilities $1/4, 1/2, 1/4$, and expectation $2(1/4) + 1(1/2) = 1$. From all-red the next count is certainly 2; from all-blue it is certainly 0. These are absorbing states. The letters here denote copied colors; numbered-parent drawings on the launch page implement the same uniform selection.

2 / hints to offer only after exploration

1. RB and BR are different ordered draws even though their red counts agree.

Student page 2: A mean is not a typical history

Core proof: finite probabilities and a limit

Let X_k be the red count after k generations from the original mixed pair, with $X_0=1$. Once a color is lost, it cannot return under this rule.

A fair copying rule can erase a color

3 / prompt and solution

Find exact probabilities for $X_k=0, 1$ and 2 for EVERY integer $k \geq 0$. Prove that the probability of remaining mixed forever is zero. Is there a finite generation by which loss is absolutely certain?

Solution: To stay mixed through k generations, the process must take the mixed-to-mixed transition each time, so $P(X_k=1)=2^{-(k)}$. Symmetry or summing the identical per-step absorption hazards gives $P(X_k=0)=P(X_k=2)=(1-2^{-(k)})/2$. The event of staying mixed forever is contained in staying mixed for each fixed k , hence its probability is at most $2^{-(k)}$ for every k and therefore zero. For every finite k , $2^{-(k)} > 0$, so there is no finite absolute deadline. The $k=0$ formulas give $(0, 1, 0)$.

3 / hints to offer only after exploration

1. Each additional mixed generation multiplies the survival probability by $1/2$.

4 / prompt and solution

Choose k so that the chance of still being mixed is below $1/100$. Compute the expected red count at that k and explain how it can coexist with almost every population having just one color. What changes if the two parent draws are WITHOUT replacement?

Solution: The least such k is 7 since $2^{-(6)}=1/64 > 1/100$ and $2^{-(7)}=1/128 < 1/100$; any $k \geq 7$ works. $E[X_k]=0 \cdot P_0 + 1 \cdot 2^{-(k)} + 2 \cdot (1-2^{-(k)})/2 = 1$ for every k . The average across histories remains one because red fixation and blue fixation balance; it need not describe any common individual population. Without replacement each parent is copied once, so the mixed pair persists forever. The unchanged expectation alone cannot distinguish these two processes.

4 / hints to offer only after exploration

1. Average across possible histories, not across the two children of one realized history.

Student page 3: What if the colors start unequally?

Continuation: three independent offspring

Now use THREE parents and THREE children, with the same independent with-replacement rule. Start with one red and two blue. Write X_k for the red count, so $X_0=1$. Counts 0 and 3 are absorbing.

A fair copying rule can erase a color

5 / prompt and solution

Find the transition probabilities from red count 1 and from red count 2. Prove that the next expected red count equals the current count. Use your rows to find the chance of remaining mixed for k generations.

Solution: From count 1, each child is red with probability $1/3$, independently. The probabilities of counts 0,1,2,3 are $(8,12,6,1)/27$. From count 2 they are $(1,6,12,8)/27$. The expectations are 1 and 2, either by the rows or by summing three child indicators with means $1/3$ or $2/3$. From either mixed count, the total mixed probability is $18/27=2/3$. Therefore mixed survival through k generations is $(2/3)^k$, despite possible movement between counts 1 and 2.

5 / hints to offer only after exploration

1. Count ordered child-color strings, then combine strings with equal red counts.

6 / prompt and solution

Prove the eventual all-red probability from the one-red start. Do not argue by red/blue symmetry: the starting population is unequal. Explain why this is a probability across histories, not a fractional final population.

Solution: Conditional expectation preserves $E[X_k]=1$ for every k . Write $r_k=P(X_k=3)$. Then $1=3r_k+E[X_k \cdot 1_{\{X_k \in \{1,2\}\}}]$, and the remaining term is between 0 and $2(2/3)^k$. Thus $r_k \rightarrow 1/3$. Absorbed-red events increase with k ; their limit is eventual all-red, so its probability is $1/3$. Mixed survival tends to zero, so eventual all-blue has probability $2/3$. Every absorbed population has three of one color; the fraction $1/3$ refers to the collection of possible histories.

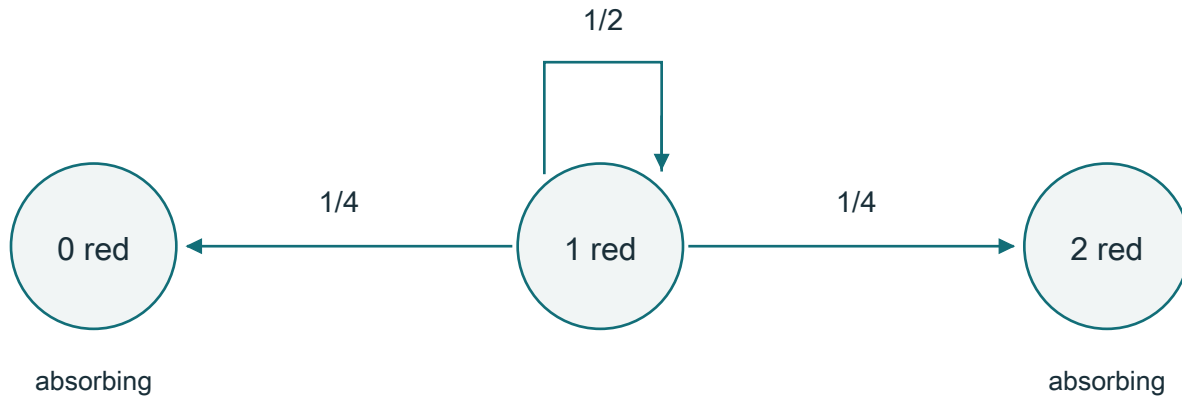
6 / hints to offer only after exploration

1. Separate the expected contribution of absorbed-red histories from the vanishing mixed histories.

A fair copying rule can erase a color

Worked visual / facilitator only

The two-parent count chain has only one transient state. Its outgoing probabilities explain both the unchanged mean and the vanishing chance of a mixed population.



Starting in the middle: $P(X_k=1)=2^{-(k)}$. The two absorbing probabilities are each $(1-2^{-(k)})/2$. Hence $E[X_k]=1$ for every finite k , even when almost all histories have lost a color.

Optional facilitator extension 1

Gate: Linearity of finite expectation and geometric probability bounds.

For a general fixed population size $N \geq 2$, with i red parents, prove eventual absorption with probability one and eventual all-red probability i/N under the same neutral sampling rule.

Solution: Given a generation with j red parents, each of N offspring has red probability j/N , hence the next expected red count is j . Thus $E[X_k]=i$ at every finite k . In any generation, the event that all N children choose one specified parent has probability $N^{-(N)}$, and makes the next generation monochromatic. Therefore the probability of still being mixed after k steps is at most $(1-N^{-(N)})^k$, which tends to zero. This is a uniform bound; independence of successive population states is not assumed. Writing $E[X_k]=N \cdot P(X_k=N) + E[X_{k-1} \cdot 1_{\text{mixed}}]$, the second term lies between 0 and N times the survival probability. Consequently eventual all-red has probability i/N ; all-blue has $1-i/N$. Endpoints $i=0, N$ are already absorbed.

Mathematics and prior use

Neutral Wright-Fisher sampling, conditional expectation, absorption and fixation probabilities without a general optional-stopping theorem.

Prior use: AP-02/03/05 examine sampling estimators and intervals. This is a generational Markov process with absorbing states. AP-26 has state recurrence, but here the transition rule is randomized reproduction and the proof uses tail probabilities.

A fair copying rule can erase a color

Source: Jeffrey Chasnov, Mathematical Biology

§5.5, printed p. 78 / PDF p. 84, opening comparison of Wright-Fisher and Moran sampling.. Use the discrete-generation Wright-Fisher rule, not the subsequent Moran/diffusion calculation. All $N=2$, $N=3$ and general finite- N absorption proofs are supplied independently. Evidence: Reopened 2026-09-26: inspected the description of N independently chosen gametes and the explicit distinction from one-birth/one-death Moran steps. No diffusion approximation is used.

<https://www.math.hkust.edu.hk/~machas/mathematical-biology.pdf>

Design labels that survive a reaction

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

First-pilot candidate for the exchange launch and invented labels; general weights and reachability formulas need algebra. This preserves actual stoichiometric conservation, not reaction kinetics.

All linear conserved labels, a complete reachable-state chain with a legal-path certificate, and a second network showing why a conserved total alone can be insufficient.

Prerequisites and preparation

Core: Counting and legal exchanges; signed linear expressions for all conserved labels. Integer intervals and a constructive path prove complete reachability.

Extensions: Nonnegative integer states, enabling conditions and a second reaction network.

Reading and representation: Read A, B and C species labels and a reversible recipe. Distinguish one C molecule from a count of its constituents.

Materials: At least 6 A, 10 B and 5 C counters in distinct shapes; reaction card; pencil. Label weights are invented, so do not preprint an atom decomposition on C. Preparation: about 7 minutes.

0-10 trade counters freely; 10-25 invent conserved labels; 25-40 use labels to exclude targets; 40-60 classify the full state chain. The changed reaction and catalyst extension can follow later.

Launch and facilitation

Trade one A and two B for one C, or reverse that trade. The number of counters changes. Can you invent a way of labeling counters whose total never changes?

Student page 1: The counter total changes. What survives?

Core: reversible legal trades

A forward move consumes 1 A and 2 B and creates 1 C. A backward move consumes 1 C and creates 1 A and 2 B. A move is legal only when all consumed counters are present. Start with $(A,B,C)=(4,7,1)$. Counts must stay nonnegative integers.

Design labels that survive a reaction

1 / prompt and solution

Explore both directions. Choose a target you can reach and record a legal sequence. Compare the total number of counters before and after a forward move. Could total counter count certify that a target is impossible?

Solution: A forward move changes (A,B,C) by $(-1,-2,+1)$, so total count falls by 2; a backward move raises it by 2. For example two forward moves reach $(2,3,3)$; one backward move reaches $(5,9,0)$. All legal proposals are checked move by move. The ordinary total is not conserved, so unequal totals alone cannot certify impossibility. Its parity is conserved, but that weaker fact is not the same as a fixed total.

1 / hints to offer only after exploration

1. Count what disappears and what is created separately.

2 / prompt and solution

Put a positive whole-number weight on each species. Design TWO different weight systems for which every legal move preserves total weight. Make the two systems more than just scaled copies. Test them on your sequence.

Solution: Weights (w_A, w_B, w_C) must obey $w_C = w_A + 2w_B$. Examples $(1,1,3)$ and $(2,1,4)$ are positive and not scaled copies. Their totals at the initial state are 14 and 19. For every move, change in total weight is $-w_A - 2w_B + w_C = 0$, and reversing changes the sign but remains zero. Arbitrary positive integer w_A, w_B determine a valid positive integer w_C . Testing examples motivates but does not replace this one-move proof.

2 / hints to offer only after exploration

1. Make the value of the ingredients equal the value of one product.

Student page 2: Turn a label into a complete certificate

Core proof: arbitrary weights and impossible targets

Allow weights to be any real numbers, including zero or negative values. A label total is $w_A A + w_B B + w_C C$. It must be preserved by the reaction wherever that reaction is legal, not just at one chosen inventory.

Design labels that survive a reaction

3 / prompt and solution

Find EVERY weight triple that is always conserved. Find two simple conserved totals from which every other linear conserved total can be built. Prove that your list is complete.

Solution: A forward move changes the label by $-w_A - 2w_B + w_C$. Thus preservation is equivalent to $w_C = w_A + 2w_B$, because the forward reaction is legal somewhere, for example at $(1, 2, 0)$. Every weight triple is $(r, s, r+2s)$, with arbitrary real r, s . The corresponding total is $r(A+C) + s(B+2C)$. Hence $A+C$ and $B+2C$ generate every linear conserved total and are independent. The one-step calculation proves necessity and sufficiency for both reaction directions.

3 / hints to offer only after exploration

1. Compute the change symbolically before choosing numerical weights.

4 / prompt and solution

Find EVERY reachable state from $(4, 7, 1)$, and draw legal arrows connecting your list. Prove completeness as well as reachability. Invent one nonnegative target that passes the $A+C$ test but fails $B+2C$, and another that does the opposite.

Solution: Initial conserved totals are $A+C=5$ and $B+2C=9$. If $C=z$, then $A=5-z$ and $B=9-2z$. Nonnegative integer states force $z=0, 1, 2, 3, 4$, giving $(5, 9, 0), (4, 7, 1), (3, 5, 2), (2, 3, 3), (1, 1, 4)$. Consecutive states are linked by a legal forward move to larger z and a legal backward move to smaller z . The initial $z=1$ therefore reaches every listed state. There are no others because the two totals and nonnegativity force the list. Target $(0, 1, 5)$ passes $A+C=5$ but has $B+2C=11$; target $(2, 1, 4)$ passes $B+2C=9$ but has $A+C=6$. These independent tests each exclude targets the other accepts.

4 / hints to offer only after exploration

1. Choose the number of C counters; the two totals determine everything else.

Student page 3: When do invariant tests finish the job?

Continuation: arbitrary inventories and a changed reaction

First keep $A+2B \leftrightarrow C$ but allow any initial nonnegative integer inventory (a, b, c) . Then change the reaction explicitly in the second task.

Design labels that survive a reaction

5 / prompt and solution

Classify EVERY reachable state from (a,b,c) . Give a legal route to each state you claim and the smallest possible number of reaction firings needed to reach it.

Solution: Write $M=a+c$ and $N=b+2c$. The complete list is $(M-z, N-2z, z)$ for every integer $0 \leq z \leq \min(M, \text{floor}(N/2))$. The initial $z=c$ lies in this interval. To reach a larger z , fire forward $z-c$ times; every intermediate step is legal because its next state has $A, B \geq 0$. To reach a smaller z , fire backward $c-z$ times, consuming an available C at each step. Each firing changes z by exactly one, so at least $|z-c|$ firings are required and these monotone paths attain that bound. The invariant calculation excludes every other state.

5 / hints to offer only after exploration

1. The permitted C counts form an unbroken integer interval containing the initial count.

6 / prompt and solution

Replace the reaction by $2A \leftrightarrow 2B$. Starting at $(A,B)=(5,0)$, find all reachable states. Why does the conserved total $A+B$ fail to decide reachability by itself? State an extra invariant and prove that your full list is reachable.

Solution: $A+B=5$ is conserved, but each move changes A by two, so A stays odd (and B even). The nonnegative possibilities are exactly $(5,0), (3,2), (1,4)$. Consecutive states are legal forward moves, and reversing is legal when there are two B counters. Target $(4,1)$ has the right total but wrong parity. In this second network, total plus parity plus nonnegativity do characterize the reachable states from this start; that conclusion follows from the explicit path, not from a general claim that invariants always suffice.

6 / hints to offer only after exploration

1. Look at the size of one step along the line of fixed total.

Optional facilitator extension 1

Gate: Reaction enabling conditions and a proof of all nonnegative integer states.

Change the reaction to $A+B \leftrightarrow 2B$. Its net change is $(-1, +1)$, and $A+B$ is conserved. Show why $(2,0)$ cannot reach $(1,1)$, even though the target has the same total and differs by an integer multiple of the change vector. Classify reachability for a general initial (a,b) .

Solution: From $(2,0)$, the forward reaction lacks B and the backward reaction lacks two B , so no move is enabled. More generally, if $b=0$ the initial state is trapped. If $b \geq 1$ and total $n=a+b$, exactly the states $(n-z, z)$ with $1 \leq z \leq n$ are reachable: forward moves increase z when $A > 0$ and $B \geq 1$; backward moves decrease z when $B \geq 2$. These unit steps traverse the whole interval but can never reach $z=0$. Thus even linear invariants and an allowed integer displacement can miss the enabling obstruction. This is a different network from the core, whose gap-free path proof established sufficiency.

Design labels that survive a reaction

Mathematics and prior use

Stoichiometric left-nullspace, nonnegative integer reachability and the difference between necessary invariants and constructive sufficiency.

Prior use: Weeks 2 and 4 use toggle/remainder invariants. This family deliberately proves sufficiency for one stoichiometric class, then exposes the limits of linear invariants in a changed reaction. AD-13 has a different monomial-move invariant.

Source: Jeffrey Chasnov, Mathematical Biology

§6.1, printed pp. 85-86 / PDF pp. 91-92, reaction signs and mass-action rates.. Use stoichiometric signed changes for $A+2B \rightleftharpoons C$. The counter network, all linear invariants and reachability classifications are new direct calculations; no rate law is needed for the discrete game. Evidence: Reopened 2026-09-26: inspected $A+B \rightarrow C$ and its signed concentration derivatives. Our different stoichiometry is explicitly stated; conserved weights follow from its change vector, not by copying source rate equations.

<https://www.math.hkust.edu.hk/~machas/mathematical-biology.pdf>

Choose when one sensor can separate two tanks

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Promising with linear equations. Equal-rate ambiguity is accessible; the final timing proof needs powers and inequalities. Exact uniqueness must not be advertised as noise robustness.

A proof of permanent ambiguity, a complete positive-state reconstruction test, and a mathematically optimal observation time under a fixed absolute-error model.

Prerequisites and preparation

Core: Fractions, powers and two linear equations; nonnegative real states. Exact bounded-error propagation and a geometric comparison for choosing the measurement time.

Extensions: A two-by-two matrix determinant and global observability versus exceptional states.

Reading and representation: Read initial amounts, per-step retained fractions, and one measured total. A tank icon is a state diagram, not a claim that ordinary gravity-driven tanks decay geometrically.

Materials: Pencil, two supplied hidden-state boxes and a sum-sensor icon; optional counters for initial split. No liquid experiment is required. Preparation: about 7 minutes.

0-10 choose hidden splits; 10-25 prove permanent ambiguity for equal rates; 25-40 reconstruct unequal rates; later 30 minutes for noisy readings and optimal observation timing.

Launch and facilitation

Two hidden amounts both shrink every step. You can see only their sum. Choose two different starting splits that the sensor cannot distinguish, then change a known decay factor and investigate what becomes visible.

Student page 1: Different tanks, the same entire record

Core: hidden states and a fixed output law

At time $n=0$ the nonnegative real amounts are a and b . Each step multiplies BOTH amounts by $1/2$. The sensor reports only their sum y_n . This geometric decay rule is supplied; it is not a law for every physical tank.

Choose when one sensor can separate two tanks

1 / prompt and solution

Choose two different nonnegative starting splits with total 8. Predict their entire sensor records and prove whether any later reading can distinguish them. Describe every initial state giving this record. Is the zero-total record exceptional?

Solution: Examples (2,6) and (4,4) both give $y_n = 8 \cdot 2^{(-n)}$ for every $n \geq 0$. In general the record depends only on $a+b$, so exactly $(a, 8-a)$, $0 \leq a \leq 8$, fit it. Waiting longer never distinguishes these states. If the total is zero, nonnegativity forces $a=b=0$, so that one record does identify its state. The sensor fails to identify every state globally, despite this exceptional unique record.

1 / hints to offer only after exploration

1. Factor the common decay term out of the total.

2 / prompt and solution

Keep the equal decay rates. You may take one extra reading at time 0: either a alone or $z=a+2b$. Choose one sensor and prove that it recovers every initial pair when combined with the total y_0 . Test your chosen split.

Solution: Measuring a gives $b=y_0-a$. Measuring $z=a+2b$ gives $b=z-y_0$ and $a=2y_0-z$. These formulas prove uniqueness and reconstruct the readings. In the weighted case nonnegative compatibility is $y_0 \leq z \leq 2y_0$ with $y_0 \geq 0$. The added sensor measures a different combination; it is not merely another copy of a later total from the old sensor.

2 / hints to offer only after exploration

1. Design a second equation that is not just a multiple of the first.

Student page 2: Different rates can reveal the hidden split

Core proof: reconstruction and feasible readings

Change the known rates: the first amount retains $1/2$ each step, the second retains $1/4$. The sensor still reports only $y_n = a(1/2)^n + b(1/4)^n$. Initial amounts remain nonnegative real numbers.

3 / prompt and solution

The readings are $y_0=8$ and $y_1=3$. Recover both starting amounts and predict y_2 . A rival proposes (6,2). Use an actual sensor reading to reject or accept that proposal.

Solution: Equations $a+b=8$ and $a/2+b/4=3$ give $2a+b=12$, hence $a=b=4$. Therefore $y_2=4/4+4/16=5/4$. The rival (6,2) gives $y_1=3+1/2=7/2$, so it fails the observed second reading despite matching y_0 . Substitution of (4,4) verifies both given readings.

Choose when one sensor can separate two tanks

3 / hints to offer only after exploration

1. Clear the fraction in the second equation before subtracting.

4 / prompt and solution

Derive a reconstruction formula for arbitrary readings y_0, y_1 . Decide exactly which pairs of readings come from nonnegative amounts. Include zero amounts and prove sufficiency.

Solution: The unique algebraic solution is $a=4y_1-y_0$ and $b=2y_0-4y_1$. It is nonnegative exactly when $y_0 \geq 0$ and $y_0/4 \leq y_1 \leq y_0/2$. Equality at the lower bound means $a=0$; at the upper bound $b=0$. If $y_0=0$, both bounds force $y_1=0$ and the state is $(0,0)$. Conversely these inequalities make the formulas nonnegative, and substitution yields the supplied readings, establishing sufficiency. A negative recovered amount signals incompatible data/model, not a second admissible solution.

4 / hints to offer only after exploration

1. Apply nonnegativity separately to the two reconstructed amounts.

Student page 3: Choose rates and timing with noise in mind

Continuation: exact deterministic error bounds

The initial total y_0 is exact. One later reading has additive error e with $|e| \leq 1/100$. This absolute error bound is the same at every observation time. All rates are known exactly; compute the inverse from the reported reading.

5 / prompt and solution

Keep the first rate $1/2$. Choose the second rate as $1/4$ or $49/100$. For a true initial state $(4,4)$, predict the time-1 reading and find the greatest possible error in either reconstructed amount. Prove a formula explaining the difference.

Solution: For distinct rates r, s , $a=(y_1-s y_0)/(r-s)$, $b=(r y_0-y_1)/(r-s)$. A reading error e gives $\Delta a=e/(r-s)$, $\Delta b=-e/(r-s)$. With $r=1/2, s=1/4$, true $y_1=3$ and the sharp bound is $(1/100)/(1/4)=1/25$. With $s=49/100$, true $y_1=99/25$ and the bound is 1. Both signs of the extreme error leave the true $(4,4)$ reconstruction nonnegative, so these maxima are physically attainable here. Unequal rates ensure exact uniqueness but a small separation amplifies noise.

5 / hints to offer only after exploration

1. Subtract the exact-total equation before propagating the reading error.

Choose when one sensor can separate two tanks

6 / prompt and solution

Now choose when to observe. Rates are $r=3/4$ and $s=1/2$. Use y_0 and one reading y_k , where k is any positive integer. Find the k that gives the smallest worst-case reconstruction-error bound, and prove it beats EVERY other allowed k .

Solution: Replace r,s in the time-1 inversion by r^k, s^k . The error in each recovered amount is at most $\varepsilon/[r^k - s^k]$, with $\varepsilon=1/100$. Let $d_k=(3/4)^k - (1/2)^k$. Then $d_{k+1} - d_k = (1/4)(1/2)^k [2 - (3/2)^k]$. This is positive at $k=1$ and negative for every integer $k \geq 2$, since $(3/2)^2 > 2$ and the power increases. Thus d_k is uniquely largest at $k=2$, with $d_2=5/16$, and the smallest bound is $16\varepsilon/5=4/125$. Waiting one extra step helps, but arbitrarily late observations make the difference vanish and conditioning worse. Sharpness is attained at interior states for which both signs of ε preserve nonnegative recovery; otherwise nonnegative constraints may further reduce the feasible error range.

6 / hints to offer only after exploration

1. Compare consecutive gaps $d_{k+1} - d_k$ rather than trying many times.

Optional facilitator extension 1

Gate: Two-by-two determinants and distinguishing every real state.

For $x_{n+1} = \text{diag}(r,s)x_n$ and output $y_n = \alpha a_n + \beta b_n$, decide exactly when the first two readings identify every real initial state. Relate this to nonnegative exceptional states.

Solution: The two-reading matrix is $[[\alpha, \beta], [\alpha r, \beta s]]$, with determinant $\alpha\beta(s-r)$. If it is nonzero, the matrix is invertible and every state is uniquely recovered. If $\alpha=0$ or $\beta=0$, an entire component is unseen at every time. If $r=s$, every later row is a scalar multiple of the first, so a nonzero kernel direction remains. Thus the condition is also necessary for identifying every state from the entire record in this diagonal model. Restricting to nonnegative states can make special boundary data unique, such as a positive-weight sum equal to zero, but it does not restore global observability when the determinant vanishes: an interior positive state can be perturbed slightly in a kernel direction.

Mathematics and prior use

Observability and conditioning of a two-state discrete-time linear system; the singular direction appears when decay rates coincide or a sensor ignores one component.

Prior use: Week 6 uses freely chosen code questions. GA-14 studies aliasing of periodic signals; AP-19 places a travel-time receiver. Here a fixed output law observes a changing linear state, and delayed readings may improve or worsen sensitivity.

Choose when one sensor can separate two tanks

Source: Åström and Murray, Feedback Systems: Output Feedback

Second-edition chapter 8, Observability and State Estimation; chapter landing-page scope.. Use state reconstruction from outputs as context. The complete discrete two-tank calculation and bounded-noise timing proof are derived here. Evidence: Checked 2026-09-26: the official chapter listing identifies observability/state estimation. Its linked chapter PDF returned a fetch error; no uninspected theorem from that PDF is invoked.

https://www.fbswiki.org/wiki/index.php/Feedback_Systems:_An_Introduction_for_Scientists_and_Engineers

Source: MIT 6.3100, DT Observer

PDF pp. 15-20, state/output representation and sensor noise; pp. 32-33 and 40, discrete-time state and output equations.. Provides primary course context for state versus measured output and noise. This worksheet uses a new exact diagonal discrete model, not the motor example. Evidence: Opened 2026-09-26: inspected the state/output equations, additive output-noise diagrams, and the discrete-time form $x[n+1]=A_d x[n]+B_d u[n]$, $y[n]=C_d x[n]+D_d u[n]$. Reconstruction and optimal-delay claims below are independently proved.

https://introcontrol.mit.edu/_static/spring25/lectures/DT_Observer.pdf

Four messages through a damaged strip

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

First-pilot candidate. Concrete strips allow broad entry; the complete counting proof needs organizing possible received words. The staged supplied code is labeled honestly so a tested example is not mistaken for an independent construction.

An attained five-bit construction and a four-bit counting obstruction, with a precise separation between correction, detection and violations of the error promise.

Prerequisites and preparation

Core: Binary patterns, counting disagreements and an adversarial guarantee. Powers of two through 32 and counting disjoint neighborhoods for the optimal-length proof.

Extensions: XOR parity checks, two-bit syndromes and a finite decoding proof.

Reading and representation: Rules can be spoken; black/white patterns also carry 0/1 labels. The location of an error is unknown. At most one includes no error.

Materials: Four message symbols, blank five-square strips, 0/1 counters, pencil. Give page 2 only after design attempts; its supplied codebook is a staged comparison, not part of the initial challenge. Preparation: about 8 minutes.

0-10 damage and compare strips; 10-30 construct a candidate five-bit code; 30-45 verify a code and decoding rule; 45-60 prove a four-bit obstruction; later 20 minutes for detection versus correction.

Launch and facilitation

Invent four message strips. Another person may secretly change one square, or leave it alone. Can the receiver recover the message every time, even when the adversary chooses the worst square?

Student page 1: Invent a code that an adversary cannot confuse

Core: choose patterns and try to break them

A message is sent as a fixed-length strip of 0s and 1s. An adversary may flip at most ONE bit, with its position hidden, or make no change. You need four distinct messages. The receiver knows the codebook, but not which message was sent.

Four messages through a damaged strip

1 / prompt and solution

Choose four five-bit codewords for your messages. Let a partner try to produce the SAME received strip from two different messages using at most one flip from each. Revise your code if that happens. Record a successful code or the strongest obstruction you found.

Solution: A valid example is 00000,11100,10011,01111, whose pair distances are 3,3,4,4,3,3. Any chosen four words work exactly when every pair differs in at least three coordinates; prompt 2 proves this criterion. If a pair differs in one coordinate, the first word itself is reachable unchanged from itself and by one flip from the second. If they differ in two, flip one differing coordinate from the first and the other from the second to obtain the same midpoint word. These give explicit obstructions to failed proposals. Do not reveal the sample code until page 2.

1 / hints to offer only after exploration

1. Compare the pairs that look most alike, not only a few random errors.

2 / prompt and solution

How far apart must EVERY pair of codewords be to make one-bit correction possible? Prove necessity by constructing an ambiguous received word when the condition fails, and prove sufficiency for every allowed error.

Solution: Hamming distance counts differing coordinates. Distance at least 3 is necessary by the distance-1 and distance-2 ambiguities described above; distance 0 already confuses messages without noise. It is sufficient because a word within one flip of two codewords would put those originals within at most two flips of each other, by changing via the common received word. Thus radius-one neighborhoods are disjoint. A decoder selects the unique codeword at distance at most one whenever such a word exists.

2 / hints to offer only after exploration

1. Join two possible originals through the common received word.

Student page 2: Audit a construction, then rule out a shorter one

Core proof: correction and a counting obstruction

Now compare with this supplied codebook: A=00000, B=11100, C=10011, D=01111. It is a construction to verify, whether or not it matches your first design. Keep the at-most-one-flip promise.

Four messages through a damaged strip

3 / prompt and solution

Audit all six pairs. Decode 10111, 01000 and 11100, explaining every correction or no-change decision. Then design a short universal decoding instruction for this codebook.

Solution: Distances AB,AC,AD,BC,BD,CD are 3,3,4,4,3,3. Received 10111 is one flip from C=10011 (position 3); 01000 is one flip from A=00000 (position 2); 11100 equals B unchanged. Each is farther than one from the other codewords by the distance proof. The universal instruction is: count disagreements with each codeword; if exactly one has distance 0 or 1, return its message. If none does, report that the one-flip promise or codebook has failed; do not invent a guaranteed message. Two such candidates cannot occur for this verified code.

3 / hints to offer only after exploration

1. The adversary is allowed to leave the strip unchanged.

4 / prompt and solution

Could ANY four-bit codebook carry four messages with the same guarantee? Count the received words possible from one codeword and use your disjointness proof. Explain why this also rules out every shorter length.

Solution: One four-bit codeword has five distinct radius-one received words: itself and four single-coordinate flips. Four messages would need $4 \times 5 = 20$ disjoint words, but the entire four-bit space has only $2^4 = 16$. Therefore length 4 is impossible for every codebook. A shorter valid code could be padded by fixed zero bits to length 4; distances would stay unchanged and the distance criterion would still give correction, contradicting impossibility. The verified five-bit construction attains the minimum length. Counting one codebook that fails would not prove this universal bound.

4 / hints to offer only after exploration

1. The neighborhoods must be disjoint for all messages at once.

Student page 3: What can the receiver honestly conclude?

Continuation: promises, detection and correction

Use the supplied five-bit codebook. A decoder is allowed to say “the error promise failed” when no codeword lies within one flip.

Four messages through a damaged strip

5 / prompt and solution

How many five-bit received words lie within one flip of this codebook? Find a received word outside all those neighborhoods and prove that it violates the promise. Then show how TWO flips can fool the decoder into returning the wrong message.

Solution: Each of four disjoint neighborhoods has $1+5=6$ words, so 24 of the 32 words are covered and 8 are outside. For example 00110 has distances 2,3,3,2 from A,B,C,D, so it cannot arise under the one-flip promise. From A=00000, flipping positions 1 and 2 produces 11000, which is only one flip from B=11100; the decoder returns B incorrectly. An uncovered word certifies a violated promise, but some two-flip violations imitate a promised transmission and cannot be recognized by this decoder.

5 / hints to offer only after exploration

1. A failed promise need not always produce an obvious alarm.

6 / prompt and solution

Change the goal: detect any single flip, but do not necessarily correct it. Construct four messages using THREE bits and prove that every one-bit change is detected. Exhibit an ambiguity showing why detection is weaker than correction.

Solution: Use 000,011,101,110, the three-bit words of even parity. Any one-bit flip changes parity, so no corrupted word belongs to the codebook and every single flip is detected. But received 001 is one flip from 000,011 and 101, so it does not identify a unique original. Three bits can therefore detect for four messages even though four bits cannot correct. Detection means noticing a change, not naming its position or recovering the original.

6 / hints to offer only after exploration

1. Can all your codewords have the same parity?

Four messages through a damaged strip

Worked visual / facilitator only

Six distance checks certify the supplied construction. Every allowed received strip has exactly one codeword within distance one.

A	0	0	0	0	0	AB: 3	AC: 3
B	1	1	1	0	0	AD: 4	BC: 4
C	1	0	0	1	1	BD: 3	CD: 3
D	0	1	1	1	1		

Each length-five codeword has $1+5=6$ allowed received words. The four disjoint neighborhoods cover 24 of the 32 possible strips. The outside word 00110 has distances 2, 3, 3, 2; a promise-respecting transmission cannot produce it.

For the extension, the five single-bit syndromes are (1,0), (1,1), (0,1), (1,0), (0,1). All single flips are detected, but repeated columns prevent unique correction. More generally, exactly the 24 error patterns with nonzero syndrome are detected.

Optional facilitator extension 1

Gate: XOR arithmetic and two-component binary checks.

For a different length-five code, define $h(w)=(w_1+w_2+w_4, w_2+w_3+w_5)$ modulo 2. Can taking all words with $h(w)=(0,0)$ correct every one-bit error? Prove your answer, and identify exactly what the check does detect.

Solution: The two checks define 8 codewords because w_1, w_2, w_3 are free and force $w_4=w_1+w_2$ and $w_5=w_2+w_3$ modulo 2. A flip in positions 1,2,3,4,5 changes h by (1,0), (1,1), (0,1), (1,0), (0,1), respectively. Every one-bit flip has nonzero syndrome and is detected, but positions 1 and 4 have identical syndromes, as do 3 and 5. In particular 00000 and 10010 are both codewords at distance 2; received 10000 is one flip from each. Hence these checks do not correct every one-bit error. More parity checks alone are not a proof unless their error signatures separate all allowed errors. More generally, for a codeword c and an XOR error pattern e , $h(c+e)=h(c)+h(e)=h(e)$. The check detects exactly the 24 nonzero-syndrome error patterns among all 32 patterns. It misses the seven nonzero kernel errors; the eighth kernel pattern is no error. Thus the exact detection claim includes, but is not limited to, every one-bit error.

Four messages through a damaged strip

Mathematics and prior use

Minimum distance, adversarial one-bit correction, sphere packing and a finite optimal code. No asymptotic Shannon capacity theorem is claimed.

Prior use: Week 6 uses coding and optimal queries. This family uses a noisy channel with unknown error position, disjoint Hamming neighborhoods and an optimal fixed length. AP-09 relates integer frequencies, not message recovery.

Source: Claude Shannon, A Mathematical Theory of Communication

Part II, §11, PDF p. 19, noisy discrete-channel representation.. Use noisy transmission as context. The four-message code, all-distance audit and sphere-packing proof are explicit finite mathematics, not a deduction from the asymptotic theorem. Evidence: Reopened 2026-09-26: inspected the distinction between sent and received signals and the effect of noise. Our noise is worst-case at most one bit, with no probability distribution assumed.

<https://people.math.harvard.edu/~ctm/home/text/others/shannon/entropy/entropy.pdf>

Design a meter that disturbs the circuit less

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Promising for algebra-ready learners. The branch diagram must be explicit. A physical voltmeter demonstration is optional future work; the present guarantees concern only the stated ideal model.

A correct loaded-circuit solution, a sharp resistance threshold for a chosen error tolerance, and a counterexample to treating agreement between meters as evidence of non-disturbance.

Prerequisites and preparation

Core: Supplied Ohm and node-balance laws, fractions and one linear equation; relative error and inequalities for a meter design. The entire activity is a paper DC model.

Extensions: Symbolic positive resistances and a two-resistor equivalent-source calculation.

Reading and representation: Read source voltage, ground and branch connections. A meter reads its own terminal voltage and simultaneously adds the specified resistance.

Materials: Printed circuit diagrams, pencil and optional calculator for checks. No powered construction is part of the activity. All ideal values and laws are supplied. Preparation: about 7 minutes.

0-10 trace branches and test voltage proposals; 10-25 attach a chosen meter; 25-45 derive a tolerance rule; 45-60 compare energy and multiple agreeing meters. General resistance formula is a later extension.

Launch and facilitation

The meter reports a voltage but also becomes a new branch. Choose a meter resistance, predict whether the report will agree with the no-meter calculation, and test your prediction using current balance.

Student page 1: The meter is another branch

Core: test a voltage using current balance

An ideal source fixes the top node at 6 V relative to ground at 0 V. A $2\ \Omega$ resistor joins the top node to X; a $1\ \Omega$ resistor joins X to ground. Supplied laws: current through resistance R is voltage drop/R; incoming current equals outgoing current at X. A meter of resistance $R > 0$ joins X to ground and reports voltage X.

1 / prompt and solution

With NO meter attached, choose a trial voltage between 0 and 6 and compute both currents. Adjust until current balances. Find the exact voltage X and explain why it is unique.

Solution: Incoming current is $(6-X)/2$ and outgoing current is $X/1$. Balance gives $6-X=2X$, hence $X=2$ V and both currents 2 A. The linear equation has nonzero coefficient 3, so this voltage is unique. Any trial voltage above 2 has outgoing current larger than incoming; below 2 has the reverse imbalance.

Design a meter that disturbs the circuit less

1 / hints to offer only after exploration

1. The source fixes the top node, not the internal node X.

2 / prompt and solution

Choose meter resistance $R=1, 2$ or $6\ \Omega$. Predict the direction of the change, then solve the LOADED node equation. Find all three currents and compare their sum at X with the no-meter result.

Solution: The equation becomes $(6-X)/2=X+X/R$. For $R=1$, $X=6/5\text{ V}$ and currents are $12/5\text{ A}$ in, $6/5\text{ A}$ through the original $1\ \Omega$ branch and $6/5\text{ A}$ through the meter. For $R=2$, $X=3/2\text{ V}$ and currents are $9/4, 3/2, 3/4\text{ A}$. For $R=6$, $X=9/5\text{ V}$ and currents are $21/10, 9/5, 3/10\text{ A}$. Each satisfies balance. In every case X falls below 2 V while the source current rises above 2 A . The meter current must be included rather than using the old node voltage as fixed.

2 / hints to offer only after exploration

1. The incoming current splits into two outgoing branches.

Student page 2: Specify how accurate the meter must be

Core proof: a sharp design inequality

Allow any real meter resistance $R>0$. Define relative loading error as $E=(2-X)/2$, comparing the loaded reading with the exact no-meter voltage 2 V . Component values and the source remain exact.

3 / prompt and solution

Find X and each branch current as formulas in R. Prove whether ANY finite positive R gives exactly the no-meter voltage. Explain the ideal limit as R becomes arbitrarily large.

Solution: Solving balance gives $X=6R/(3R+2)$, original-branch current $I_1=X$, meter current $I_m=6/(3R+2)$, and incoming current $I_{in}=6(R+1)/(3R+2)=I_1+I_m$. The voltage shortfall is $2-X=4/(3R+2)>0$ for finite $R>0$, so no such meter reads exactly 2. As $R\rightarrow\infty$, $X\rightarrow 2$, $I_m\rightarrow 0$ and $I_{in}\rightarrow 2$. This is the ideal zero-loading limit. $R=0$ is excluded from the formula derivation because the meter would be a short circuit, not a positive resistor.

3 / hints to offer only after exploration

1. Solve the loaded equation before taking a limit.

Design a meter that disturbs the circuit less

4 / prompt and solution

Choose a tolerance $0 < \epsilon < 1$. Find the smallest R that guarantees $E \leq \epsilon$, proving both directions. Test a 1% tolerance. Does meeting this design bound account for every possible real-world measurement error?

Solution: $E = 2/(3R+2)$. Since the denominator is positive, $E \leq \epsilon$ is equivalent to $R \geq 2(1-\epsilon)/(3\epsilon)$. Equality attains the tolerance, so the threshold is sharp. For $\epsilon = 1/100$ the minimum is 66Ω and the reading is $99/50 = 1.98 \text{ V}$. The bound concerns loading in this exact ideal circuit; it does not account for uncertain source/resistor values, meter calibration, or effects absent from the model.

4 / hints to offer only after exploration

1. Keep relative error separate from the voltage difference in volts.

Student page 3: Agreement is not the same as non-disturbance

Continuation: energy and several meters

Supplied power laws: the source supplies $6 I_{\text{in}}$ watts, and a resistor dissipates $I^2 R$, equivalently voltage-drop²/ R . Multiple meters are independent branches electrically, not independent copies of an undisturbed quantity.

5 / prompt and solution

For $R = 2 \Omega$, audit source power against the sum of all three resistor dissipations. Then model $R = 0$ as an ideal wire from X to ground, separately from the positive- R formula. Find its node voltage, currents and resistor-power balance.

Solution: At $R = 2$, source power is $6(9/4) = 27/2 \text{ W}$. Top 2Ω dissipation is $(9/4)^2 \cdot 2 = 81/8 \text{ W}$; original 1Ω gives $(3/2)^2 = 9/4 \text{ W}$; meter gives $(3/4)^2 \cdot 2 = 9/8 \text{ W}$. Their sum is $108/8 = 27/2 \text{ W}$. An ideal short sets $X = 0$: top current is 3 A , original 1Ω current is 0 , and wire current is 3 A by balance. Source power is 18 W , all dissipated in the top 2Ω resistor; the ideal wire has zero voltage drop and no dissipation. Never compute wire current by dividing 0 by 0 .

5 / hints to offer only after exploration

1. Use the branch currents already verified, and keep the zero-resistance case separate.

Design a meter that disturbs the circuit less

6 / prompt and solution

Attach TWO meters, each $R=2\ \Omega$, across X and ground. They agree with each other: do they reproduce the no-meter voltage? Generalize to m identical meters of resistance $R>0$, where m is a positive integer, and design R so the combined loading error is at most ϵ .

Solution: Two $2\ \Omega$ meters add two outgoing currents $X/2$, so balance is $(6-X)/2=2X$ and $X=6/5\text{ V}$. Both report this same loaded value, not the no-meter 2 V . With m meters, $(6-X)/2=X+mX/R$, so $X=6R/(3R+2m)$ and relative error is $2m/(3R+2m)$. Thus $E\leq\epsilon$ is equivalent to $R\geq 2m(1-\epsilon)/(3\epsilon)$. At 1% tolerance each of two identical meters must have at least $132\ \Omega$. Agreement arises because the meters share a node; it does not show that their extra branches are negligible.

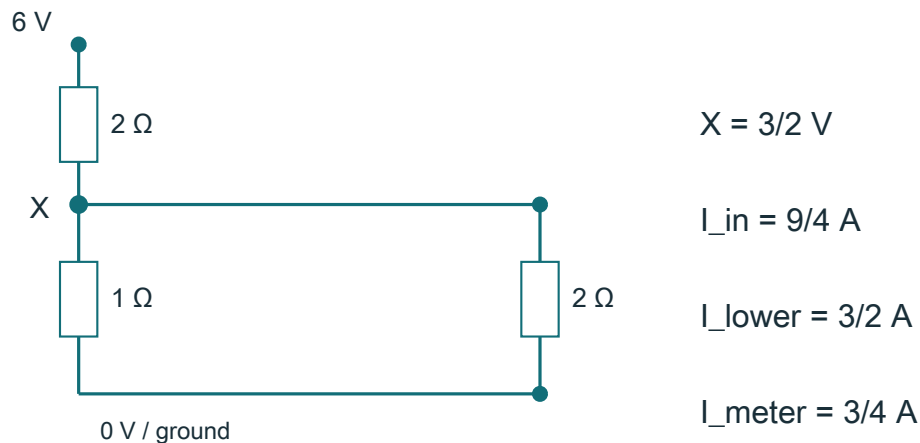
6 / hints to offer only after exploration

1. Add every meter branch to the node equation, even if the readings are identical.

Design a meter that disturbs the circuit less

Worked visual / facilitator only

Loaded circuit with $R=2\ \Omega$. The listed I values are currents, while X is a voltage. All branches meet at the filled junctions.



Source power: $6 \cdot (9/4) = 27/2\text{ W}$. Dissipation: upper $81/8\text{ W}$, lower $9/4\text{ W}$, meter $9/8\text{ W}$; these sum to $27/2\text{ W}$. The voltage $3/2\text{ V}$ is smaller than the unloaded 2 V even though the meter reports its own loaded voltage exactly.

Optional facilitator extension 1

Gate: Symbolic algebra with strictly positive resistances.

Replace 6 V by $V > 0$, the upper resistor by $a > 0$ and the lower by $b > 0$. Derive the no-meter voltage V_0 , the loaded voltage with a meter $R > 0$, and a relative-error rule. Identify a single resistance that controls the loading.

Solution: No-meter balance $(V-X)/a = X/b$ gives $V_0 = Vb/(a+b)$. With the meter, $(V-X)/a = X/b + X/R$ gives $X = VbR/[ab + R(a+b)]$. Put $R_* = ab/(a+b) > 0$. Then $X = V_0 R/(R + R_*)$, and relative error is $R_*/(R + R_*)$. Hence the sharp tolerance condition is $R \geq R_*(1-\epsilon)/\epsilon$ for $0 < \epsilon < 1$. This proves an equivalent-source description for this two-resistor circuit directly; it does not prove the general Thevenin theorem for arbitrary networks.

Design a meter that disturbs the circuit less

Mathematics and prior use

Kirchhoff node balance, sensor loading, a quantitative approximation to a noninvasive limit, and energy conservation as an independent audit.

Prior use: AP-10 uses linear spring networks and AP-11 uses flux conservation. This circuit adds a measurement branch, so the learner must recompute the object being observed; it is not just another series/parallel reduction exercise.

Source: David Tong, Electromagnetism

§4.1.3, printed pp. 74-77 / PDF pp. 8-11; Ohm law (4.4) and Joule heating.. Supply ideal resistive component laws and node-current conservation. The meter-loading network, tolerance design and multiple-meter analysis are new exact instances. Evidence: Reopened 2026-09-26: inspected the stated extra assumption of Ohm behavior and the I^2R dissipation law. A general meter or component is not claimed to obey every ideal value.

<https://davidtong.org/pdfs/teaching/electromagnetism/electro3.pdf>